

Topological Energy Condensate Theory (TECT): 31

Jusang Lee*
Independent Researcher, Incheon, Korea
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I. NEXT EXACT ENLARGED QUARTIC REDUCED SHELL KERNEL BEYOND THE FAILED (u, v, κ) TRUNCATION

II. CURRENT STAGE

We freeze the currently best operational backend and the corrected local seed as

$$\boxed{(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096).} \quad (1)$$

The broad branch search is no longer the issue. The actual obstruction is that the compressed quartic shell kernel

$$U_{\alpha\beta}^{\text{simple}} = u \delta_{\alpha\beta} + v + \kappa (\hat{k}_\alpha \cdot \hat{k}_\beta)^2 \quad (2)$$

is too small to reproduce the corrected projected quartic data.

III. 1. EXACT PAIR-CLASS DECOMPOSITION ON THE BCC SHELL

Let S_* be the selected commensurate BCC shell, and let

$$\hat{k}_\alpha \in S_*, \quad \alpha = 1, \dots, N_*.$$

For the BCC shell, ordered shell pairs (α, β) fall into four symmetry classes:

1. diagonal:

$$\alpha = \beta;$$

2. opposite:

$$\hat{k}_\beta = -\hat{k}_\alpha;$$

3. perpendicular:

$$\hat{k}_\alpha \cdot \hat{k}_\beta = 0;$$

4. oblique:

$$|\hat{k}_\alpha \cdot \hat{k}_\beta| = \frac{1}{2}.$$

These four classes exhaust all pair possibilities on the BCC shell.

Define the corresponding class projectors

$$P_{\alpha\beta}^{(d)} := \mathbf{1}_{\{\alpha=\beta\}}, \quad P_{\alpha\beta}^{(\text{opp})} := \mathbf{1}_{\{\hat{k}_\beta = -\hat{k}_\alpha\}}, \quad (3)$$

$$P_{\alpha\beta}^{(\perp)} := \mathbf{1}_{\{\hat{k}_\alpha \cdot \hat{k}_\beta = 0\}}, \quad P_{\alpha\beta}^{(\text{obl})} := \mathbf{1}_{\{|\hat{k}_\alpha \cdot \hat{k}_\beta| = 1/2\}}. \quad (4)$$

By construction,

$$\boxed{P^{(d)} + P^{(\text{opp})} + P^{(\perp)} + P^{(\text{obl})} = \mathbf{1}} \quad (5)$$

on ordered shell-pair space.

* jusanglee@kakao.com

IV. 2. MINIMAL ENLARGED EXACT QUARTIC KERNEL

The minimal enlarged quartic shell kernel is therefore

$$U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_{\perp} P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})}. \quad (6)$$

Equivalently,

$$U_{\alpha\beta}^{(4c)} = \begin{cases} u_d, & \alpha = \beta, \\ u_{\text{opp}}, & \hat{k}_{\beta} = -\hat{k}_{\alpha}, \\ u_{\perp}, & \hat{k}_{\alpha} \cdot \hat{k}_{\beta} = 0, \\ u_{\text{obl}}, & |\hat{k}_{\alpha} \cdot \hat{k}_{\beta}| = 1/2. \end{cases} \quad (7)$$

This is the smallest shell kernel that is fully resolved at the level of the four observed quartic class averages

$$(U_d, U_{\text{opp}}, U_{\perp}, U_{\text{obl}}).$$

V. 3. WHY THE (u, v, κ) TRUNCATION MUST FAIL

For the old kernel (2), the four class values are not independent:

$$U_d = u + v + \kappa, \quad U_{\text{opp}} = v + \kappa, \quad U_{\perp} = v, \quad U_{\text{obl}} = v + \frac{\kappa}{4}. \quad (8)$$

Hence the old kernel imposes the exact algebraic identity

$$4U_{\text{obl}} - 3U_{\perp} - U_{\text{opp}} = 0. \quad (9)$$

Therefore the defect of the old truncation is precisely

$$\Delta_{\text{ker}} := 4U_{\text{obl}} - 3U_{\perp} - U_{\text{opp}}. \quad (10)$$

So once $\Delta_{\text{ker}} \neq 0$, the compressed kernel $u\delta_{\alpha\beta} + v + \kappa(\hat{k}_{\alpha} \cdot \hat{k}_{\beta})^2$ is mathematically incapable of reproducing the projected quartic data.

VI. 4. EXACT COEFFICIENT IDENTIFICATION FROM THE PROJECTED DATA

The enlarged kernel (6) is directly calibrated by the extracted class means:

$$u_d = U_d, \quad u_{\text{opp}} = U_{\text{opp}}, \quad u_{\perp} = U_{\perp}, \quad u_{\text{obl}} = U_{\text{obl}}. \quad (11)$$

Thus, for the corrected branch,

$$u_d = 2493.1087739398513, \quad u_{\text{opp}} = 14874.171297437068, \quad u_{\perp} = 9027.17290917827, \quad u_{\text{obl}} = 2244.8894311890813. \quad (12)$$

Likewise, for the uncorrected branch,

$$u_d = 1064.2600262852873, \quad u_{\text{opp}} = 8617.71246090361, \quad u_{\perp} = 5121.74549258667, \quad u_{\text{obl}} = 1329.8059441886778. \quad (13)$$

Hence the four-class kernel is not a heuristic ansatz: it is the exact minimal interpolation of the already extracted quartic class data.

VII. 5. REDUCED QUARTIC FORM ON THE ORBIT-AMPLITUDE SECTOR

Let the shell modes be grouped into the three orbit sectors

$$A, \quad E, \quad O,$$

and let the corresponding real orbit amplitudes be

$$a_A, \quad a_E, \quad a_O.$$

Introduce

$$r_A := a_A^2, \quad r_E := a_E^2, \quad r_O := a_O^2.$$

Then the quartic reduced contribution takes the form

$$F_4^{\text{red}} = \frac{1}{4} \sum_{\alpha, \beta \in S_*} U_{\alpha\beta}^{(4c)} |\psi_\alpha|^2 |\psi_\beta|^2. \quad (14)$$

After orbit reduction, this becomes

$$F_4^{\text{red}}(r_A, r_E, r_O) = \frac{1}{4} \sum_{X, Y \in \{A, E, O\}} M_{XY} r_X r_Y, \quad (15)$$

where

$$M_{XY} = N_{XY}^{(d)} u_d + N_{XY}^{(\text{opp})} u_{\text{opp}} + N_{XY}^{(\perp)} u_{\perp} + N_{XY}^{(\text{obl})} u_{\text{obl}}. \quad (16)$$

Here $N_{XY}^{(c)}$ is the purely combinatorial count of ordered shell pairs with first index in orbit X , second index in orbit Y , and pair-class c .

Thus the next reduced closure problem is finite and explicit: one must compute the orbit-count matrices

$$N_{XY}^{(d)}, \quad N_{XY}^{(\text{opp})}, \quad N_{XY}^{(\perp)}, \quad N_{XY}^{(\text{obl})},$$

and then assemble the 3×3 reduced quartic matrix M_{XY} .

VIII. 6. RESIDUAL-LEVEL REDUCED SYSTEM

Since the correct reduced object is the projected residual map rather than a naive reduced free-energy critical point, the quartic contribution should enter the residual equations as

$$G_X^{(4)}(a_A, a_E, a_O) = \frac{\partial F_4^{\text{red}}}{\partial a_X}, \quad X \in \{A, E, O\}. \quad (17)$$

Using $r_X = a_X^2$, one obtains

$$G_X^{(4)} = a_X \sum_{Y \in \{A, E, O\}} M_{XY} r_Y. \quad (18)$$

Therefore the corrected reduced residual system should be upgraded from the failed three-parameter quartic compression to the class-resolved finite system

$$G_X(a_A, a_E, a_O) = G_X^{(2)} + a_X \sum_Y M_{XY} r_Y + \cdots, \quad X = A, E, O. \quad (19)$$

IX. 7. SHARP LOGICAL CONSEQUENCE

The next exact nonredundant step is therefore not a new backend scan and not a new branch search. It is the finite class-resolved upgrade:

$$\boxed{(u, v, \kappa) \longrightarrow (u_d, u_{\text{opp}}, u_{\perp}, u_{\text{obl}}) \longrightarrow M_{XY} \longrightarrow G_X^{(4)}} \quad (20)$$

If the residual misfit remains large even after this four-class upgrade, then the next obstruction is no longer pair-class compression but genuine higher shell-tensor structure, i.e. dependence on full orbit-labeled quartic couplings rather than only pair classes.

X. FINAL SUMMARY

The failed kernel

$$U_{\alpha\beta} = u\delta_{\alpha\beta} + v + \kappa(\hat{k}_{\alpha} \cdot \hat{k}_{\beta})^2$$

is a three-parameter submanifold inside the full four-class BCC shell-pair space.

Because the BCC shell has four distinct pair classes

$$(d, \text{opp}, \perp, \text{obl}),$$

the minimal exact enlarged reduced quartic kernel is

$$U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_{\perp} P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})}.$$

The old kernel is equivalent to the restricted relations

$$U_d = u + v + \kappa, \quad U_{\text{opp}} = v + \kappa, \quad U_{\perp} = v, \quad U_{\text{obl}} = v + \kappa/4,$$

and therefore necessarily imposes

$$4U_{\text{obl}} - 3U_{\perp} - U_{\text{opp}} = 0.$$

Since the extracted corrected quartic data violate this relation strongly, the simple (u, v, κ) truncation is exactly ruled out.

Hence the next mathematically correct reduced closure is:

$$(u, v, \kappa) \rightarrow (u_d, u_{\text{opp}}, u_{\perp}, u_{\text{obl}}) \rightarrow M_{XY} \rightarrow G_X^{(4)}.$$

This is the minimal exact enlarged quartic reduced shell kernel consistent with the current corrected backend stage.

XI. TECT MICROSCOPIC PDE HONESTY AUDIT AND NO-LOOP FREEZE

XII. GOAL

Before continuing any further reduction or numerical insertion, we freeze the exact logical hierarchy of the TECT PDE program.

The purpose is:

$$\boxed{\text{separate once and for all the full exact target PDE, the strongest current working microscopic PDE, and the reduced residu}} \quad (21)$$

This prevents future loopbacks caused by mixing different layers of description.

XIII. 1. THE THREE NON-EQUIVALENT LAYERS

There are three distinct objects in the current program.

A. Layer I: full complete microscopic TECT action

This is the true final target. It is the action whose Euler–Lagrange equations would define the actual microscopic PDE of TECT without further ontological placeholders.

At the present stage, this object is *not yet uniquely derived unconditionally*. The notes state explicitly that the current derivation has not yet reached the final unconditional-complete stage.

B. Layer II: strongest explicit working microscopic branch

What is currently available is the strongest explicit working microscopic branch:

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \Psi]. \quad (22)$$

This is the sharpest currently explicit microscopic realization, but it is still a *working microscopic ansatz*, not yet the proven final complete TECT action.

C. Layer III: residual-based reduced shell extractor

The currently implemented backend extractor acts on the best confirmed numerical BCC branch and returns

$$h_{\alpha}, \quad U_{\alpha\beta}, \quad (\sigma_0, \eta), \quad (U_d, U_{\text{opp}}, U_{\perp}, U_{\text{obl}}),$$

and optionally compressed coefficients such as

$$(u, v, \kappa, \Delta_{\text{ker}}).$$

These quantities are not yet theorem-level coefficients of a fully closed microscopic energy functional. They are residual-based effective reduced coefficients of the current solver family.

Therefore the quartic kernel issue belongs to Layer III, not directly to Layer I.

XIV. 2. WHAT IS ALREADY EXACT AND WHAT IS NOT

A. Already closed at theorem/reduction level

The reduced infrared Dirac-type principal symbol is already fixed:

$$A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2). \quad (23)$$

Also fixed is the finite extraction target:

$$L(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_{\parallel}, \alpha, \beta). \quad (24)$$

So the infrared form is closed, while the microscopic coefficient point is not yet extracted from the final complete action.

B. Not yet exact

What is still open is the actual complete microscopic coefficient point

$$(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O) \quad \text{or equivalently} \quad (\lambda_{\parallel}, \alpha, \beta)$$

derived from the full background-dependent TECT Hessian.

Also still open are the explicit constitutive choices entering the full complete microscopic theory, such as the refined derivative sector, the higher projector invariants, and the extra mixing terms beyond the current explicit branch.

XV. 3. ALL APPROXIMATIONS / TRUNCATIONS / FROZEN ASSUMPTIONS USED SO FAR

We now list them sharply.

A. A. Working-action truncation

The current explicit PDE is not the final exact TECT PDE. It is the strongest explicit working microscopic ansatz (22).

So every calculation performed inside this branch is conditional on the statement:

$$\boxed{\text{the true complete microscopic TECT action may contain additional UV slots beyond the current } L_{\text{mic}}.}$$

B. B. Minimal complete-action ansatz with named open slots

To avoid hidden omissions, the correct object to freeze is not only the working branch, but the *minimal final-complete action ansatz*:

$$\boxed{S_{\text{TECT}}^{\text{comp}} = \int dt d^3x \left(\mathcal{L}_{\Psi}^{\text{core}} + \mathcal{L}_P^{\text{geom}} + \mathcal{L}_{\sigma}^{\text{vol}} + \mathcal{L}_X^{\text{II}} + \mathcal{L}_{\text{mix}}^{\text{comp}} \right).} \quad (25)$$

Here the explicit pieces are:

$$\mathcal{L}_{\Psi}^{\text{core}} = |\partial_t \Psi|^2 - \mu^2 \Psi^\dagger \Psi + Z \partial_i \Psi^\dagger \partial_i \Psi - Y (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) - u (\Psi^\dagger \Psi)^2 + \mathcal{L}_{\Psi}^{\text{ref}}, \quad (26)$$

$$\mathcal{L}_P^{\text{geom}} = \frac{\kappa_P}{2} \text{Tr}(\partial_\mu P \partial^\mu P) + \sum_{n \geq 2} a_n \mathcal{I}_n[P, \partial P, \partial^2 P, \dots], \quad (27)$$

$$\mathcal{L}_{\sigma}^{\text{vol}} = \frac{1}{2} (\partial_t \sigma)^2 - \frac{1}{2} M_{\sigma}^2 \sigma^2 - \frac{1}{2} c_{\sigma}^2 (\nabla \sigma)^2, \quad (28)$$

$$\mathcal{L}_X^{\text{II}} = \frac{1}{2} M_X^2 X_i^a X_i^a + \mathcal{L}_{X, \text{kin}}^{\text{ref}}, \quad (29)$$

$$\mathcal{L}_{\text{mix}}^{\text{comp}} = \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \Psi] + \mathcal{L}_{\text{mix}}^{\text{extra}}. \quad (30)$$

This is the key no-loop freeze: all still-open UV structure must appear explicitly as

$$\mathcal{L}_{\Psi}^{\text{ref}}, \quad \mathcal{I}_n, \quad \mathcal{L}_{X, \text{kin}}^{\text{ref}}, \quad \mathcal{L}_{\text{mix}}^{\text{extra}},$$

rather than being hidden implicitly.

C. C. Heavy-field elimination / Schur-complement reduction

Whenever we pass to the low shell carrier, we are using a reduced operator of the form

$$\boxed{H_{\text{eff}}^{\text{comp}}(k) = H_{\text{low, low}}(k) - H_{\text{low, rem}}(k) H_{\text{rem, rem}}(k)^{-1} H_{\text{rem, low}}(k).} \quad (31)$$

This is exact only after the full complete Hessian is known. Inside the current branch, it is exact relative to the chosen block decomposition, but still conditional on the working action.

D. D. Branch choice truncation

At the current sharpest explicit realization, the longitudinal sector sits on Branch B:

$$\Lambda_{\parallel}^{(0)} = 0, \quad \lambda_c \neq 0.$$

This is not yet proved to be the unique decision of complete TECT. So Branch B is the strongest explicit current realization, not yet the final unconditional decision of the full theory.

E. E. Solver-family background freeze

The current best operational backend point is frozen as

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62),$$

and the current corrected local seed is frozen as

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096).$$

These numbers define the strongest current numerical branch of the present solver family. They are operationally important, but they are not yet theorem-level exact constants of the final complete microscopic theory.

F. F. Residual-based extraction instead of free-energy extraction

The current solver exposes the stationary residual $R[\Psi]$, not an independently closed full microscopic free-energy functional.

Therefore the implemented projection extractor is residual-based. This means the extracted shell coefficients are *effective reduced coefficients of the current solver family*, not yet exact coefficients of a fully closed microscopic energy functional.

G. G. Quartic shell-kernel compression

The simplest quartic shell compression

$$U_{\alpha\beta} = u\delta_{\alpha\beta} + v + \kappa(\hat{k}_{\alpha} \cdot \hat{k}_{\beta})^2$$

has been decisively falsified by the extracted class averages.

Thus the failure of (u, v, κ) does *not* mean that the microscopic PDE was wrong. It means only that the reduced quartic shell kernel basis at Layer III was too small.

XVI. 4. THE MAXIMALLY HONEST PDE WE SHOULD NOW CALL THE “ACTUAL TARGET PDE”

To avoid wasting time, the object we should now solve and audit is the following explicit *named-slot complete PDE system*.

a. Condensate equation

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - 2u(\Psi^\dagger \Psi) \Psi + \mathcal{F}_\Psi^{\text{ref}} + \alpha_X \frac{\delta(X_i^a J_i^a)}{\delta \Psi^\dagger} + \beta_X \frac{\delta(X_i^a K_i^a)}{\delta \Psi^\dagger} + \frac{\delta \mathcal{L}_{\text{mix}}^{\text{extra}}}{\delta \Psi^\dagger}. \quad (32)$$

b. Projector equation

$$\kappa_P \partial_\mu \partial^\mu P + \mathcal{F}_P^{\text{geom}} + \beta_X \frac{\delta(X_i^a K_i^a)}{\delta P} + \frac{\delta \mathcal{L}_{\text{mix}}^{\text{extra}}}{\delta P} = 0. \quad (33)$$

c. *Volumetric equation*

$$\partial_t^2 \sigma - c_\sigma^2 \nabla^2 \sigma + M_\sigma^2 \sigma + \frac{\delta \mathcal{L}_{\text{mix}}^{\text{extra}}}{\delta \sigma} = 0. \quad (34)$$

d. *Class-II carrier equation*

$$M_X^2 X_i^a + \frac{\delta \mathcal{L}_{X,\text{kin}}^{\text{ref}}}{\delta X_i^a} + \alpha_X J_i^a[\Psi] + \beta_X K_i^a[P, \Psi] + \frac{\delta \mathcal{L}_{\text{mix}}^{\text{extra}}}{\delta X_i^a} = 0. \quad (35)$$

This is the strongest honest “exact target PDE” currently available: it contains every known mandatory slot, and every unknown part is explicitly named rather than silently omitted.

XVII. 5. WHY THE 4-CLASS SHELL KERNEL DOES NOT CONTRADICT THIS

The four-class enlargement

$$(U_d, U_{\text{opp}}, U_\perp, U_{\text{obl}})$$

belongs to the reduced shell quartic extraction layer.

The current notes already say that the simple (u, v, κ) quartic kernel fails strongly, and therefore the quartic reduced kernel basis must be enlarged beyond that truncation.

This does not alter (32)–(35). It only says that after solving or evaluating the current solver-family branch, the projected quartic shell matrix cannot be compressed to three parameters.

So:

$$\boxed{\text{4-class exact shell kernel} = \text{refinement of the reduced shell model},}$$

not

$$\boxed{\text{a contradiction of the microscopic PDE layer.}}$$

XVIII. 6. NO-LOOP PROTOCOL FROM NOW ON

To avoid future time loss, the program should be frozen in the following order.

A. Step 1: freeze the PDE hierarchy

We permanently distinguish:

1. **final exact target PDE**: the named-slot system (32)–(35);
2. **current explicit working branch**: (22);
3. **current numerical operational branch**: frozen backend plus corrected seed;
4. **reduced shell effective model**: extracted from the residual-based projector.

B. Step 2: never again compress an unknown slot silently

Every unclosed structure must remain visibly labeled as one of:

$$\mathcal{L}_\Psi^{\text{ref}}, \quad \mathcal{I}_n, \quad \mathcal{L}_{X,\text{kin}}^{\text{ref}}, \quad \mathcal{L}_{\text{mix}}^{\text{extra}}.$$

C. Step 3: keep the reduced quartic problem in its own layer

The quartic shell-basis enlargement is a Layer-III problem:

$$U_{\alpha\beta} \rightarrow (U_d, U_{\text{opp}}, U_{\perp}, U_{\text{obl}}) \rightarrow \text{enlarged reduced kernel.}$$

It must not be mistaken for a modification of the microscopic PDE itself.

D. Step 4: the only genuinely irreducible unresolved task

The true final unresolved task remains:

$$\boxed{\text{derive the full background-dependent Hessian of the complete action and extract } (\lambda_{\parallel}, \alpha, \beta)} \quad (36)$$

or equivalently

$$(\Lambda_{\parallel}^{(0)}, \lambda_c, \Lambda_E, \Lambda_O).$$

XIX. 7. FINAL FROZEN STATEMENT

From now on, the mathematically correct and maximally honest status statement is:

$$\boxed{\text{The reduced infrared PDE form is closed.}} \quad (37)$$

$$\boxed{\text{The strongest explicit current microscopic PDE is a working ansatz, not yet the final complete TECT PDE.}} \quad (38)$$

$$\boxed{\text{The current backend numbers are operational branch data of the present solver family.}} \quad (39)$$

$$\boxed{\text{The quartic } (u, v, \kappa) \text{ failure is a reduced shell-kernel failure, not by itself a PDE failure.}} \quad (40)$$

$$\boxed{\text{The final no-loop target is the named-slot complete PDE plus explicit Hessian coefficient extraction.}} \quad (41)$$

XX. FINAL SUMMARY

The correct no-loop conclusion is the following.

$$\boxed{\text{The reduced infrared PDE form is already closed.}} \quad (42)$$

$$\boxed{\text{The strongest explicit currently available microscopic PDE is } L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \Psi]} \quad (43)$$

but this is still a working microscopic ansatz, not yet the uniquely derived final complete TECT PDE.

$$\boxed{\text{The honest exact target PDE should now be frozen as the named-slot complete system}} \quad (44)$$

$$(\Psi, P, \sigma, X_i^a)$$

with explicit open sectors

$$\mathcal{L}_\Psi^{\text{ref}}, \quad \mathcal{I}_n[P, \partial P, \dots], \quad \mathcal{L}_{X, \text{kin}}^{\text{ref}}, \quad \mathcal{L}_{\text{mix}}^{\text{extra}}.$$

The current backend point and corrected seed are operational data of the present solver family,

 (45)

not yet theorem-level constants of the final complete microscopic theory.

The failure of the quartic (u, v, κ) kernel is a reduced shell-kernel failure,

 (46)

not by itself a failure of the microscopic PDE layer.

Therefore the correct frozen research order is:

named-slot complete PDE $\longrightarrow$ actual/working branch solution $\longrightarrow$ background-dependent Hessian $\longrightarrow L(k) \rightarrow K_i \rightarrow P_* K_i$

while the quartic-shell enlargement

$$(u, v, \kappa) \rightarrow (4\text{-class or larger})$$

is handled strictly as a reduced-model refinement layer.

This is the maximally rigorous freeze that prevents future loopbacks.

XXI. FROZEN STATUS DASHBOARD

Layer — Object — Status — Meaning

- Layer I — Final complete TECT action / microscopic PDE — Open — The true final microscopic PDE of TECT.
It is not yet uniquely and unconditionally closed.
- Layer II — Strongest explicit working microscopic branch — Frozen working — The sharpest currently explicit microscopic realization used in actual calculations.
- Layer III — Current backend numerical branch — Frozen operational — The best confirmed branch inside the present solver family.
- Layer IV — Reduced shell / residual extractor — Active reduced — The projected effective extraction layer. It must not be confused with the microscopic PDE itself.

XXII. EXACT / WORKING / REDUCED AUDIT TABLE

Object — Status — Sharp interpretation

- Reduced infrared symbol $A_1(p) = \lambda_{\parallel} p_{\parallel} \sigma_3 + \alpha(p_1 \sigma_1 + p_2 \sigma_2) + \beta(-p_2 \sigma_1 + p_1 \sigma_2)$ — Closed — The infrared Dirac-type form is already fixed at the theorem/reduction level.
- Current explicit microscopic branch $L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \Psi]$ — Frozen working — This is the strongest explicit current branch, but still not the final uniquely derived complete TECT action.
- Backend point $(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62)$ and corrected seed $\delta^{\text{final}} = (0.0096, -0.0288, 0.0096)$ — Frozen operational — These are operational data of the present solver family, not yet theorem-level constants of the final complete theory.
- Compressed quartic shell kernel $u\delta_{\alpha\beta} + v + \kappa(\hat{k}_{\alpha} \cdot \hat{k}_{\beta})^2$ — Rejected as insufficient — Its failure is a reduced shell-kernel failure, not by itself a failure of the microscopic PDE layer.

XXIII. NAMED OPEN SLOTS OF THE COMPLETE PDE

Open slot — Role — Status — Rule from now on

- $\mathcal{L}_\Psi^{\text{ref}}$ — Refined derivative sector of the condensate field Ψ — Open — Must always be shown explicitly; never hidden implicitly.
- $\mathcal{I}_n[P, \partial P, \partial^2 P, \dots]$ — Higher geometric projector invariants — Open — Must remain visible as the true geometric uncertainty sector.
- $\mathcal{L}_{X,\text{kin}}^{\text{ref}}$ — Kinetic/derivative refinement of the Class II carrier X_i^a — Open — Must remain explicitly named until closed or ruled out.
- $\mathcal{L}_{\text{mix}}^{\text{extra}}$ — Extra microscopic mixing terms among Ψ, P, σ, X — Open — All unknown mixing structure must be placed here rather than silently omitted.

XXIV. FROZEN COMPLETE PDE TEMPLATE

The maximally honest current target PDE should be frozen as the following named-slot complete system.

A. Condensate equation

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - 2u(\Psi^\dagger \Psi) \Psi + \mathcal{F}_\Psi^{\text{ref}} + \alpha_X \frac{\delta(X_i^a J_i^a)}{\delta \Psi^\dagger} + \beta_X \frac{\delta(X_i^a K_i^a)}{\delta \Psi^\dagger} + \frac{\delta \mathcal{L}_{\text{mix}}^{\text{extra}}}{\delta \Psi^\dagger}. \quad (47)$$

B. Projector equation

$$\kappa_P \partial_\mu \partial^\mu P + \mathcal{F}_P^{\text{geom}} + \beta_X \frac{\delta(X_i^a K_i^a)}{\delta P} + \frac{\delta \mathcal{L}_{\text{mix}}^{\text{extra}}}{\delta P} = 0. \quad (48)$$

C. Volumetric equation

$$\partial_t^2 \sigma - c_\sigma^2 \nabla^2 \sigma + M_\sigma^2 \sigma + \frac{\delta \mathcal{L}_{\text{mix}}^{\text{extra}}}{\delta \sigma} = 0. \quad (49)$$

D. Class-II carrier equation

$$M_X^2 X_i^a + \frac{\delta \mathcal{L}_{X,\text{kin}}^{\text{ref}}}{\delta X_i^a} + \alpha_X J_i^a[\Psi] + \beta_X K_i^a[P, \Psi] + \frac{\delta \mathcal{L}_{\text{mix}}^{\text{extra}}}{\delta X_i^a} = 0. \quad (50)$$

XXV. FROZEN GOAL STACK

Tag — Meaning — Current content — Completion criterion

- C — Core final goal — Final complete microscopic PDE together with actual coefficient extraction from the background-dependent Hessian — Complete the chain $L(k) \rightarrow K_i \rightarrow P_* K_i P_* \rightarrow (\lambda_\parallel, \alpha, \beta)$ from the complete action.
- M — Medium-term goal — Freeze the named-slot complete PDE and classify every open UV slot — Produce a closed audit of which slots are exact, working, open, removable, or numerically insertable.
- S — Short-term goal — Start the sectorwise Hessian audit on the frozen working branch (Ψ, P, σ, X_i^a) — Write the explicit first- and second-order contribution table of each sector to the low shell carrier and remote block.

XXVI. CONCLUSION PROTOCOL FOR FUTURE CHATS

From now on, every conclusion section should end with the following dashboard:

$$\boxed{\text{Current status: } \dots} \quad (51)$$

$$\boxed{\text{Goal C: } \dots} \quad (52)$$

$$\boxed{\text{Goal M: } \dots} \quad (53)$$

$$\boxed{\text{Goal S: } \dots} \quad (54)$$

This freezes the project state in a no-loop format.

XXVII. FINAL SUMMARY

The project state is now frozen in a no-loop table form.

$$\boxed{\text{Layer I: final complete TECT PDE} = \text{open},} \quad (55)$$

$$\boxed{\text{Layer II: strongest explicit microscopic PDE} = \text{frozen working branch},} \quad (56)$$

$$\boxed{\text{Layer III: current backend branch} = \text{frozen operational branch},} \quad (57)$$

$$\boxed{\text{Layer IV: reduced shell extractor} = \text{active reduced layer}.} \quad (58)$$

All unclosed microscopic structure is now forced to remain visible in the named open slots

$$\mathcal{L}_{\Psi}^{\text{ref}}, \quad \mathcal{I}_n[P, \partial P, \dots], \quad \mathcal{L}_{X, \text{kin}}^{\text{ref}}, \quad \mathcal{L}_{\text{mix}}^{\text{extra}}.$$

The frozen goal hierarchy is:

$$\boxed{\text{Goal C: final complete microscopic PDE} + \text{actual Hessian coefficient extraction},}$$

$$\boxed{\text{Goal M: complete named-slot PDE audit},}$$

$$\boxed{\text{Goal S: sectorwise Hessian contribution table on the working branch}.}$$

Therefore the next mathematically correct step is not another broad reduction, but the explicit slot-by-slot Hessian audit of

$$(\Psi, P, \sigma, X_i^a)$$

on the frozen working branch.

XXVIII. SECTORWISE HESSIAN AUDIT OF THE FROZEN COMPLETE PDE

XXIX. GOAL

The next nonredundant step is to stop speaking about the complete PDE only at the action level and to write the actual *sectorwise Hessian audit*.

The purpose is to determine, on the frozen complete PDE template, which microscopic Hessian blocks contribute at leading order to

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O,$$

and which blocks are only remote-gap / mass / subleading correction sectors.

XXX. 1. FROZEN COMPLETE BLOCH HESSIAN

We take as the frozen complete microscopic Hessian

$$\mathcal{H}_{\text{Bloch}}^{\text{comp}}(k) = \begin{pmatrix} H_{\Psi\Psi}(k) & H_{\Psi P}(k) & H_{\Psi\sigma}(k) & H_{\Psi X}(k) \\ H_{P\Psi}(k) & H_{PP}(k) & H_{P\sigma}(k) & H_{PX}(k) \\ H_{\sigma\Psi}(k) & H_{\sigma P}(k) & H_{\sigma\sigma}(k) & H_{\sigma X}(k) \\ H_{X\Psi}(k) & H_{XP}(k) & H_{X\sigma}(k) & H_{XX}(k) \end{pmatrix}. \quad (59)$$

The isolated low shell carrier lives inside the Ψ -sector. Everything else is remote:

$$\mathcal{H}_{\text{low}} \subset \mathcal{H}_{\Psi}, \quad \mathcal{H}_{\text{rem}} = \mathcal{H}_P \oplus \mathcal{H}_{\sigma} \oplus \mathcal{H}_X. \quad (60)$$

Therefore the exact shell-effective operator is

$$H_{\text{eff}}(k) = H_{\text{low,low}}(k) - H_{\text{low,rem}}(k) H_{\text{rem,rem}}(k)^{-1} H_{\text{rem,low}}(k). \quad (61)$$

XXXI. 2. BLOCKWISE EXPANSION AROUND THE FROZEN BACKGROUND

Write the first-order transport expansion of each relevant block as

$$H_{AB}(k_* + p) = H_{AB}^{(0)} + p_\mu \partial^\mu H_{AB}|_{k_*} + O(|p|^2). \quad (62)$$

Equivalently, in defect/worldvolume notation,

$$\mathcal{H}_{AB} = \mathcal{H}_{AB}^{(0)}(\xi) - i \mathcal{V}_{AB}^\mu(\xi) \partial_{y^\mu} + \mathcal{W}_{AB}^{\mu\nu}(\xi) \partial_{y^\mu} \partial_{y^\nu} + \cdots. \quad (63)$$

Only the first-order kernels

$$\mathcal{V}_{AB}^\mu$$

contribute to the principal graded $2 + 2$ transport symbol.

XXXII. 3. CANONICAL COEFFICIENT AUDIT TARGET

At the current sharpest structural stage, the leading coefficient audit is:

$$\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O, \quad (64)$$

with subleading or completion-level corrections possibly coming from

$$OD, \quad DO, \quad DD.$$

Then

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O. \quad (65)$$

XXXIII. 4. SECTORWISE HESSIAN TABLE

We now classify each microscopic sector.

A. 4A. The $\Psi\Psi$ block

a. Role. $H_{\Psi\Psi}$ contains the bare shell carrier, the bare same-shell dispersion, and the direct non-mediated first-order transport already present in

$$\mathcal{L}_{\Psi}^{\text{core}}.$$

b. What it can do. It can contribute to:

1. the low-low carrier benchmark $H_{\text{low,low}}$;
2. same-shell non-enriched benchmark gaps in the remote sector;
3. direct first-order transport if $\mathcal{L}_{\Psi}^{\text{ref}}$ contains admissible grading-odd derivative slots.

c. Sharp status. On the current strongest explicit working branch, $H_{\Psi\Psi}$ alone is *not* the trusted source of the full coefficient point; rather, it provides the carrier scaffold on which the Schur-renormalized transport acts.

So:

$H_{\Psi\Psi}$: carrier scaffold + direct benchmark block, not the currently dominant explicit witness source for $(\lambda_c, \Lambda_E, \Lambda_O)$.	(66)
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B. 4B. The PP block

a. Role. H_{PP} is the projector-geometry Hessian generated by

$$\mathcal{L}_P^{\text{geom}} = \frac{\kappa_P}{2} \text{Tr}(\partial_\mu P \partial^\mu P) + \sum_{n \geq 2} a_n \mathcal{I}_n.$$

b. What it can do. It can contribute to:

1. remote geometric gaps;
2. direct geometric transport kernels $V^\mu(\xi) = \theta'(\xi) \mathcal{M}^\mu(\theta)$;
3. mediated corrections to the shell carrier through $H_{\Psi P} H_{PP}^{-1} H_{P\Psi}$.

c. Sharp status. PP is structurally capable of feeding both the longitudinal and transverse channels, but in the current strongest explicit branch this contribution is not yet source-by-source extracted. Hence:

H_{PP} : structurally relevant, numerically/executably open at the full source-by-source level.	(67)
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C. 4C. The $\sigma\sigma$ block

a. Role. $H_{\sigma\sigma}$ is the heavy volumetric Hessian.

b. What it can do. At leading order it primarily supplies:

1. remote spectral gap;
2. wrong-internal splitting scale;
3. indirect correction through $H_{\Psi\sigma} H_{\sigma\sigma}^{-1} H_{\sigma\Psi}$.

c. Sharp status. Unless $\mathcal{L}_{\text{mix}}^{\text{extra}}$ contains a derivative-active σ -carrier slot, the σ -sector is not expected to be the leading principal-symbol generator of

$$\lambda_c, \quad \Lambda_E, \quad \Lambda_O.$$

Hence:

$$\boxed{H_{\sigma\sigma} : \text{primarily remote-gap support, not the leading explicit principal-symbol witness sector.}} \quad (68)$$

D. 4D. The XX block

a. Role. H_{XX} is the heavy Class II mediator denominator. At the current working truncation,

$$H_{XX} \approx M_X^2 \mathbf{1}_X$$

up to possible refinements from $\mathcal{L}_{X,\text{kin}}^{\text{ref}}$.

b. What it can do. It does not by itself generate shell transport. Its role is to weight the Schur complement:

$$\Delta_X = -H_{\Psi X} H_{XX}^{-1} H_{X\Psi}. \quad (69)$$

Thus:

$$\boxed{H_{XX} : \text{denominator / suppression scale, not a numerator witness by itself.}} \quad (70)$$

E. 4E. The ΨX and $X\Psi$ blocks

a. Role. These are the currently strongest explicit witness-producing blocks. On the frozen working branch,

$$\boxed{H_{\Psi X} = \alpha_X \mathcal{J}(k) + \beta_X \mathcal{K}(k), \quad H_{X\Psi} = H_{\Psi X}^\dagger.} \quad (71)$$

After integrating out X ,

$$\Delta_X^{(2)} = -\frac{1}{M_X^2} (\alpha_X \delta J + \beta_X \delta K) (\alpha_X \delta J + \beta_X \delta K)^\dagger. \quad (72)$$

Splitting this,

$$\Delta_X^{(2)} = -\frac{\alpha_X^2}{M_X^2} \delta J \delta J^\dagger - \frac{\alpha_X \beta_X}{M_X^2} (\delta J \delta K^\dagger + \delta K \delta J^\dagger) - \frac{\beta_X^2}{M_X^2} \delta K \delta K^\dagger. \quad (73)$$

b. Leading audit consequence. The current sharpest identification is:

1. the JJ piece feeds the primitive longitudinal slot;
2. the $JK + KJ$ piece feeds the transverse E/O slots;
3. the pure KK piece is transverse at principal order and does not generate λ_c .

Therefore:

$$\boxed{\Psi X / X\Psi \text{ is the leading explicit current witness sector.}} \quad (74)$$

In coefficient language,

$$\boxed{c_{\parallel} = \frac{\alpha_X^2}{M_X^2} j^2, \quad c_E = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad c_O = \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (75)}$$

Hence the working-branch witness formulas are

$$\boxed{\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \frac{\alpha_X^2}{M_X^2} j^2, \quad \alpha = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_E^{(0)}, \quad \beta = I_{\theta} \frac{\alpha_X \beta_X}{M_X^2} (j \cdot k) \Lambda_O^{(0)}. \quad (76)}$$

F. 4F. The $\Psi P, P\Psi, \Psi\sigma, \sigma\Psi, PX, XP, P\sigma, \sigma P, X\sigma, \sigma X$ blocks

a. Role. These are cross-remote or low-remote mixing blocks beyond the simplest explicit Class II witness route.
b. Sharp status. At the current sharpest explicit working-branch level, these are not yet the primary audited source of the coefficient point. They should be treated as:

1. completion-level correction channels;
2. possible source terms for the true full-complete Hessian;
3. subleading unless $\mathcal{L}_{\text{mix}}^{\text{extra}}$ activates them at first derivative order.

So:

all remaining off-diagonal blocks are currently completion channels, not the leading explicit branch witness.

 (77)

XXXIV. 5. REDUCED COEFFICIENT ROUTING TABLE

The sectorwise routing is therefore:

$H_{\Psi\Psi} \rightarrow$ carrier scaffold / benchmark,

 (78)

$H_{\Psi X}, H_{X\Psi}, H_{XX} \rightarrow$ leading explicit Schur witness route,

 (79)

$H_{PP} \rightarrow$ geometric completion route,

 (80)

$H_{\sigma\sigma} \rightarrow$ remote-gap support route,

 (81)

all other cross blocks $\rightarrow$ completion-level corrections.

 (82)

More sharply:

$\mathcal{H}_{OO}$ (fed primarily by the JJ route) $\rightarrow \lambda_c$,

 (83)

$\mathcal{H}_{TD} + \mathcal{H}_{DT}$ (fed primarily by the $JK + KJ$ route) $\rightarrow \Lambda_E, \Lambda_O$.

 (84)

XXXV. 6. WHAT IS ACTUALLY THE NEXT EXECUTABLE CALCULATION

The next executable task is no longer to discuss the action abstractly.

It is:

1. write the explicit background-linearized kernels

$$\delta J, \quad \delta K$$

on the frozen corrected branch;

2. identify their OO projection and read off

$$\lambda_c = 2 \lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[V_{OO}]}{\theta'^3};$$

3. identify their $TD + DT$ projection and read off

$$\Lambda_E = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}];$$

4. only after that, return to the residual-level quartic enlarged-kernel rerun.

XXXVI. 7. HONEST STATUS OF THIS STEP

This sectorwise Hessian audit closes the *routing problem*: we now know which microscopic blocks are supposed to feed which reduced coefficients.

What is still open is not the routing logic but the literal evaluation of the background-dependent differentiated kernels on the actual frozen corrected branch.

XXXVII. FINAL SUMMARY

The sectorwise Hessian audit is now fixed.

The complete microscopic Hessian is

$$\mathcal{H}_{\text{Bloch}}^{\text{comp}}(k),$$

with low shell carrier inside the Ψ -sector and remote sector

$$(P, \sigma, X).$$

At the current sharpest explicit working-branch level:

$$H_{\Psi\Psi} \text{ provides the carrier scaffold and benchmark structure,}$$

$$H_{\Psi X}, H_{X\Psi}, H_{XX} \text{ provide the leading explicit witness route through the Schur correction,}$$

$$H_{PP} \text{ is the geometric completion route,}$$

$$H_{\sigma\sigma} \text{ is primarily the remote-gap support route,}$$

and

$$\text{all other off-diagonal remote blocks are completion-level correction channels.}$$

The coefficient routing is:

$$\mathcal{H}_{OO} \rightarrow \lambda_c, \quad \mathcal{H}_{TD} + \mathcal{H}_{DT} \rightarrow \Lambda_E, \Lambda_O.$$

On the current explicit Class II witness route:

$$JJ \rightarrow \lambda_c, \quad JK + KJ \rightarrow \Lambda_E, \Lambda_O, \quad KK \not\rightarrow \lambda_c \text{ at principal order.}$$

Therefore the next executable microscopic step is:

$$\text{differentiate the frozen corrected-branch currents } \delta J, \delta K, \text{ project them to } OO \text{ and } TD + DT, \text{ and read off } (\lambda_c, \Lambda_E, \Lambda_O).$$

XXXVIII. EXPLICIT CALCULATION OF THE CLASS-II TRANSPORT KERNEL ON THE FROZEN CORRECTED BRANCH

XXXIX. FROZEN BRANCH

We keep fixed the operational working point

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096), \quad (85)$$

so that the corrected local branch is

$$\Psi_{\text{corr}} = \Psi_{\text{BCC}} + 0.0096 \Phi_A - 0.0288 \Phi_E + 0.0096 \Phi_O. \quad (86)$$

XL. 1. STARTING POINT: THE STRONGEST EXPLICIT WORKING MICROSCOPIC BRANCH

The frozen explicit working branch is

$$L_{\text{mic}} = L_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \Psi]. \quad (87)$$

The heavy mediator equation is algebraic:

$$X_i^a = -\frac{1}{M_X^2} \left(\alpha_X J_i^a + \beta_X K_i^a \right). \quad (88)$$

Substituting back gives the exact tree-level effective interaction

$$L_{\text{eff}}^{(X)} = -\frac{\alpha_X^2}{2M_X^2} J_i^a J_i^a - \frac{\alpha_X \beta_X}{M_X^2} J_i^a K_i^a - \frac{\beta_X^2}{2M_X^2} K_i^a K_i^a. \quad (89)$$

Thus the current exact source decomposition is

$$\boxed{JJ, \quad JK + KJ, \quad KK.} \quad (90)$$

XLI. 2. BACKGROUND EXPANSION AND PRINCIPAL TRANSPORT PIECE

Define the total mediator current

$$\boxed{\mathcal{J}_i^a := \alpha_X J_i^a + \beta_X K_i^a.} \quad (91)$$

Expand around the corrected defect/BCC background:

$$\mathcal{J}_i^a[\Phi_0 + \delta\Phi] = \bar{\mathcal{J}}_i^a + \mathcal{J}_{i,\text{lin}}^a[\delta\Phi] + O(\delta\Phi^2). \quad (92)$$

After integrating out X , the transport-relevant effective term is

$$\delta L_{\text{eff}}^{\text{med}} = -\frac{1}{2M_X^2} (\bar{\mathcal{J}}_i^a + \mathcal{J}_{i,\text{lin}}^a) (\bar{\mathcal{J}}_i^a + \mathcal{J}_{i,\text{lin}}^a). \quad (93)$$

The principal first-order transport symbol comes from the background-linear cross term:

$$\boxed{\delta L_{\text{eff}}^{(1)} = -\frac{1}{M_X^2} \bar{\mathcal{J}}_i^a \mathcal{J}_{i,\text{lin}}^a.} \quad (94)$$

Using (91), this becomes

$$\delta L_{\text{eff}}^{(1)} = -\frac{\alpha_X^2}{M_X^2} \bar{J}_i^a J_{i,\text{lin}}^a - \frac{\alpha_X \beta_X}{M_X^2} \left(\bar{J}_i^a K_{i,\text{lin}}^a + \bar{K}_i^a J_{i,\text{lin}}^a \right) - \frac{\beta_X^2}{M_X^2} \bar{K}_i^a K_{i,\text{lin}}^a. \quad (95)$$

This is the exact source-by-source first-order transport decomposition on the frozen working branch.

XLII. 3. SOURCE CLASSIFICATION

At the current strongest working-branch level, the source classification is:

$$\boxed{\text{principal longitudinal source} = JJ,} \quad (96)$$

$$\boxed{\text{principal transverse source} = JK + KJ,} \quad (97)$$

$$\boxed{KK = \text{same-shell residual} / \text{complementary correction sector.}} \quad (98)$$

More sharply, at principal shell order,

$$\boxed{\lambda_{\parallel}^{(JJ)} \neq 0 \text{ is allowed,} \quad \alpha^{(JJ)} = \beta^{(JJ)} = 0,} \quad (99)$$

while

$$\boxed{JK + KJ \implies (\Lambda_E, \Lambda_O) \text{ only at principal shell order,}} \quad (100)$$

and

$$\boxed{c_{\parallel}^{(JK)} = 0, \quad c_{\parallel}^{(KK)} = 0 \quad \text{at principal shell order.}} \quad (101)$$

XLIII. 4. PROJECTION FORMULAS FOR THE ONE-DEFECT COEFFICIENT POINT

The longitudinal coefficient is read from the OO -block:

$$\boxed{V_{OO}^{\mu} = \Pi_Q \mathcal{V}_{OO}^{\mu} \Pi_Q, \quad \lambda_c = 2 \lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[V_{OO}]}{\theta'(\xi)^3}.} \quad (102)$$

The transverse even/odd coefficients are read from the mixed $TD + DT$ -block:

$$\boxed{\mathcal{M}_{TD+DT}^{\mu} = \Pi_Q (\mathcal{V}_{TD}^{\mu} + \mathcal{V}_{DT}^{\mu}) \Pi_Q,} \quad (103)$$

$$\boxed{\Lambda_E = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].} \quad (104)$$

Thus the exact microscopic chain is

$$\boxed{\mathcal{H}_{OO} \rightarrow \lambda_c \rightarrow \lambda_{\parallel}, \quad \mathcal{H}_{TD}, \mathcal{H}_{DT} \rightarrow (\Lambda_E, \Lambda_O) \rightarrow (\alpha, \beta).} \quad (105)$$

XLIV. 5. BRANCH-B COEFFICIENT SHEET: EXPLICIT OVERLAP INSERTION

At Branch B level the exact overlap dictionary is

$$\boxed{\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \lambda_c, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.} \quad (106)$$

The explicit defect overlaps are already known:

$$\boxed{I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa.} \quad (107)$$

Therefore the reduced coefficients are explicitly

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3,} \quad (108)$$

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa.} \quad (109)$$

Hence the transverse norm is

$$v_{\perp} = \sqrt{\alpha^2 + \beta^2} = \frac{\sqrt{3}\pi}{16} |\theta_0 \kappa| \sqrt{\Lambda_E^2 + \Lambda_O^2}. \quad (110)$$

The longitudinal magnitude is

$$|v_{\parallel}| = |\lambda_{\parallel}| = \frac{3\sqrt{3}\pi}{512} |\lambda_c| |\theta_0^3 \kappa^3|. \quad (111)$$

Therefore the absolute anisotropy ratio is

$$\frac{|v_{\parallel}|}{v_{\perp}} = \frac{3}{32} (\theta_0 \kappa)^2 \frac{|\lambda_c|}{\sqrt{\Lambda_E^2 + \Lambda_O^2}}. \quad (112)$$

XLV. 6. WORKING-BRANCH LONGITUDINAL FORMULA IN TERMS OF THE BLOCH CALIBRATION CONSTANT

The current notes give the exact Branch-B one-mediator formula

$$\lambda_{c;OO} = -\frac{\alpha_X \beta_X}{M_X^2} C_{\parallel}. \quad (113)$$

Hence

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3, \quad I_3 = I_{\theta^3}. \quad (114)$$

Substituting the explicit overlap gives

$$\lambda_{\parallel} = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} C_{\parallel} \theta_0^3 \kappa^3. \quad (115)$$

So the practical Gate-2 criterion is

$$\alpha_X \beta_X \neq 0, \quad C_{\parallel} \neq 0, \quad \theta_0 \kappa \neq 0 \implies \lambda_{\parallel} \neq 0. \quad (116)$$

Moreover, the exact calibration identity is

$$C_{\parallel} = -\frac{M_X^2}{\alpha_X \beta_X I_3} \left[(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22} \right]. \quad (117)$$

Hence

$$C_{\parallel} \neq 0 \iff (\hat{n} \cdot \vec{W}_+)_{11} \neq (\hat{n} \cdot \vec{W}_+)_{22}. \quad (118)$$

XLVI. 7. WHAT IS NOW ACTUALLY COMPUTED, AND WHAT REMAINS OPEN

What is now explicitly computed is:

1. the exact mediator-eliminated transport decomposition,
2. the exact projection rules for $\lambda_c, \Lambda_E, \Lambda_O$,

3. the explicit Branch-B coefficient sheet

$$\lambda_{\parallel}, \alpha, \beta$$

with the defect-overlap factors inserted,

4. the exact practical longitudinal formula in terms of $C_{\parallel}$,

5. the exact transverse norm and anisotropy ratio in terms of

$$\lambda_c, \Lambda_E, \Lambda_O.$$

What remains genuinely open is not the algebraic reduction, but the actual defect-background evaluation of the unreduced microscopic current vertices

$$J_i^a[\Psi], \quad K_i^a[P, \Psi],$$

equivalently the explicit evaluation of

$$\mathcal{V}_{OO}^{\mu}, \quad \mathcal{V}_{TD}^{\mu}, \quad \mathcal{V}_{DT}^{\mu},$$

or, in the safest longitudinal working convention, the direct projected Bloch quantity

$$(\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22}.$$

XLVII. FINAL SUMMARY

On the frozen corrected branch

$$\Psi_{\text{corr}} = \Psi_{\text{BCC}} + 0.0096 \Phi_A - 0.0288 \Phi_E + 0.0096 \Phi_O,$$

the heavy Class-II mediator can be eliminated exactly at tree level, giving

$$L_{\text{eff}}^{(X)} = -\frac{\alpha_X^2}{2M_X^2} JJ - \frac{\alpha_X \beta_X}{M_X^2} JK - \frac{\beta_X^2}{2M_X^2} KK.$$

The principal first-order transport symbol comes from the background-linear cross term

$$-\frac{1}{M_X^2} \bar{\mathcal{J}} \mathcal{J}_{\text{lin}}, \quad \mathcal{J} = \alpha_X J + \beta_X K,$$

so the exact source decomposition is

$$JJ, \quad JK + KJ, \quad KK.$$

At the current sharpest working-branch level,

$$\boxed{JJ \text{ is the unique principal longitudinal source,}}$$

$$\boxed{JK + KJ \text{ is the principal transverse source,}}$$

and

$$\boxed{KK \text{ is the same-shell residual/complementary correction sector.}}$$

The coefficient point is extracted by

$$\lambda_c = 2 \lim_{\theta' \rightarrow 0} \frac{\mathfrak{P}_{\parallel}[V_{OO}]}{\theta'(\xi)^3}, \quad \Lambda_E = \mathfrak{P}_E[\mathcal{M}_{TD+DT}^{\text{inv}}], \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}].$$

With the explicit overlap values

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256}\theta_0^3\kappa^3, \quad I_\theta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa,$$

the reduced Branch-B coefficients are already in fully explicit form:

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512}\lambda_c\theta_0^3\kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16}\Lambda_E\theta_0\kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16}\Lambda_O\theta_0\kappa.$$

Equivalently, in the working one-mediator longitudinal notation,

$$\lambda_{c;OO} = -\frac{\alpha_X\beta_X}{M_X^2}C_{\parallel}, \quad \lambda_{\parallel} = \frac{3\sqrt{3}\pi}{512}\frac{\alpha_X\beta_X}{M_X^2}C_{\parallel}\theta_0^3\kappa^3.$$

Thus the longitudinal bottleneck is now exactly

$$C_{\parallel} \neq 0?,$$

while the transverse bottleneck is the explicit evaluation of

$$\Lambda_E, \quad \Lambda_O$$

from the $TD + DT$ projected one-defect Hessian kernels.

So the algebraic reduction is no longer open. The remaining frontier is the literal evaluation of the background-dependent microscopic current kernels on the frozen corrected branch.

XLVIII. FINITE MATRIX-ELEMENT FORMULAS FOR THE ACTUAL CORRECTED-BRANCH AUDIT

XLIX. GOAL

The remaining frontier is no longer structural. It is the explicit finite extraction of

$$C_{\parallel}, \quad \Lambda_E, \quad \Lambda_O,$$

on the frozen corrected branch.

Equivalently, we now want a formula that can be implemented directly as a finite harmonic sum.

L. 1. FROZEN CARRIER BASIS AND SHELL-CORE BLOCK

Fix the folded Bragg point

$$k_* = \frac{G_*}{2},$$

and the shell-core basis

$$|+, a\rangle, \quad |-, a\rangle, \quad a = 1, 2.$$

The finite shell matrix is block decomposed as

$$H_{\text{shell}}(p) = \begin{pmatrix} E_+(p) & B(p) & C_+(p) \\ B(p)^\dagger & E_-(p) & C_-(p) \\ C_+(p)^\dagger & C_-(p)^\dagger & H_{ss}(p) \end{pmatrix}, \quad (119)$$

with

$$(E_+(p))_{ab} = \langle +, a | H_{\text{shell}}(p) | +, b \rangle, \quad (E_-(p))_{ab} = \langle -, a | H_{\text{shell}}(p) | -, b \rangle. \quad (120)$$

At $p = 0$, diagonalize the two 2×2 valley blocks:

$$E_+(0)\eta_+ = \varepsilon_+\eta_+, \quad E_-(0)\eta_- = \varepsilon_-\eta_-, \quad \|\eta_\pm\| = 1, \quad (121)$$

with

$$\eta_+ = \begin{pmatrix} A_1^{(+)} \\ A_2^{(+)} \end{pmatrix}, \quad \eta_- = \begin{pmatrix} A_1^{(-)} \\ A_2^{(-)} \end{pmatrix}.$$

Then the Bloch asymmetry scalar is

$$\Delta_{00} = \overline{A_1^{(+)}} A_1^{(-)} - \overline{A_2^{(+)}} A_2^{(-)} = \eta_+^\dagger \Sigma_3 \eta_-. \quad (122)$$

LI. 2. FINITE FOURIER FORM OF THE INTERVALLEY J -MATRIX

Let the carrier-periodic internal wavefunctions be expanded as

$$u_a^\pm(x) = \sum_{G \in \mathcal{S}_N} c_{a,G}^{(\pm)} e^{iG \cdot x}, \quad a = 1, 2. \quad (123)$$

Let the intervalley J -kernel have Fourier coefficients

$$J_j(x) = \sum_Q \hat{J}_j(Q; 0) e^{iQ \cdot x}. \quad (124)$$

The reduced intervalley J -matrix is

$$(\mathcal{U}_j(0))_{ab} = \int_{\text{cell}} d^3x e^{-iG_* \cdot x} \overline{u_a^+(x)} J_j(x) u_b^-(x). \quad (125)$$

Substituting (123)–(124), the cell integral gives the exact finite convolution

$$(\mathcal{U}_j(0))_{ab} = V_{\text{cell}} \sum_{G, G' \in \mathcal{S}_N} \overline{c_{a,G}^{(+)}} \hat{J}_j(G_* + G - G'; 0) c_{b,G'}^{(-)}. \quad (126)$$

This is the literal finite matrix-element formula for the longitudinal intervalley current overlap.

LII. 3. LONGITUDINAL SINGLET AND $C_\parallel$

Define

$$u_\parallel = \frac{1}{2} \text{Tr}(\Sigma_3 \hat{n}^j \mathcal{U}_j(0)) = \frac{1}{2} [(\hat{n} \cdot \mathcal{U})_{11} - (\hat{n} \cdot \mathcal{U})_{22}]. \quad (127)$$

Then

$$L_* = m_\parallel u_\parallel. \quad (128)$$

Using (126),

$$u_\parallel = \frac{V_{\text{cell}}}{2} \sum_{G, G' \in \mathcal{S}_N} \left(\overline{c_{1,G}^{(+)}} c_{1,G'}^{(-)} - \overline{c_{2,G}^{(+)}} c_{2,G'}^{(-)} \right) \hat{n}^j \hat{J}_j(G_* + G - G'; 0). \quad (129)$$

Hence

$$L_* = \frac{m_{\parallel} V_{\text{cell}}}{2} \sum_{G, G' \in \mathcal{S}_N} \left(\overline{c_{1,G}^{(+)} c_{1,G'}^{(-)}} - \overline{c_{2,G}^{(+)} c_{2,G'}^{(-)}} \right) \hat{n}^j \hat{\mathbb{J}}_j(G_* + G - G'; 0). \quad (130)$$

Now the practical current-branch longitudinal witness is

$$\lambda_{\parallel} = \frac{1}{2} \Delta W_{\parallel}, \quad \Delta W_{\parallel} = (\hat{n} \cdot \vec{W}_+)_{11} - (\hat{n} \cdot \vec{W}_+)_{22}. \quad (131)$$

On the Branch-B one-mediator calibration sheet,

$$\lambda_{\parallel} = -\frac{\alpha_X \beta_X}{2M_X^2} C_{\parallel} I_3, \quad I_3 = I_{\theta'^3}. \quad (132)$$

Therefore

$$C_{\parallel} = -\frac{M_X^2}{\alpha_X \beta_X I_3} \Delta W_{\parallel}. \quad (133)$$

So the exact finite longitudinal task is reduced to either:

$$\boxed{\text{compute } \Delta W_{\parallel}}, \quad (134)$$

or equivalently

$$\boxed{\text{compute } u_{\parallel} \text{ from the finite sum (129)}}. \quad (135)$$

LIII. 4. EQUIVALENT UNCONDITIONAL JJ-FINITE SUM

The unconditional longitudinal overlap can also be written directly as

$$\mathcal{M}_{ab}^{(JJ)} = V_{\text{cell}} \sum_{G, G' \in \mathcal{S}_N} \overline{c_{a,G}^{(+)}} \hat{V}_{\parallel}^{(JJ)}(G_* + G - G') c_{b,G'}^{(-)}. \quad (136)$$

Thus

$$m_3 = -\frac{V_{\text{cell}}}{2\Gamma_J} \sum_{G, G' \in \mathcal{S}_N} \left(\overline{c_{1,G}^{(+)} c_{1,G'}^{(-)}} - \overline{c_{2,G}^{(+)} c_{2,G'}^{(-)}} \right) \hat{V}_{\parallel}^{(JJ)}(G_* + G - G'). \quad (137)$$

So the unconditional longitudinal frontier is literally one finite harmonic sum.

LIV. 5. FINITE TRANSVERSE MATRICES

Define the derivative-resolved transverse shell matrices

$$\mathcal{T}_{\nu} = \partial_{p_{\nu}} \hat{\Sigma}_{JK}(G_*; p) \Big|_0 + \partial_{p_{\nu}} \hat{\Sigma}_{KK}(G_*; p) \Big|_0, \quad \nu = 1, 2. \quad (138)$$

Projecting them to the 2×2 carrier gives

$$(\mathcal{T}_{\nu})_{ab} = V_{\text{cell}} \sum_{G, G' \in \mathcal{S}_N} \overline{c_{a,G}^{(+)}} \hat{\mathcal{T}}_{\nu}(G_* + G - G'; 0) c_{b,G'}^{(-)}, \quad (139)$$

where

$$\widehat{\mathfrak{D}}_\nu := \partial_{p_\nu} \widehat{\Sigma}_{JK} \Big|_0 + \partial_{p_\nu} \widehat{\Sigma}_{KK} \Big|_0.$$

The exact transport-side transverse witnesses are

$$c_E = \frac{1}{4} \text{Tr}(\sigma_1 \mathcal{T}_1 + \sigma_2 \mathcal{T}_2), \quad c_O = \frac{1}{4} \text{Tr}(\sigma_2 \mathcal{T}_1 - \sigma_1 \mathcal{T}_2). \quad (140)$$

Writing

$$\mathcal{T}_\nu = \begin{pmatrix} t_{11}^{(\nu)} & t_{12}^{(\nu)} \\ t_{21}^{(\nu)} & t_{22}^{(\nu)} \end{pmatrix},$$

these become explicitly

$$c_E = \frac{1}{4} \left[t_{12}^{(1)} + t_{21}^{(1)} + i(t_{21}^{(2)} - t_{12}^{(2)}) \right], \quad (141)$$

$$c_O = \frac{1}{4} \left[i(t_{21}^{(1)} - t_{12}^{(1)}) - (t_{12}^{(2)} + t_{21}^{(2)}) \right]. \quad (142)$$

Hence the unconditional transverse pass condition

$$(c_E, c_O) \neq (0, 0)$$

is reduced to explicit finite sums over the carrier coefficients $c_{a,G}^{(\pm)}$ and the derivative kernels $\widehat{\mathfrak{D}}_\nu$.

IV. 6. FOLDED-BRAGG CLASS II COMPRESSION

Inside the explicit folded-Bragg Class II working branch, the derivative bookkeeping compresses further to

$$\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \widehat{J}_i^A(Q) \widehat{K}_{i,\alpha}^A(Q'), \quad \alpha = 1, 2. \quad (143)$$

For the periodic locked background

$$\widehat{J}_i^A(Q) = iQ_i A_Q j^A. \quad (144)$$

Therefore

$$\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} iQ_i A_Q j^A \widehat{K}_{i,\alpha}^A(Q'). \quad (145)$$

In the minimal first-shell-dominant reduction,

$$\widehat{T}_{(II)}^\alpha(G_*) \propto iA_{G_*}(C_\alpha \cdot G_*), \quad \alpha = 1, 2, \quad (146)$$

so the strongest practical explicit transverse test is

$$A_{G_*}(C_1 \cdot G_*) \neq 0, \quad A_{G_*}(C_2 \cdot G_*) \neq 0. \quad (147)$$

LVI. 7. REDUCED COEFFICIENT SHEET

Once $\lambda_c, \Lambda_E, \Lambda_O$ are extracted, the reduced PDE coefficients are

$$\boxed{\lambda_{\parallel} = \frac{\lambda_c}{2} I_{\theta'^3}, \quad \alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O.} \quad (148)$$

Using

$$I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3, \quad I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

we get

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \lambda_c \theta_0^3 \kappa^3,} \quad (149)$$

$$\boxed{\alpha = -\frac{\sqrt{3}\pi}{16} \Lambda_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \Lambda_O \theta_0 \kappa.} \quad (150)$$

LVII. 8. WHAT IS NOW GENUINELY LEFT

The algebra is no longer open.

The only remaining microscopic insertion is:

1. compute the carrier Fourier coefficients $c_{a,G}^{(\pm)}$ from the actual corrected branch;
2. compute the harmonic kernels

$$\widehat{\mathbf{J}}_j(Q; 0), \quad \widehat{V}_{\parallel}^{(JJ)}(Q), \quad \widehat{\mathfrak{D}}_{\nu}(Q; 0), \quad \widehat{K}_{i,\alpha}^A(Q);$$

3. evaluate the finite sums (129), (137), (139), or the compressed convolution (145).

So the honest frontier is now a literal finite harmonic computation, not a structural ambiguity.

LVIII. FINAL SUMMARY

The remaining corrected-branch microscopic audit is now reduced to explicit finite harmonic sums. For the longitudinal sector, the key intervalley current matrix is

$$(\mathcal{U}_j(0))_{ab} = V_{\text{cell}} \sum_{G, G' \in \mathcal{S}_N} \overline{c_{a,G}^{(+)}} \widehat{\mathbf{J}}_j(G_* + G - G'; 0) c_{b,G'}^{(-)},$$

which gives

$$u_{\parallel} = \frac{V_{\text{cell}}}{2} \sum_{G, G'} \left(\overline{c_{1,G}^{(+)}} c_{1,G'}^{(-)} - \overline{c_{2,G}^{(+)}} c_{2,G'}^{(-)} \right) \widehat{n}^j \widehat{\mathbf{J}}_j(G_* + G - G'; 0),$$

and hence

$$L_* = m_{\parallel} u_{\parallel}.$$

Equivalently, the unconditional longitudinal coefficient is one finite harmonic sum:

$$m_3 = -\frac{V_{\text{cell}}}{2\Gamma_J} \sum_{G, G'} \left(\overline{c_{1,G}^{(+)}} c_{1,G'}^{(-)} - \overline{c_{2,G}^{(+)}} c_{2,G'}^{(-)} \right) \widehat{V}_{\parallel}^{(JJ)}(G_* + G - G').$$

For the transverse sector, the derivative shell matrices are

$$(\mathcal{T}_\nu)_{ab} = V_{\text{cell}} \sum_{G, G'} \overline{c_{a, G}^{(+)}} \widehat{\mathfrak{D}}_\nu(G_* + G - G'; 0) c_{b, G'}^{(-)}, \quad \nu = 1, 2,$$

and the exact witnesses are

$$c_E = \frac{1}{4} \text{Tr}(\sigma_1 \mathcal{T}_1 + \sigma_2 \mathcal{T}_2), \quad c_O = \frac{1}{4} \text{Tr}(\sigma_2 \mathcal{T}_1 - \sigma_1 \mathcal{T}_2).$$

Inside the explicit folded-Bragg Class II branch, the transverse coefficient compresses to

$$\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \widehat{J}_i^A(Q) \widehat{K}_{i, \alpha}^A(Q'),$$

which, in the first-shell-dominant reduction, becomes

$$\widehat{T}_{(II)}^\alpha(G_*) \propto i A_{G_*}(C_\alpha \cdot G_*).$$

Therefore the project has now reached the stage where the remaining frontier is a literal finite harmonic computation on the frozen corrected branch. No structural theorem is missing anymore.

LIX. CORRECTED-BRANCH FIRST-SHELL INSERTION

LX. GOAL

We now insert the frozen corrected branch directly into the first-shell witness formulas. The corrected operational branch is

$$\boxed{\Psi_{\text{corr}} = \Psi_{\text{BCC}} + 0.0096 \Phi_A - 0.0288 \Phi_E + 0.0096 \Phi_O.} \quad (151)$$

The purpose of this step is to convert the abstract finite-harmonic witness formulas into corrected first-shell coefficient formulas.

LXI. 1. CORRECTED FIRST-SHELL HARMONIC AMPLITUDE

Write the selected shell harmonic of a field Ψ at G_* as

$$A_{G_*}[\Psi] := \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x e^{-iG_* \cdot x} \Psi(x). \quad (152)$$

Applying this to (151),

$$\boxed{A_{G_*}^{\text{corr}} = A_{G_*}^{\text{BCC}} + 0.0096 A_{G_*}^{[A]} - 0.0288 A_{G_*}^{[E]} + 0.0096 A_{G_*}^{[O]}.} \quad (153)$$

If the corrected branch is viewed as a small local refinement, the same formula may be written as the first-order response expansion

$$\boxed{A_{G_*}^{\text{corr}} = A_{G_*}^{(0)} + \delta_A^{\text{final}} \partial_{\delta_A} A_{G_*}|_0 + \delta_E^{\text{final}} \partial_{\delta_E} A_{G_*}|_0 + \delta_O^{\text{final}} \partial_{\delta_O} A_{G_*}|_0 + O(\delta^2),} \quad (154)$$

with

$$(\delta_A^{\text{final}}, \delta_E^{\text{final}}, \delta_O^{\text{final}}) = (0.0096, -0.0288, 0.0096).$$

LXII. 2. CORRECTED LONGITUDINAL INPUT

For the explicit canonical witness branch,

$$T_* = T^1, \quad R_\perp = 1, \quad \varphi = 0,$$

the scalar longitudinal input is already isolated as

$$u_\parallel(T^1) = \frac{1}{2} \Re(\kappa_J), \quad L_*(T^1) = \frac{m_\parallel}{2} \Re(\kappa_J). \quad (155)$$

The pure- JJ longitudinal tensor collapses to

$$C_{jm}^{(JJ)} = \Xi_{JJ} \delta_{jm}, \quad \Xi_{JJ} = \mathcal{N}_{JJ} L_*,$$

so the first-shell Fourier coefficient is

$$\widehat{\mathcal{A}}_{JJ}(G_*) = i \mathcal{N}_{JJ} m_\parallel u_\parallel |G_*| A_{G_*}. \quad (156)$$

Therefore the corrected branch gives

$$\widehat{\mathcal{A}}_{JJ}^{\text{corr}}(G_*) = i \mathcal{N}_{JJ}^{\text{corr}} m_\parallel^{\text{corr}} u_\parallel^{\text{corr}} |G_*| A_{G_*}^{\text{corr}}. \quad (157)$$

At first corrected order,

$$\mathcal{N}_{JJ}^{\text{corr}} = \mathcal{N}_{JJ}^{(0)} + 0.0096 \mathcal{N}_{JJ}^{[A]} - 0.0288 \mathcal{N}_{JJ}^{[E]} + 0.0096 \mathcal{N}_{JJ}^{[O]} + O(\delta^2), \quad (158)$$

$$m_\parallel^{\text{corr}} = m_\parallel^{(0)} + 0.0096 m_\parallel^{[A]} - 0.0288 m_\parallel^{[E]} + 0.0096 m_\parallel^{[O]} + O(\delta^2), \quad (159)$$

$$u_\parallel^{\text{corr}} = u_\parallel^{(0)} + 0.0096 u_\parallel^{[A]} - 0.0288 u_\parallel^{[E]} + 0.0096 u_\parallel^{[O]} + O(\delta^2). \quad (160)$$

Thus the corrected longitudinal witness coefficient is

$$\lambda_\parallel^{\text{corr}} = -\frac{3\sqrt{3}\pi}{512} \Gamma_{JJ}^{\text{corr}} m_\parallel^{\text{corr}} u_\parallel^{\text{corr}} \theta_0^3 \kappa^3, \quad (161)$$

with

$$\Gamma_{JJ}^{\text{corr}} \propto \alpha_X \mathcal{N}_{JJ}^{\text{corr}} |G_*| A_{G_*}^{\text{corr}} \quad (162)$$

in the first-shell dominant reduction.

Combining (161) and (162),

$$\lambda_\parallel^{\text{corr}} \propto \alpha_X \mathcal{N}_{JJ}^{\text{corr}} |G_*| A_{G_*}^{\text{corr}} m_\parallel^{\text{corr}} u_\parallel^{\text{corr}} \theta_0^3 \kappa^3. \quad (163)$$

This is now the corrected first-shell longitudinal formula.

LXIII. 3. CORRECTED TRANSVERSE INSERTION

The exact explicit folded-Bragg Class II transverse overlap is

$$\widehat{T}_{(II)}^\alpha(G_*) = -\frac{\alpha_X \beta_X}{M_X^2} \sum_A \sum_{Q+Q'=G_*} \widehat{J}_i^A(Q) \widehat{K}_{i,\alpha}^A(Q'), \quad \alpha = 1, 2. \quad (164)$$

In the minimal first-shell-dominant reduction,

$$\widehat{T}_{(II)}^\alpha(G_*) \propto i A_{G_*}(C_\alpha \cdot G_*). \quad (165)$$

Therefore the corrected branch induces

$$\boxed{\widehat{T}_{(II),\text{corr}}^\alpha(G_*) \propto iA_{G_*}^{\text{corr}}(C_\alpha^{\text{corr}} \cdot G_*), \quad \alpha = 1, 2.} \quad (166)$$

Now expand the transverse shell projections:

$$\boxed{(C_\alpha \cdot G_*)_{\text{corr}} = (C_\alpha \cdot G_*)^{(0)} + 0.0096 \Xi_\alpha^{[A]} - 0.0288 \Xi_\alpha^{[E]} + 0.0096 \Xi_\alpha^{[O]} + O(\delta^2),} \quad (167)$$

where

$$\Xi_\alpha^{[r]} := \partial_{\delta_r}(C_\alpha \cdot G_*)|_0, \quad r \in \{A, E, O\}.$$

Hence the corrected transverse witnesses are

$$\boxed{\alpha^{\text{corr}} = I_\theta c_E^{\text{corr}}, \quad \beta^{\text{corr}} = I_\theta c_O^{\text{corr}},} \quad (168)$$

with

$$\boxed{c_E^{\text{corr}} \propto A_{G_*}^{\text{corr}}(C_1^{\text{corr}} \cdot G_*), \quad c_O^{\text{corr}} \propto A_{G_*}^{\text{corr}}(C_2^{\text{corr}} \cdot G_*).} \quad (169)$$

Using

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

we get

$$\boxed{\alpha^{\text{corr}} \propto -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_{G_*}^{\text{corr}}(C_1^{\text{corr}} \cdot G_*),} \quad (170)$$

$$\boxed{\beta^{\text{corr}} \propto -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_{G_*}^{\text{corr}}(C_2^{\text{corr}} \cdot G_*).} \quad (171)$$

Thus the corrected practical transverse test is

$$\boxed{A_{G_*}^{\text{corr}}(C_1^{\text{corr}} \cdot G_*) \neq 0 \quad \text{or} \quad A_{G_*}^{\text{corr}}(C_2^{\text{corr}} \cdot G_*) \neq 0.} \quad (172)$$

LXIV. 4. CANONICAL EXPLICIT ONE-CHANNEL INSERTION

On the explicit canonical branch used throughout the acceptance-stage notes, the simplest nondegenerate working insertion is

$$C_2^{\text{corr}} \cdot G_* = 0, \quad C_1^{\text{corr}} \cdot G_* \neq 0,$$

so that

$$\boxed{\beta^{\text{corr}} = 0, \quad \alpha^{\text{corr}} \neq 0.} \quad (173)$$

In the same explicit canonical witness normalization, the finite-audit demo yields

$$\Delta_{00} = 0.8396344637 + 0.0226429117i, \quad u_\parallel(T^1) = 0.5, \quad L_*(T^1) = 0.5,$$

$$\lambda_\parallel = 0.031883191949430, \quad \alpha = 0.340087380793916, \quad \beta = 0,$$

with positive Gate-2 and Gate-3 witness margins. This should be interpreted as a normalization sanity check of the explicit canonical witness branch, not yet as the final corrected-branch full microscopic insertion.

LXV. 5. FIRST CORRECTED-BRANCH FACTORIZED AUDIT SHEET

Collecting the previous formulas, the corrected first-shell audit closes on the following four scalars:

$$\boxed{A_{G_*}^{\text{corr}}, \quad \mathcal{N}_{JJ}^{\text{corr}}, \quad m_{\parallel}^{\text{corr}} u_{\parallel}^{\text{corr}}, \quad (C_{\alpha}^{\text{corr}} \cdot G_*)}, \quad \alpha = 1, 2. \quad (174)$$

The corrected reduced coefficients are then factorized as

$$\boxed{\lambda_{\parallel}^{\text{corr}} \propto A_{G_*}^{\text{corr}} \mathcal{N}_{JJ}^{\text{corr}} m_{\parallel}^{\text{corr}} u_{\parallel}^{\text{corr}} \theta_0^3 \kappa^3}, \quad (175)$$

$$\boxed{\alpha^{\text{corr}} \propto A_{G_*}^{\text{corr}} (C_1^{\text{corr}} \cdot G_*) \theta_0 \kappa, \quad \beta^{\text{corr}} \propto A_{G_*}^{\text{corr}} (C_2^{\text{corr}} \cdot G_*) \theta_0 \kappa.} \quad (176)$$

Therefore the corrected-branch first-shell nondegeneracy criterion is:

$$\boxed{A_{G_*}^{\text{corr}} \neq 0, \quad \mathcal{N}_{JJ}^{\text{corr}} m_{\parallel}^{\text{corr}} u_{\parallel}^{\text{corr}} \neq 0, \quad (C_1^{\text{corr}} \cdot G_*) \neq 0 \text{ or } (C_2^{\text{corr}} \cdot G_*) \neq 0.} \quad (177)$$

LXVI. 6. HONEST STATUS OF WHAT REMAINS

The structural reduction is no longer open.

What remains open is only the literal corrected-branch numerical insertion of the four scalars

$$A_{G_*}^{\text{corr}}, \quad \mathcal{N}_{JJ}^{\text{corr}}, \quad m_{\parallel}^{\text{corr}} u_{\parallel}^{\text{corr}}, \quad (C_{\alpha}^{\text{corr}} \cdot G_*).$$

Once they are inserted, the first-shell corrected witness coefficients are fully determined.

LXVII. FINAL SUMMARY

The frozen corrected branch

$$\Psi_{\text{corr}} = \Psi_{\text{BCC}} + 0.0096 \Phi_A - 0.0288 \Phi_E + 0.0096 \Phi_O$$

has now been inserted into the first-shell witness formulas.

Its selected shell harmonic is

$$A_{G_*}^{\text{corr}} = A_{G_*}^{\text{BCC}} + 0.0096 A_{G_*}^{[A]} - 0.0288 A_{G_*}^{[E]} + 0.0096 A_{G_*}^{[O]},$$

and its transverse shell projections satisfy

$$(C_{\alpha} \cdot G_*)_{\text{corr}} = (C_{\alpha} \cdot G_*)^{(0)} + 0.0096 \Xi_{\alpha}^{[A]} - 0.0288 \Xi_{\alpha}^{[E]} + 0.0096 \Xi_{\alpha}^{[O]} + O(\delta^2).$$

Therefore the corrected first-shell longitudinal witness is factorized as

$$\lambda_{\parallel}^{\text{corr}} \propto A_{G_*}^{\text{corr}} \mathcal{N}_{JJ}^{\text{corr}} m_{\parallel}^{\text{corr}} u_{\parallel}^{\text{corr}} \theta_0^3 \kappa^3,$$

while the corrected transverse witnesses are factorized as

$$\alpha^{\text{corr}} \propto A_{G_*}^{\text{corr}} (C_1^{\text{corr}} \cdot G_*) \theta_0 \kappa, \quad \beta^{\text{corr}} \propto A_{G_*}^{\text{corr}} (C_2^{\text{corr}} \cdot G_*) \theta_0 \kappa.$$

So the corrected-branch first-shell nondegeneracy problem is now reduced to the explicit insertion of only four corrected scalars:

$$A_{G_*}^{\text{corr}}, \quad \mathcal{N}_{JJ}^{\text{corr}}, \quad m_{\parallel}^{\text{corr}} u_{\parallel}^{\text{corr}}, \quad (C_{\alpha}^{\text{corr}} \cdot G_*), \quad \alpha = 1, 2.$$

No structural ambiguity remains at the first-shell level. The only remaining task is the literal numerical or symbolic insertion of these corrected scalar data.

LXVIII. EXACT ORBIT-BASIS RESPONSE MATRIX FOR THE CORRECTED FIRST-SHELL AUDIT

LXIX. GOAL

We now reduce the corrected-branch insertion to an explicit orbit-basis response matrix. The corrected orbit perturbation vector is

$$\delta_{\text{corr}} = \begin{pmatrix} \delta_A \\ \delta_E \\ \delta_O \end{pmatrix} = \begin{pmatrix} 0.0096 \\ -0.0288 \\ 0.0096 \end{pmatrix}. \quad (178)$$

The purpose is to determine which witness scalars are exactly closed at the (A, E, O) -orbit level, and which ones require finer sign-resolved suborbit data.

LXX. 1. EXACT BCC ORBIT VALUES

For the selected first BCC shell, the 12 shell directions split exactly into

$$\mathcal{A} \text{ (2 patches)}, \quad \mathcal{E} \text{ (2 patches)}, \quad \mathcal{O} \text{ (8 patches)}.$$

The exact orbit values are:

$$c^2 = \begin{cases} 1 & \alpha \in \mathcal{A}, \\ 0 & \alpha \in \mathcal{E}, \\ \frac{1}{4} & \alpha \in \mathcal{O}, \end{cases} \quad (179)$$

and

$$P_2(c_\alpha) = \begin{cases} 1 & \alpha \in \mathcal{A}, \\ -\frac{1}{2} & \alpha \in \mathcal{E}, \\ -\frac{1}{8} & \alpha \in \mathcal{O}. \end{cases} \quad (180)$$

The transverse shell coordinates are sign-sensitive:

$$t_{\alpha 1} = 0, \quad t_{\alpha 2} = 0 \quad (\alpha \in \mathcal{A}), \quad (181)$$

$$t_{\alpha 1} = \pm 1, \quad t_{\alpha 2} = 0 \quad (\alpha \in \mathcal{E}), \quad (182)$$

$$t_{\alpha 1} = \pm \frac{1}{2}, \quad t_{\alpha 2} = \pm \frac{1}{\sqrt{2}} \quad (\alpha \in \mathcal{O}). \quad (183)$$

Thus scalar orbit reductions built from c_α^2 or $P_2(c_\alpha)$ close exactly at the (A, E, O) -level, whereas transverse vector reductions built from $t_{\alpha 1}, t_{\alpha 2}$ do not.

LXXI. 2. EXACT SCALAR ORBIT-RESPONSE ROWS

A. Selected shell harmonic

Choose G_* in the axial orbit $\mathcal{A}$. Then the selected first-shell harmonic is simply the axial orbit amplitude:

$$A_{G_*} = A_A. \quad (184)$$

Hence the exact orbit-response row of A_{G_*} is

$$r^{(A_G)} = (1, 0, 0). \quad (185)$$

B. c^2 -weighted scalar channel

Any scalar source proportional to c_α^2 has exact orbit row

$$\boxed{r^{(c^2)} = \left(1, 0, \frac{1}{4}\right)}. \quad (186)$$

This includes, in particular, the orbit profile entering the exact linear growth correction

$$\delta\sigma_\alpha \propto c_\alpha^2,$$

and more generally any first-shell longitudinal/scalar channel whose orbit dependence is of pure c_α^2 -type.

C. Quadrupolar anisotropy channel

Any scalar source proportional to $P_2(c_\alpha)$ has exact orbit row

$$\boxed{r^{(P_2)} = \left(1, -\frac{1}{2}, -\frac{1}{8}\right)}. \quad (187)$$

LXXII. 3. EXACT ORBIT RESPONSE MATRIX

Collect the exact scalar rows into

$$\boxed{\mathbb{R}_{\text{orb}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & \frac{1}{4} \\ 1 & -\frac{1}{2} & -\frac{1}{8} \end{pmatrix}}. \quad (188)$$

Multiplying by the corrected branch perturbation vector (178) gives

$$\boxed{\mathbb{R}_{\text{orb}} \delta_{\text{corr}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & \frac{1}{4} \\ 1 & -\frac{1}{2} & -\frac{1}{8} \end{pmatrix} \begin{pmatrix} 0.0096 \\ -0.0288 \\ 0.0096 \end{pmatrix} = \begin{pmatrix} 0.0096 \\ 0.0120 \\ 0.0228 \end{pmatrix}}. \quad (189)$$

So the corrected branch induces the following exact first-shell scalar responses:

$$\boxed{\delta A_{G_*} = 0.0096}, \quad (190)$$

$$\boxed{\delta S_{c^2} = 0.0120}, \quad (191)$$

$$\boxed{\delta S_{P_2} = 0.0228}. \quad (192)$$

LXXIII. 4. CONSEQUENCE FOR THE LONGITUDINAL CORRECTED WITNESS

At first-shell level, the corrected longitudinal factorized witness was

$$\lambda_{\parallel}^{\text{corr}} \propto A_{G_*}^{\text{corr}} \mathcal{N}_{JJ}^{\text{corr}} m_{\parallel}^{\text{corr}} u_{\parallel}^{\text{corr}} \theta_0^3 \kappa^3.$$

The orbit reduction now implies:

1. the selected harmonic factor $A_{G_*}^{\text{corr}}$ is exactly controlled by the axial response row $r^{(A_G)}$;
2. any pure c_α^2 -type contribution inside $\mathcal{N}_{JJ}^{\text{corr}}$ is exactly controlled by $r^{(c^2)}$;
3. any pure $P_2(c_\alpha)$ -type anisotropy correction is exactly controlled by $r^{(P_2)}$.

Therefore the exact orbit-reduced corrected longitudinal ansatz becomes

$$\lambda_{\parallel}^{\text{corr}} \propto \left(A_{G_*}^{(0)} + 0.0096 \right) \left(\mathcal{N}_{JJ}^{(0)} + 0.0120 \mathcal{N}_{JJ}^{[c^2]} + 0.0228 \mathcal{N}_{JJ}^{[P_2]} \right) \left(m_{\parallel} u_{\parallel} \right)_{\text{corr}} \theta_0^3 \kappa^3, \quad (193)$$

where the last overlap factor $(m_{\parallel} u_{\parallel})_{\text{corr}}$ is still an actual Bloch overlap scalar and is not fixed by orbit data alone.

LXXIV. 5. WHY THE TRANSVERSE WITNESS DOES *NOT* CLOSE ON (A, E, O) ALONE

The corrected transverse witness was

$$\alpha^{\text{corr}} \propto A_{G_*}^{\text{corr}} (C_1^{\text{corr}} \cdot G_*) \theta_0 \kappa, \quad \beta^{\text{corr}} \propto A_{G_*}^{\text{corr}} (C_2^{\text{corr}} \cdot G_*) \theta_0 \kappa.$$

However, $C_1 \cdot G_*$ and $C_2 \cdot G_*$ are carried by the transverse shell coordinates

$$t_{\alpha 1}, \quad t_{\alpha 2},$$

whose values change sign within the $\mathcal{E}$ and $\mathcal{O}$ orbits.

Therefore the orbit-average vector

$$(\delta_A, \delta_E, \delta_O)$$

is insufficient to determine them.

a. Conclusion. The scalar orbit reduction is exact for

$$A_{G_*}, \quad c_\alpha^2\text{-type channels}, \quad P_2(c_\alpha)\text{-type channels},$$

but the transverse witnesses require a sign-resolved suborbit vector, for example

$$\delta_{\text{sub}} = (\delta_A, \delta_{E+}, \delta_{E-}, \delta_{O,1+}, \delta_{O,1-}, \delta_{O,2+}, \delta_{O,2-})^T, \quad (194)$$

or equivalently the explicit corrected Bloch coefficients $c_{a,G}^{(\pm)}$.

So the exact first-shell transverse frontier is not a missing theorem, but the fact that $(C_1 \cdot G_*)$ and $(C_2 \cdot G_*)$ are sign-sensitive observables.

LXXV. 6. ACTUAL CORRECTED-BRANCH CLOSURE STATE

The current closure state is therefore:

$$\text{scalar witness sector} \implies \text{exactly reduced to the orbit response matrix } \mathbb{R}_{\text{orb}}, \quad (195)$$

while

$$\text{transverse witness sector} \implies \text{requires sign-resolved suborbit or explicit Bloch data.} \quad (196)$$

Thus the remaining first-shell task is now sharply split into:

1. orbit-exact scalar insertion, already closed;
2. sign-resolved transverse insertion, still to be evaluated.

LXXVI. FINAL SUMMARY

For the corrected branch

$$\delta_{\text{corr}} = (\delta_A, \delta_E, \delta_O)^T = (0.0096, -0.0288, 0.0096)^T,$$

the exact first-shell scalar orbit-response matrix is

$$\mathbb{R}_{\text{orb}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & \frac{1}{4} \\ 1 & -\frac{1}{2} & -\frac{1}{8} \end{pmatrix}.$$

Therefore

$$\mathbb{R}_{\text{orb}} \delta_{\text{corr}} = \begin{pmatrix} 0.0096 \\ 0.0120 \\ 0.0228 \end{pmatrix}.$$

This means:

$$\boxed{\delta A_{G_*} = 0.0096,}$$

$$\boxed{\delta S_{c^2} = 0.0120,}$$

$$\boxed{\delta S_{P_2} = 0.0228.}$$

Hence the corrected first-shell scalar witness sector is now exactly closed at the (A, E, O) -orbit level. By contrast, the transverse witnesses

$$(C_1 \cdot G_*), \quad (C_2 \cdot G_*)$$

are carried by the sign-sensitive coordinates

$$t_{\alpha 1}, \quad t_{\alpha 2},$$

so they do not close on the orbit-average vector $(\delta_A, \delta_E, \delta_O)$ alone.

Therefore the true remaining first-shell frontier is:

$$\boxed{\text{orbit-exact scalar insertion: closed,}}$$

$$\boxed{\text{sign-resolved transverse insertion: still requires suborbit or explicit Bloch data.}}$$

LXXVII. EXACT SUBORBIT RESPONSE MATRIX AND THE VANISHING OF THE LINEAR CORRECTED TRANSVERSE SHIFT

LXXVIII. GOAL

We now pass from the coarse orbit basis

$$(A, E, O)$$

to the minimal sign-resolved first-shell suborbit basis needed for the transverse witnesses

$$(C_1 \cdot G_*), \quad (C_2 \cdot G_*).$$

The purpose is to determine whether the frozen corrected local seed

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096)$$

actually modifies the first-shell transverse witness at linear order.

LXXIX. 1. MINIMAL FIRST-SHELL SUBORBIT BASIS

Choose the shell axis

$$\hat{n} = \frac{(1, 1, 0)}{\sqrt{2}}, \quad e_1 = \frac{(1, -1, 0)}{\sqrt{2}}, \quad e_2 = (0, 0, 1).$$

Then the 12 BCC shell patches split into the following minimal suborbit basis:

$$\mathcal{A}_+ : \frac{(1, 1, 0)}{\sqrt{2}}, \quad \mathcal{A}_- : -\frac{(1, 1, 0)}{\sqrt{2}},$$

$$\mathcal{E}_+ : \frac{(1, -1, 0)}{\sqrt{2}}, \quad \mathcal{E}_- : -\frac{(1, -1, 0)}{\sqrt{2}},$$

and four oblique sign-resolved groups:

$$\mathcal{O}_{++}, \quad \mathcal{O}_{+-}, \quad \mathcal{O}_{-+}, \quad \mathcal{O}_{--},$$

defined by the signs of the two products

$$c_\alpha t_{\alpha 1}, \quad c_\alpha t_{\alpha 2}.$$

A convenient explicit realization is:

$$\mathcal{O}_{++} = \left\{ \frac{(1, 0, 1)}{\sqrt{2}}, \frac{(-1, 0, -1)}{\sqrt{2}} \right\}, \tag{197}$$

$$\mathcal{O}_{+-} = \left\{ \frac{(1, 0, -1)}{\sqrt{2}}, \frac{(-1, 0, 1)}{\sqrt{2}} \right\}, \tag{198}$$

$$\mathcal{O}_{-+} = \left\{ \frac{(0, 1, 1)}{\sqrt{2}}, \frac{(0, -1, -1)}{\sqrt{2}} \right\}, \tag{199}$$

$$\mathcal{O}_{--} = \left\{ \frac{(0, 1, -1)}{\sqrt{2}}, \frac{(0, -1, 1)}{\sqrt{2}} \right\}. \tag{200}$$

Thus the minimal sign-resolved first-shell vector is

$$\delta_{\text{sub}} = \begin{pmatrix} \delta_{A+} \\ \delta_{A-} \\ \delta_{E+} \\ \delta_{E-} \\ \delta_{O++} \\ \delta_{O+-} \\ \delta_{O-+} \\ \delta_{O--} \end{pmatrix}. \tag{201}$$

LXXX. 2. EXACT SUBORBIT VALUES

For the axial and equatorial patches one has:

$$\mathcal{A}_\pm : \quad c = \pm 1, \quad t_1 = 0, \quad t_2 = 0,$$

$$\mathcal{E}_\pm : \quad c = 0, \quad t_1 = \pm 1, \quad t_2 = 0.$$

For the four oblique suborbits, the products relevant for the transverse source are constant:

$$ct_1 = \begin{cases} +\frac{1}{4} & \mathcal{O}_{++}, \mathcal{O}_{+-}, \\ -\frac{1}{4} & \mathcal{O}_{-+}, \mathcal{O}_{--}, \end{cases} \quad (202)$$

$$ct_2 = \begin{cases} +\frac{1}{2\sqrt{2}} & \mathcal{O}_{++}, \mathcal{O}_{-+}, \\ -\frac{1}{2\sqrt{2}} & \mathcal{O}_{+-}, \mathcal{O}_{--}. \end{cases} \quad (203)$$

This is the exact minimal resolution needed because the patchwise Class II transverse couplings satisfy

$$\Gamma_\alpha^1 \propto c_\alpha (Bt_{\alpha 1} - Ct_{\alpha 2}), \quad \Gamma_\alpha^2 \propto c_\alpha (Bt_{\alpha 2} + Ct_{\alpha 1}),$$

so only the products $c_\alpha t_{\alpha 1}$ and $c_\alpha t_{\alpha 2}$ matter at first-shell level.

LXXXI. 3. EXACT SUBORBIT RESPONSE ROWS

Define the suborbit response rows.

A. Selected harmonic row

If $G_* \in \mathcal{A}_+$, then

$$r_{\text{sub}}^{(A_G)} = (1, 0, 0, 0, 0, 0, 0, 0). \quad (204)$$

B. c^2 -scalar row

Since

$$c^2 = 1 \text{ on } \mathcal{A}_\pm, \quad 0 \text{ on } \mathcal{E}_\pm, \quad \frac{1}{4} \text{ on all } \mathcal{O}\text{-suborbits},$$

the exact row is

$$r_{\text{sub}}^{(c^2)} = \left(1, 1, 0, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right). \quad (205)$$

C. Quadrupolar row

Since

$$P_2(c) = 1 \text{ on } \mathcal{A}_\pm, \quad -\frac{1}{2} \text{ on } \mathcal{E}_\pm, \quad -\frac{1}{8} \text{ on all } \mathcal{O}\text{-suborbits},$$

the exact row is

$$r_{\text{sub}}^{(P_2)} = \left(1, 1, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{8}, -\frac{1}{8}, -\frac{1}{8}, -\frac{1}{8} \right). \quad (206)$$

D. Transverse rows

From (202)–(203), the exact transverse rows are

$$r_{\text{sub}}^{(ct_1)} = \left(0, 0, 0, 0, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4} \right), \quad (207)$$

$$r_{\text{sub}}^{(ct_2)} = \left(0, 0, 0, 0, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}} \right). \quad (208)$$

LXXXII. 4. ORBIT-SYMMETRIC CORRECTED SEED LIFTED TO SUBORBIT BASIS

The frozen corrected seed is orbit-symmetric:

$$\delta_A = 0.0096, \quad \delta_E = -0.0288, \quad \delta_O = 0.0096.$$

Therefore its lift to the suborbit basis is

$$\delta_{\text{sub}}^{\text{corr}} = \begin{pmatrix} 0.0096 \\ 0.0096 \\ -0.0288 \\ -0.0288 \\ 0.0096 \\ 0.0096 \\ 0.0096 \\ 0.0096 \end{pmatrix}. \quad (209)$$

LXXXIII. 5. ACTUAL LINEAR CALCULATION

A. Scalar responses

Apply the scalar rows to (209):

$$r_{\text{sub}}^{(A_G)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0.0096, \quad (210)$$

$$r_{\text{sub}}^{(c^2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0.0096 + 0.0096 + \frac{1}{4}(4 \times 0.0096) = 0.0216, \quad (211)$$

$$r_{\text{sub}}^{(P_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0.0096 + 0.0096 - \frac{1}{2}(-0.0288 - 0.0288) - \frac{1}{8}(4 \times 0.0096) = 0.0456. \quad (212)$$

These are the exact lifted suborbit scalar responses.

B. Transverse responses

Now apply the transverse rows:

$$r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = \frac{1}{4}(0.0096 + 0.0096 - 0.0096 - 0.0096) = 0, \quad (213)$$

$$r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = \frac{1}{2\sqrt{2}}(0.0096 - 0.0096 + 0.0096 - 0.0096) = 0. \quad (214)$$

Hence

$$\boxed{r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0.} \quad (215)$$

LXXXIV. 6. CONSEQUENCE FOR THE CORRECTED FIRST-SHELL TRANSVERSE WITNESS

At first-shell level the transverse witness is proportional to

$$A_{G_*}(C_1 \cdot G_*), \quad A_{G_*}(C_2 \cdot G_*).$$

The above computation proves that the *linear correction* to the sign-sensitive pieces

$$(C_1 \cdot G_*), \quad (C_2 \cdot G_*)$$

from the current orbit-symmetric seed vanishes identically at first-shell level.

Therefore:

$$\boxed{\delta(C_1 \cdot G_*)_{\text{corr}}^{\text{1st shell}} = 0, \quad \delta(C_2 \cdot G_*)_{\text{corr}}^{\text{1st shell}} = 0.} \quad (216)$$

Equivalently,

$$\boxed{\alpha^{\text{corr}} - \alpha^{(0)} \text{ and } \beta^{\text{corr}} - \beta^{(0)} \text{ receive no linear first-shell contribution from } (\delta_A, \delta_E, \delta_O).} \quad (217)$$

LXXXV. 7. HONEST INTERPRETATION

This does *not* mean that the total transverse witness vanishes. It means only:

1. the present geometry-preserving corrected seed changes the scalar/longitudinal shell data;
2. but its first-shell transverse correction vanishes at linear order because the seed is orbit-symmetric inside each sign-sensitive suborbit pair;
3. therefore any nonzero corrected transverse shift must come from one of the following:
 - (a) pre-existing base-branch transverse background,
 - (b) genuine sign-resolved suborbit splitting beyond the (A, E, O) ansatz,
 - (c) higher-shell or non-first-shell contributions,
 - (d) nonlinear $O(\delta^2)$ terms.

So the corrected-branch first-shell transverse frontier is now sharply localized.

LXXXVI. 8. PRACTICAL NEXT STEP

The next executable transverse step is therefore not another orbit-average correction. It is one of the following two:

$$\boxed{\text{(i) extend the reduced seed from } (A, E, O) \text{ to a sign-resolved suborbit seed,}} \quad (218)$$

or

$$\boxed{\text{(ii) evaluate the explicit Bloch coefficients } c_{a,G}^{(\pm)} \text{ and the derivative kernels directly.}} \quad (219)$$

LXXXVII. FINAL SUMMARY

The minimal sign-resolved first-shell suborbit basis is

$$(\mathcal{A}_+, \mathcal{A}_-, \mathcal{E}_+, \mathcal{E}_-, \mathcal{O}_{++}, \mathcal{O}_{+-}, \mathcal{O}_{-+}, \mathcal{O}_{--}),$$

with exact transverse rows

$$r_{\text{sub}}^{(ct_1)} = \left(0, 0, 0, 0, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right),$$

$$r_{\text{sub}}^{(ct_2)} = \left(0, 0, 0, 0, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}\right).$$

The frozen corrected seed lifts to

$$\delta_{\text{sub}}^{\text{corr}} = (0.0096, 0.0096, -0.0288, -0.0288, 0.0096, 0.0096, 0.0096, 0.0096)^T.$$

Applying the transverse rows gives the exact result

$$\boxed{r_{\text{sub}}^{(ct_1)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0, \quad r_{\text{sub}}^{(ct_2)} \cdot \delta_{\text{sub}}^{\text{corr}} = 0.}$$

Therefore the present orbit-symmetric corrected seed produces no linear first-shell correction to the transverse witnesses

$$(C_1 \cdot G_*), \quad (C_2 \cdot G_*).$$

So the corrected first-shell transverse frontier is no longer ambiguous:

it must come from base-branch transverse data, sign-resolved suborbit splitting, higher-shell input, or $O(\delta^2)$ terms.

LXXXVIII. QUADRATIC CORRECTED-BRANCH TRANSVERSE SHIFT

LXXXIX. GOAL

We now compute the first nontrivial $O(\delta^2)$ transverse correction generated by the frozen corrected orbit seed

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096)^T.$$

The exact first-shell Class II transverse witness has the form

$$\boxed{\widehat{T}_{(II)}^\alpha(G_*) \propto i A_{G_*} (C_\alpha \cdot G_*), \quad \alpha = 1, 2.} \tag{220}$$

The key issue is therefore to distinguish:

1. the corrected variation of the selected harmonic A_{G_*} ,
2. the corrected variation of the sign-sensitive transverse factor

$$X_\alpha := C_\alpha \cdot G_*.$$

XC. 1. EXACT EXPANSION STRUCTURE

Write the corrected expansions around the base branch $\delta = 0$ as

$$A_{G_*}(\delta) = A_0 + a_A \delta_A + a_E \delta_E + a_O \delta_O + \frac{1}{2} \delta^T H^{(A)} \delta + O(\delta^3), \quad (221)$$

and

$$X_\alpha(\delta) = X_\alpha^{(0)} + \ell_\alpha \cdot \delta + \frac{1}{2} \delta^T H^{(\alpha)} \delta + O(\delta^3), \quad \alpha = 1, 2. \quad (222)$$

From the previous sign-resolved suborbit calculation, the linear correction of the sign-sensitive transverse factor vanishes on the present orbit-symmetric corrected seed:

$$\boxed{\ell_\alpha \cdot \delta_{\text{corr}} = 0, \quad \alpha = 1, 2.} \quad (223)$$

Therefore

$$\boxed{X_\alpha(\delta_{\text{corr}}) = X_\alpha^{(0)} + Q_\alpha(\delta_{\text{corr}}) + O(\delta^3),} \quad (224)$$

where

$$\boxed{Q_\alpha(\delta) := \frac{1}{2} \delta^T H^{(\alpha)} \delta.} \quad (225)$$

Likewise define

$$\boxed{Q_A(\delta) := \frac{1}{2} \delta^T H^{(A)} \delta.} \quad (226)$$

XCI. 2. EXPANSION OF THE TOTAL TRANSVERSE WITNESS

Using (220), the corrected transverse shell coefficient becomes

$$\hat{T}_{(II),\text{corr}}^\alpha(G_*) \propto i A_{G_*}(\delta_{\text{corr}}) X_\alpha(\delta_{\text{corr}}) \quad (227)$$

$$= i \left[A_0 + (a \cdot \delta_{\text{corr}}) + Q_A(\delta_{\text{corr}}) + O(\delta^3) \right] \left[X_\alpha^{(0)} + Q_\alpha(\delta_{\text{corr}}) + O(\delta^3) \right]. \quad (228)$$

Hence, up to quadratic order,

$$\boxed{\hat{T}_{(II),\text{corr}}^\alpha(G_*) = i \left[A_0 X_\alpha^{(0)} + (a \cdot \delta_{\text{corr}}) X_\alpha^{(0)} + A_0 Q_\alpha(\delta_{\text{corr}}) + X_\alpha^{(0)} Q_A(\delta_{\text{corr}}) \right] + O(\delta^3).} \quad (229)$$

This formula is the exact corrected-branch transverse bookkeeping up to $O(\delta^2)$.

XCII. 3. HONEST INTERPRETATION OF THE FIRST NONZERO ORDER

Equation (229) implies three distinct cases.

a. Case I: pre-existing base transverse background. If

$$X_\alpha^{(0)} \neq 0,$$

then the leading corrected shift is already linear:

$$\boxed{\delta \hat{T}_{(II)}^\alpha = i(a \cdot \delta_{\text{corr}}) X_\alpha^{(0)} + O(\delta^2).} \quad (230)$$

b. Case II: base transverse factor vanishes, selected harmonic survives. If

$$X_\alpha^{(0)} = 0, \quad A_0 \neq 0,$$

then the first nonzero corrected contribution is genuinely quadratic:

$$\boxed{\widehat{T}_{(II),\text{corr}}^\alpha(G_*) = iA_0 Q_\alpha(\delta_{\text{corr}}) + O(\delta^3).} \quad (231)$$

c. Case III: both base factors vanish. If

$$A_0 = 0, \quad X_\alpha^{(0)} = 0,$$

then the first nonzero contribution appears only at cubic or higher order.

Thus the statement

“the transverse correction starts at $O(\delta^2)$ ”

is only valid in the special branch

$$X_\alpha^{(0)} = 0.$$

It is not a general statement.

XCIII. 4. EXPLICIT QUADRATIC MONOMIAL CALCULATION FOR THE FROZEN CORRECTED SEED

Now insert

$$\delta_A = 0.0096, \quad \delta_E = -0.0288, \quad \delta_O = 0.0096.$$

The exact quadratic monomials are

$$\boxed{\delta_A^2 = 9.216 \times 10^{-5}, \quad \delta_E^2 = 8.2944 \times 10^{-4}, \quad \delta_O^2 = 9.216 \times 10^{-5},} \quad (232)$$

$$\boxed{\delta_A \delta_E = -2.7648 \times 10^{-4}, \quad \delta_A \delta_O = 9.216 \times 10^{-5}, \quad \delta_E \delta_O = -2.7648 \times 10^{-4}.} \quad (233)$$

Therefore

$$Q_\alpha(\delta_{\text{corr}}) = \frac{1}{2} \left[H_{AA}^{(\alpha)} \delta_A^2 + H_{EE}^{(\alpha)} \delta_E^2 + H_{OO}^{(\alpha)} \delta_O^2 + 2H_{AE}^{(\alpha)} \delta_A \delta_E + 2H_{AO}^{(\alpha)} \delta_A \delta_O + 2H_{EO}^{(\alpha)} \delta_E \delta_O \right] \quad (234)$$

$$= \boxed{4.608 \times 10^{-5} H_{AA}^{(\alpha)} + 4.1472 \times 10^{-4} H_{EE}^{(\alpha)} + 4.608 \times 10^{-5} H_{OO}^{(\alpha)} - 2.7648 \times 10^{-4} H_{AE}^{(\alpha)} + 9.216 \times 10^{-5} H_{AO}^{(\alpha)} - 2.7648 \times 10^{-4} H_{EO}^{(\alpha)}} \quad (235)$$

Similarly,

$$\boxed{Q_A(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(A)} + 4.1472 \times 10^{-4} H_{EE}^{(A)} + 4.608 \times 10^{-5} H_{OO}^{(A)} - 2.7648 \times 10^{-4} H_{AE}^{(A)} + 9.216 \times 10^{-5} H_{AO}^{(A)} - 2.7648 \times 10^{-4} H_{EO}^{(A)}} \quad (236)$$

These are the exact $O(\delta^2)$ corrected-branch insertions.

XCIV. 5. TRANSVERSE REDUCED COEFFICIENTS

The reduced coefficients are

$$\boxed{\alpha^{\text{corr}} = I_\theta c_E^{\text{corr}}, \quad \beta^{\text{corr}} = I_\theta c_O^{\text{corr}},} \quad (237)$$

with

$$I_\theta = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa.$$

Hence, up to $O(\delta^2)$,

$$\alpha^{\text{corr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left[A_0X_1^{(0)} + (a \cdot \delta_{\text{corr}})X_1^{(0)} + A_0Q_1(\delta_{\text{corr}}) + X_1^{(0)}Q_A(\delta_{\text{corr}})\right] + O(\delta^3), \quad (238)$$

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left[A_0X_2^{(0)} + (a \cdot \delta_{\text{corr}})X_2^{(0)} + A_0Q_2(\delta_{\text{corr}}) + X_2^{(0)}Q_A(\delta_{\text{corr}})\right] + O(\delta^3). \quad (239)$$

XCV. 6. WHAT IS ACTUALLY FIXED NOW

The corrected-branch transverse frontier is now sharply localized.

1. The sign-sensitive linear correction to $X_\alpha = C_\alpha \cdot G_*$ vanishes exactly for the present orbit-symmetric seed.
2. The actual $O(\delta^2)$ correction is encoded in the Hessian coefficients $H_{rs}^{(\alpha)}$.
3. If the base branch already has $X_\alpha^{(0)} \neq 0$, then the leading corrected shift is linear through A_{G_*} .
4. Only if $X_\alpha^{(0)} = 0$ does the corrected transverse sector genuinely start at quadratic order.

Therefore the remaining microscopic question is no longer structural. It is the actual evaluation of

$$X_\alpha^{(0)}, \quad a \cdot \delta_{\text{corr}}, \quad H_{rs}^{(\alpha)}, \quad H_{rs}^{(A)}.$$

XCVI. FINAL SUMMARY

The first-shell transverse witness has the factorized form

$$\hat{T}_{(II)}^\alpha(G_*) \propto iA_{G_*}(C_\alpha \cdot G_*), \quad \alpha = 1, 2.$$

For the frozen corrected seed

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096)^T,$$

the sign-sensitive transverse factor satisfies

$$\ell_\alpha \cdot \delta_{\text{corr}} = 0.$$

Thus

$$C_\alpha \cdot G_* = X_\alpha^{(0)} + Q_\alpha(\delta_{\text{corr}}) + O(\delta^3), \quad Q_\alpha(\delta) = \frac{1}{2}\delta^T H^{(\alpha)} \delta.$$

The exact quadratic insertions are

$$\delta_A^2 = 9.216 \times 10^{-5}, \quad \delta_E^2 = 8.2944 \times 10^{-4}, \quad \delta_O^2 = 9.216 \times 10^{-5},$$

$$\delta_A \delta_E = -2.7648 \times 10^{-4}, \quad \delta_A \delta_O = 9.216 \times 10^{-5}, \quad \delta_E \delta_O = -2.7648 \times 10^{-4}.$$

Therefore

$$Q_\alpha(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(\alpha)} + 4.1472 \times 10^{-4} H_{EE}^{(\alpha)} + 4.608 \times 10^{-5} H_{OO}^{(\alpha)} - 2.7648 \times 10^{-4} H_{AE}^{(\alpha)} + 9.216 \times 10^{-5} H_{AO}^{(\alpha)} - 2.7648 \times 10^{-4} H_{EO}^{(\alpha)}.$$

Hence the corrected transverse reduced coefficients are

$$\alpha^{\text{corr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left[A_0X_1^{(0)} + (a \cdot \delta_{\text{corr}})X_1^{(0)} + A_0Q_1(\delta_{\text{corr}}) + X_1^{(0)}Q_A(\delta_{\text{corr}})\right] + O(\delta^3),$$

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16}\theta_0\kappa\left[A_0X_2^{(0)} + (a \cdot \delta_{\text{corr}})X_2^{(0)} + A_0Q_2(\delta_{\text{corr}}) + X_2^{(0)}Q_A(\delta_{\text{corr}})\right] + O(\delta^3).$$

So the exact conclusion is:

the sign-sensitive transverse factor has no linear corrected shift,

but

the total transverse coefficient can still change linearly if the base branch already has $X_\alpha^{(0)} \neq 0$.

Only in the special case

$$X_\alpha^{(0)} = 0$$

does the corrected transverse sector start genuinely at

$$O(\delta^2).$$

XC VII. BASE-BRANCH VERDICT FOR $X_\alpha^{(0)} = (C_\alpha \cdot G_*)^{(0)}$ AND THE CORRECTED TRANSVERSE EXPANSION

XC VIII. GOAL

We now determine whether the base-branch transverse factors

$$X_\alpha^{(0)} := (C_\alpha \cdot G_*)^{(0)}, \quad \alpha = 1, 2,$$

are zero or nonzero on the explicit canonical branch.

This is the key missing input needed to interpret the corrected transverse expansion

$$\alpha^{\text{corr}}, \quad \beta^{\text{corr}}.$$

XC IX. 1. EXACT EXPLICIT CANONICAL BRANCH DATA

The explicit canonical pure-even branch is fixed by

$$\boxed{T_\star = T^1, \quad \eta = z_0, \quad \Lambda_O^{(0)} = 0, \quad C_\parallel = C_E = C_O = 1.} \tag{240}$$

On this branch, the reduced coefficients are already given explicitly as

$$\boxed{\lambda_\parallel = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3,} \tag{241}$$

$$\boxed{\alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad \beta = 0.} \tag{242}$$

Therefore, as long as

$$\beta_X \neq 0, \quad \theta_0 \kappa \neq 0,$$

one has

$$\boxed{\alpha \neq 0, \quad \beta = 0.} \tag{243}$$

C. 2. RELATION TO THE FIRST-SHELL TRANSVERSE FACTORS

Inside the explicit folded-Bragg Class II branch, the first-shell transverse shell coefficient satisfies

$$\boxed{\hat{T}_{(II)}^\alpha(G_*) \propto iA_{G_*}(C_\alpha \cdot G_*), \quad \alpha = 1, 2.} \quad (244)$$

After Bloch calibration,

$$\alpha = C_E g_E, \quad \beta = C_O g_O.$$

Since in the canonical explicit branch

$$C_E = C_O = 1,$$

the Bloch layer does not kill any nonzero transport witness.

Hence

$$\alpha \neq 0$$

means the $\alpha = 1$ transverse witness survives, while

$$\beta = 0$$

means the $\alpha = 2$ transverse witness is absent on this explicit branch.

Assuming the selected shell harmonic is nonzero,

$$A_{G_*} \neq 0,$$

equation (244) gives the base-branch verdict

$$\boxed{X_1^{(0)} = (C_1 \cdot G_*)^{(0)} \neq 0, \quad X_2^{(0)} = (C_2 \cdot G_*)^{(0)} = 0.} \quad (245)$$

So the explicit canonical pure-even branch is precisely a branch with one surviving even transverse factor and vanishing odd transverse factor.

CI. 3. CORRECTED EXPANSION REVISITED WITH THE ACTUAL BASE-BRANCH VERDICT

From the previous corrected-branch bookkeeping, write

$$A_{G_*}(\delta) = A_0 + a \cdot \delta + Q_A(\delta) + O(\delta^3), \quad (246)$$

and

$$X_\alpha(\delta) = X_\alpha^{(0)} + Q_\alpha(\delta) + O(\delta^3), \quad Q_\alpha(\delta) = \frac{1}{2} \delta^T H^{(\alpha)} \delta, \quad (247)$$

because the sign-sensitive linear correction vanishes:

$$\ell_\alpha \cdot \delta_{\text{corr}} = 0.$$

The total transverse witness is

$$\hat{T}_{(II),\text{corr}}^\alpha(G_*) = i \left[A_0 X_\alpha^{(0)} + (a \cdot \delta_{\text{corr}}) X_\alpha^{(0)} + A_0 Q_\alpha(\delta_{\text{corr}}) + X_\alpha^{(0)} Q_A(\delta_{\text{corr}}) \right] + O(\delta^3). \quad (248)$$

Now insert the base-branch verdict (245).

A. Even transverse coefficient

For $\alpha = 1$, since $X_1^{(0)} \neq 0$,

$$\widehat{T}_{(II),\text{corr}}^1(G_*) = i \left[A_0 X_1^{(0)} + (a \cdot \delta_{\text{corr}}) X_1^{(0)} + A_0 Q_1(\delta_{\text{corr}}) + X_1^{(0)} Q_A(\delta_{\text{corr}}) \right] + O(\delta^3). \quad (249)$$

Therefore the reduced even transverse coefficient satisfies

$$\alpha^{\text{corr}} = \alpha^{(0)} \left[1 + \frac{a \cdot \delta_{\text{corr}}}{A_0} + \frac{Q_A(\delta_{\text{corr}})}{A_0} + \frac{Q_1(\delta_{\text{corr}})}{X_1^{(0)}} \right] + O(\delta^3), \quad (250)$$

where

$$\alpha^{(0)} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 X_1^{(0)}$$

up to the fixed branch normalization.

So α^{corr} generally has a linear corrected shift through the selected harmonic A_{G_*} , because the base branch already has $X_1^{(0)} \neq 0$.

B. Odd transverse coefficient

For $\alpha = 2$, since $X_2^{(0)} = 0$,

$$\widehat{T}_{(II),\text{corr}}^2(G_*) = i A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3). \quad (251)$$

Hence

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3). \quad (252)$$

So on the explicit canonical pure-even branch, the odd transverse coefficient indeed starts at

$$O(\delta^2)$$

for the present orbit-symmetric corrected seed.

CII. 4. QUADRATIC INSERTION FOR THE ODD TRANSVERSE CHANNEL

For the frozen corrected seed

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096)^T,$$

the quadratic form is already known:

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)} \quad (253)$$

Therefore the first nonzero odd transverse correction is explicitly

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 \left[4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)} \right] \quad (254)$$

CIII. 5. HONEST SCOPE OF THIS VERDICT

The base-branch verdict (245) is exact for the explicit canonical pure-even witness branch. However, for the full intended corrected actual branch, the actual finite branch insertion of

$$A_{G_*}(C_1 \cdot G_*), \quad A_{G_*}(C_2 \cdot G_*)$$

is still listed in the notes as not yet explicitly inserted.

Therefore the correct strongest statement is:

1. on the explicit canonical pure-even branch,

$$X_1^{(0)} \neq 0, \quad X_2^{(0)} = 0;$$

2. hence

$$\alpha^{\text{corr}}$$

receives a linear corrected shift in general;

3. while

$$\beta^{\text{corr}}$$

starts genuinely at $O(\delta^2)$ for the present orbit-symmetric corrected seed;

4. the full intended corrected branch still requires its own finite overlap insertion for a theorem-level actual pass stamp.

CIV. FINAL SUMMARY

On the explicit canonical pure-even branch,

$$\alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad \beta = 0,$$

while the first-shell transverse shell coefficient satisfies

$$\hat{T}_{(II)}^\alpha(G_*) \propto i A_{G_*}(C_\alpha \cdot G_*).$$

Therefore, assuming $A_{G_*} \neq 0$, the base-branch transverse factors are

$$\boxed{X_1^{(0)} = (C_1 \cdot G_*)^{(0)} \neq 0, \quad X_2^{(0)} = (C_2 \cdot G_*)^{(0)} = 0.}$$

Hence the corrected transverse expansion splits sharply into two cases:

$$\boxed{\alpha^{\text{corr}} = \alpha^{(0)} \left[1 + \frac{a \cdot \delta_{\text{corr}}}{A_0} + \frac{Q_A(\delta_{\text{corr}})}{A_0} + \frac{Q_1(\delta_{\text{corr}})}{X_1^{(0)}} \right] + O(\delta^3),}$$

so the even transverse coefficient can shift already at linear order, whereas

$$\boxed{\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3),}$$

so the odd transverse coefficient starts genuinely at $O(\delta^2)$ for the present orbit-symmetric corrected seed.

With

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096)^T,$$

the quadratic insertion is

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)}.$$

So the corrected-branch transverse frontier is now fully localized: the even transverse channel is already alive on the base canonical branch, while the odd transverse channel first appears through the quadratic corrected Hessian response.

CV. EXACT EXTRACTION OF $a \cdot \delta_{\text{corr}}$ AND THE HESSIAN RESPONSE COEFFICIENTS $H_{rs}^{(2)}$

CVI. GOAL

We now compute the exact meaning of the two remaining ingredients in the corrected transverse expansion:

$$a \cdot \delta_{\text{corr}}, \quad H_{rs}^{(2)}.$$

The corrected local branch is frozen by

$$\boxed{\Psi(\delta) = \Psi_{\text{BCC}} + \delta_A \Phi_A + \delta_E \Phi_E + \delta_O \Phi_O, \quad \delta_{\text{corr}} = (0.0096, -0.0288, 0.0096).} \quad (255)$$

This is the geometry-preserving residual-based local correction ansatz. :contentReference[oaicite:1]index=1

CVII. 1. EXACT DEFINITION OF THE SELECTED HARMONIC

Define the selected shell harmonic functional

$$\boxed{A_{G_*}[\Psi] := \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x \, e^{-iG_* \cdot x} \Psi(x).} \quad (256)$$

Since A_{G_*} is linear in Ψ , and $\Psi(\delta)$ is affine in δ , we get the exact expansion

$$\boxed{A_{G_*}(\delta) = A_{G_*}^{(0)} + \sum_{r \in \{A, E, O\}} a_r \delta_r,} \quad (257)$$

where

$$\boxed{a_r := \partial_{\delta_r} A_{G_*}(\delta)|_{\delta=0} = \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x \, e^{-iG_* \cdot x} \Phi_r(x) = \widehat{\Phi}_r(G_*).} \quad (258)$$

Therefore the second derivatives vanish identically:

$$\boxed{H_{rs}^{(A)} := \partial_{\delta_r} \partial_{\delta_s} A_{G_*}(\delta)|_{\delta=0} = 0 \quad \forall r, s \in \{A, E, O\}.} \quad (259)$$

So, for the frozen orbit ansatz, there is no quadratic selected-harmonic correction at all.

CVIII. 2. EXACT ORBIT-BASIS EVALUATION OF a_r

If G_* is chosen in the axial orbit $\mathcal{A}$, then by orbit support one has

$$\boxed{a_A = \widehat{\Phi}_A(G_*), \quad a_E = \widehat{\Phi}_E(G_*) = 0, \quad a_O = \widehat{\Phi}_O(G_*) = 0.} \quad (260)$$

Hence

$$\boxed{a \cdot \delta_{\text{corr}} = a_A \delta_A + a_E \delta_E + a_O \delta_O = a_A (0.0096).} \quad (261)$$

Under the canonical orbit normalization used in the earlier orbit-response reduction,

$$a_A = 1,$$

so one gets the exact numerical insertion

$$\boxed{a \cdot \delta_{\text{corr}} = 0.0096.} \quad (262)$$

Thus the selected shell harmonic shifts linearly, and only linearly, on the frozen corrected branch.

CIX. 3. EXACT DEFINITION OF THE TRANSVERSE RESPONSE OBSERVABLE

Now define the sign-sensitive odd transverse observable

$$\boxed{X_2(\delta) := (C_2 \cdot G_*)(\delta).} \quad (263)$$

At the exact transport-extraction level, this is not a free parameter. It is the odd transverse projector of the mixed transport kernel:

$$\boxed{X_2(\delta) = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)],} \quad (264)$$

where

$$\mathcal{M}_{TD+DT}^\mu(\delta) := \Pi_Q(\mathcal{V}_{TD}^\mu(\delta) + \mathcal{V}_{DT}^\mu(\delta))\Pi_Q, \quad \Lambda_O = \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}]. \quad (265)$$

This extraction protocol is already structurally fixed. :contentReference[oaicite:2]index=2

CX. 4. EXACT MEANING OF $H_{rs}^{(2)}$

By definition, the corrected odd transverse Hessian response coefficients are

$$\boxed{H_{rs}^{(2)} := \partial_{\delta_r} \partial_{\delta_s} X_2(\delta)|_{\delta=0} = \partial_{\delta_r} \partial_{\delta_s} \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)]|_{\delta=0}.} \quad (266)$$

If one freezes the shell frame $(\hat{n}, e_1, e_2)$, the tangent Pauli basis Σ_a , and the tangent projector Π_Q at the base branch, then $\mathfrak{P}_O$ is linear and background-independent at this order, so

$$\boxed{H_{rs}^{(2)} = \mathfrak{P}_O[\partial_{\delta_r} \partial_{\delta_s} \mathcal{M}_{TD+DT}^{\text{inv}}(\delta)|_{\delta=0}].} \quad (267)$$

This is the exact operator-theoretic definition of the quadratic odd transverse response.

CXI. 5. ONE-MEDIATOR SCHUR-COMPLEMENT REALIZATION

On the current refined locked + Class II working branch, the heavy mediator produces the Schur correction

$$\Delta_X^{(2)} = -\frac{1}{M_X^2} (\alpha_X \delta J + \beta_X \delta K) (\alpha_X \delta J + \beta_X \delta K)^\dagger. \quad (268)$$

Equivalently,

$$\Delta_X^{(2)} = -\frac{\alpha_X^2}{M_X^2} \delta J \delta J^\dagger - \frac{\alpha_X \beta_X}{M_X^2} (\delta J \delta K^\dagger + \delta K \delta J^\dagger) - \frac{\beta_X^2}{M_X^2} \delta K \delta K^\dagger. \quad (269)$$

The sharp current identification is:

$$JJ \rightarrow \lambda_c, \quad JK + KJ \rightarrow (\Lambda_E, \Lambda_O),$$

while the pure K -sector is transverse at principal order. :contentReference[oaicite:3]index=3

Therefore the odd transverse response is controlled, at leading working-branch level, by the mixed $JK + KJ$ channel:

$$H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_{rs} (\delta J \delta K^\dagger + \delta K \delta J^\dagger)_{\delta=0} \right] + \text{subleading completion terms}. \quad (270)$$

CXII. 6. EXPLICIT DERIVATIVE EXPANSION OF $H_{rs}^{(2)}$

Introduce the shorthand

$$J_0 := \delta J|_{\delta=0}, \quad K_0 := \delta K|_{\delta=0}, \quad J_r := \partial_{\delta_r} \delta J|_0, \quad K_r := \partial_{\delta_r} \delta K|_0,$$

$$J_{rs} := \partial_{\delta_r} \partial_{\delta_s} \delta J|_0, \quad K_{rs} := \partial_{\delta_r} \partial_{\delta_s} \delta K|_0.$$

Then

$$\begin{aligned} \partial_{rs} (\delta J \delta K^\dagger + \delta K \delta J^\dagger) \Big|_0 &= J_{rs} K_0^\dagger + J_r K_s^\dagger + J_s K_r^\dagger + J_0 K_{rs}^\dagger \\ &\quad + K_{rs} J_0^\dagger + K_r J_s^\dagger + K_s J_r^\dagger + K_0 J_{rs}^\dagger. \end{aligned} \quad (271)$$

Therefore the exact working-branch quadratic odd transverse response is

$$H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{rs} K_0^\dagger + J_r K_s^\dagger + J_s K_r^\dagger + J_0 K_{rs}^\dagger + K_{rs} J_0^\dagger + K_r J_s^\dagger + K_s J_r^\dagger + K_0 J_{rs}^\dagger \right] + \dots \quad (272)$$

This is the exact coefficient formula that had remained implicit.

CXIII. 7. FINITE HEAVY-EIGENBASIS MATRIX-ELEMENT FORM

The operator formula above can be rewritten as a finite heavy-overlap computation.

Let ϕ_a be the low carrier basis and χ_μ the heavy eigenbasis with

$$\mathbb{O}_H \chi_\mu = E_\mu \chi_\mu, \quad E_\mu \geq \Delta_H > 0.$$

Then the standard second-order Schur formula gives

$$(\mathcal{Y}_{\text{eff}}^{(2)})_{ab} = -\sum_\mu \frac{K_{a\mu} \overline{K_{b\mu}}}{E_\mu}. \quad (273)$$

More generally, for separate perturbation channels,

$$(\mathcal{Y}_{\text{eff}}^{(2)})_{ab} = - \sum_{I,J} \epsilon_I \epsilon_J \sum_{\mu} \frac{K_{a\mu}^{(I)} \overline{K_{b\mu}^{(J)}}}{E_{\mu}}. \quad (274)$$

Thus the executable matrix-element realization of $H_{rs}^{(2)}$ is

$$H_{rs}^{(2)} = \mathfrak{P}_O \left[- \sum_{\mu} \frac{K_{\mu}^{(r)} K_{\mu}^{(s)\dagger} + K_{\mu}^{(s)} K_{\mu}^{(r)\dagger}}{E_{\mu}} + \text{direct second-variation insertions} \right]. \quad (275)$$

Here the “direct second-variation insertions” are exactly the J_{rs}, K_{rs} terms from (272).

So the genuinely remaining task is now finite and explicit: compute the projected mixed first and second variations of the microscopic J - and K -kernels along the three orbit directions

$$\Phi_A, \quad \Phi_E, \quad \Phi_O.$$

CXIV. 8. CORRECTED INSERTION FORMULAS

Since $H_{rs}^{(A)} = 0$, the corrected transverse expansion simplifies compared to the generic form:

$$\alpha^{\text{corr}} = \alpha^{(0)} \left[1 + \frac{a \cdot \delta_{\text{corr}}}{A_0} + \frac{Q_1(\delta_{\text{corr}})}{X_1^{(0)}} \right] + O(\delta^3), \quad (276)$$

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3), \quad (277)$$

where

$$Q_{\alpha}(\delta) = \frac{1}{2} \sum_{r,s \in \{A,E,O\}} H_{rs}^{(2)} \delta_r \delta_s. \quad (278)$$

For the frozen corrected seed,

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096)^T,$$

so

$$a \cdot \delta_{\text{corr}} = 0.0096 \quad (\text{canonical orbit normalization}). \quad (279)$$

Thus the only nontrivial unresolved coefficients are the six symmetric entries

$$H_{AA}^{(2)}, \quad H_{EE}^{(2)}, \quad H_{OO}^{(2)}, \quad H_{AE}^{(2)}, \quad H_{AO}^{(2)}, \quad H_{EO}^{(2)}.$$

CXV. 9. HONEST CLOSURE STATEMENT

What is closed now is no longer merely the abstract expansion. It is the exact extraction meaning of the coefficients:

$$a_r = \widehat{\Phi}_r(G_*), \quad H_{rs}^{(A)} = 0, \quad (280)$$

and

$$H_{rs}^{(2)} = \partial_{\delta_r} \partial_{\delta_s} \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)]|_0, \quad (281)$$

with its explicit one-mediator working-branch realization (272) and heavy-matrix-element realization (275).

So the remaining frontier is purely finite: evaluate these six second-response overlap coefficients on the frozen corrected branch.

CXVI. FINAL SUMMARY

For the frozen corrected orbit ansatz

$$\Psi(\delta) = \Psi_{\text{BCC}} + \delta_A \Phi_A + \delta_E \Phi_E + \delta_O \Phi_O,$$

the selected shell harmonic

$$A_{G_*}[\Psi] = \frac{1}{V_{\text{cell}}} \int_{\text{cell}} d^3x e^{-iG_* \cdot x} \Psi(x)$$

is exactly affine in δ . Therefore

$$\boxed{a_r = \partial_{\delta_r} A_{G_*}|_0 = \widehat{\Phi_r}(G_*), \quad H_{rs}^{(A)} = 0.}$$

If G_* lies in the axial orbit, then

$$a_A = \widehat{\Phi_A}(G_*), \quad a_E = a_O = 0,$$

so in canonical orbit normalization

$$\boxed{a \cdot \delta_{\text{corr}} = 0.0096.}$$

The odd transverse second-response coefficients are defined exactly by

$$\boxed{H_{rs}^{(2)} = \partial_{\delta_r} \partial_{\delta_s} \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)]|_0.}$$

On the current one-mediator working branch, this becomes

$$\boxed{H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{rs} K_0^\dagger + J_r K_s^\dagger + J_s K_r^\dagger + J_0 K_{rs}^\dagger + K_{rs} J_0^\dagger + K_r J_s^\dagger + K_s J_r^\dagger + K_0 J_{rs}^\dagger \right] + \dots}$$

Equivalently, in the heavy-eigenbasis representation,

$$\boxed{H_{rs}^{(2)} = \mathfrak{P}_O \left[-\sum_{\mu} \frac{K_{\mu}^{(r)} K_{\mu}^{(s)\dagger} + K_{\mu}^{(s)} K_{\mu}^{(r)\dagger}}{E_{\mu}} + \text{direct second-variation insertions} \right].}$$

So the corrected expansion is now sharply simplified:

$$\alpha^{\text{corr}} = \alpha^{(0)} \left[1 + \frac{0.0096}{A_0} + \frac{Q_1(\delta_{\text{corr}})}{X_1^{(0)}} \right] + O(\delta^3),$$

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3),$$

with

$$Q_{\alpha}(\delta) = \frac{1}{2} \sum_{r,s} H_{rs}^{(2)} \delta_r \delta_s.$$

Hence the only nontrivial unresolved corrected-transverse data are the six symmetric coefficients

$$H_{AA}^{(2)}, \quad H_{EE}^{(2)}, \quad H_{OO}^{(2)}, \quad H_{AE}^{(2)}, \quad H_{AO}^{(2)}, \quad H_{EO}^{(2)}.$$

The remaining frontier is therefore a finite second-response overlap computation, not a structural ambiguity.

CXVII. EXPLICIT MATRIX-ELEMENT TABLE FOR THE SIX SYMMETRIC COEFFICIENTS $H_{rs}^{(2)}$

CXVIII. GOAL

We now write the six independent second-response coefficients

$$H_{AA}^{(2)}, \quad H_{EE}^{(2)}, \quad H_{OO}^{(2)}, \quad H_{AE}^{(2)}, \quad H_{AO}^{(2)}, \quad H_{EO}^{(2)}$$

explicitly in the frozen orbit basis.

The exact definition is

$$\boxed{H_{rs}^{(2)} = \partial_{\delta_r} \partial_{\delta_s} \mathfrak{P}_O[\mathcal{M}_{TD+DT}^{\text{inv}}(\delta)] \Big|_{\delta=0}, \quad r, s \in \{A, E, O\},} \quad (282)$$

and the leading working-branch one-mediator realization is

$$\boxed{H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{rs} K_0^\dagger + J_r K_s^\dagger + J_s K_r^\dagger + J_0 K_{rs}^\dagger + K_{rs} J_0^\dagger + K_r J_s^\dagger + K_s J_r^\dagger + K_0 J_{rs}^\dagger \right] + \Delta_{rs}^{\text{comp}}.} \quad (283)$$

Here:

$$J_0 := \delta J|_0, \quad K_0 := \delta K|_0,$$

$$J_r := \partial_{\delta_r} \delta J|_0, \quad K_r := \partial_{\delta_r} \delta K|_0,$$

$$J_{rs} := \partial_{\delta_r} \partial_{\delta_s} \delta J|_0, \quad K_{rs} := \partial_{\delta_r} \partial_{\delta_s} \delta K|_0,$$

and

$$\Delta_{rs}^{\text{comp}}$$

collects completion-level corrections beyond the minimal $TD + DT$ route, namely OD, DO, DD and any additional refined slots.

CXIX. 1. DIAGONAL COEFFICIENTS

For $r = s$, the mixed first-variation terms combine pairwise. Thus the diagonal entries become:

A. A-direction

$$\boxed{H_{AA}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{AA} K_0^\dagger + 2J_A K_A^\dagger + J_0 K_{AA}^\dagger + K_{AA} J_0^\dagger + 2K_A J_A^\dagger + K_0 J_{AA}^\dagger \right] + \Delta_{AA}^{\text{comp}}.} \quad (284)$$

B. E-direction

$$\boxed{H_{EE}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{EE} K_0^\dagger + 2J_E K_E^\dagger + J_0 K_{EE}^\dagger + K_{EE} J_0^\dagger + 2K_E J_E^\dagger + K_0 J_{EE}^\dagger \right] + \Delta_{EE}^{\text{comp}}.} \quad (285)$$

C. O-direction

$$H_{OO}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{OO} K_0^\dagger + 2J_O K_O^\dagger + J_0 K_{OO}^\dagger + K_{OO} J_0^\dagger + 2K_O J_O^\dagger + K_0 J_{OO}^\dagger \right] + \Delta_{OO}^{\text{comp}}. \quad (286)$$

CXX. 2. OFF-DIAGONAL COEFFICIENTS

For $r \neq s$, the two first-variation channels remain distinct.

A. AE-mixed response

$$H_{AE}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{AE} K_0^\dagger + J_A K_E^\dagger + J_E K_A^\dagger + J_0 K_{AE}^\dagger + K_{AE} J_0^\dagger + K_A J_E^\dagger + K_E J_A^\dagger + K_0 J_{AE}^\dagger \right] + \Delta_{AE}^{\text{comp}}. \quad (287)$$

B. AO-mixed response

$$H_{AO}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{AO} K_0^\dagger + J_A K_O^\dagger + J_O K_A^\dagger + J_0 K_{AO}^\dagger + K_{AO} J_0^\dagger + K_A J_O^\dagger + K_O J_A^\dagger + K_0 J_{AO}^\dagger \right] + \Delta_{AO}^{\text{comp}}. \quad (288)$$

C. EO-mixed response

$$H_{EO}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{EO} K_0^\dagger + J_E K_O^\dagger + J_O K_E^\dagger + J_0 K_{EO}^\dagger + K_{EO} J_0^\dagger + K_E J_O^\dagger + K_O J_E^\dagger + K_0 J_{EO}^\dagger \right] + \Delta_{EO}^{\text{comp}}. \quad (289)$$

CXXI. 3. SYMMETRY PROPERTIES

By definition,

$$H_{rs}^{(2)} = H_{sr}^{(2)},$$

so only the six entries above are independent.

The decomposition also separates three different microscopic roles:

1. direct second variation terms

$$J_{rs} K_0^\dagger, \quad J_0 K_{rs}^\dagger, \quad K_{rs} J_0^\dagger, \quad K_0 J_{rs}^\dagger;$$

2. bilinear first-variation terms

$$J_r K_s^\dagger, \quad J_s K_r^\dagger, \quad K_r J_s^\dagger, \quad K_s J_r^\dagger;$$

3. completion terms

$$\Delta_{rs}^{\text{comp}},$$

which summarize OD, DO, DD and any refined slots beyond the minimal $TD + DT$ branch.

Thus each $H_{rs}^{(2)}$ can be viewed as

$$H_{rs}^{(2)} = H_{rs}^{(2),\text{dir}} + H_{rs}^{(2),\text{bilin}} + \Delta_{rs}^{\text{comp}}. \quad (290)$$

CXXII. 4. COMPACT TABLE FORM

The six coefficients are summarized as follows:

Coefficient	direct second variation	bilinear first variation	completion
$H_{AA}^{(2)}$	$J_{AA}K_0^\dagger + J_0K_{AA}^\dagger + K_{AA}J_0^\dagger + K_0J_{AA}^\dagger$	$2J_AK_A^\dagger + 2K_AJ_A^\dagger$	$\Delta_{AA}^{\text{comp}}$
$H_{EE}^{(2)}$	$J_{EE}K_0^\dagger + J_0K_{EE}^\dagger + K_{EE}J_0^\dagger + K_0J_{EE}^\dagger$	$2J_EK_E^\dagger + 2K_EJ_E^\dagger$	$\Delta_{EE}^{\text{comp}}$
$H_{OO}^{(2)}$	$J_{OO}K_0^\dagger + J_0K_{OO}^\dagger + K_{OO}J_0^\dagger + K_0J_{OO}^\dagger$	$2J_OK_O^\dagger + 2K_OJ_O^\dagger$	$\Delta_{OO}^{\text{comp}}$
$H_{AE}^{(2)}$	$J_{AE}K_0^\dagger + J_0K_{AE}^\dagger + K_{AE}J_0^\dagger + K_0J_{AE}^\dagger$	$J_AK_E^\dagger + J_EK_A^\dagger + K_AJ_E^\dagger + K_EJ_A^\dagger$	$\Delta_{AE}^{\text{comp}}$
$H_{AO}^{(2)}$	$J_{AO}K_0^\dagger + J_0K_{AO}^\dagger + K_{AO}J_0^\dagger + K_0J_{AO}^\dagger$	$J_AK_O^\dagger + J_OK_A^\dagger + K_AJ_O^\dagger + K_OJ_A^\dagger$	$\Delta_{AO}^{\text{comp}}$
$H_{EO}^{(2)}$	$J_{EO}K_0^\dagger + J_0K_{EO}^\dagger + K_{EO}J_0^\dagger + K_0J_{EO}^\dagger$	$J_EK_O^\dagger + J_OK_E^\dagger + K_EJ_O^\dagger + K_OJ_E^\dagger$	$\Delta_{EO}^{\text{comp}}$

All rows are understood inside the common prefactor

$$-\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[\cdots].$$

CXXIII. 5. INSERTION INTO THE CORRECTED ODD TRANSVERSE COEFFICIENT

The corrected odd transverse coefficient is

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3), \quad (291)$$

with

$$Q_2(\delta) = \frac{1}{2} \sum_{r,s \in \{A,E,O\}} H_{rs}^{(2)} \delta_r \delta_s. \quad (292)$$

For the frozen corrected seed

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096)^T,$$

this becomes

$$\begin{aligned} Q_2(\delta_{\text{corr}}) &= 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} \\ &\quad - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)}. \end{aligned} \quad (293)$$

Thus each microscopic row in the table above is now directly weighted in the corrected β -shift.

CXXIV. 6. WHAT IS ACTUALLY LEFT

The remaining task is no longer to guess the structure of $H_{rs}^{(2)}$.

It is to evaluate, for each ordered pair $(r, s) \in \{A, E, O\}^2$,

1. the direct second-variation matrix elements

$$J_{rs}, \quad K_{rs},$$

2. the first-variation matrix elements

$$J_r, \quad K_r,$$

3. and the completion corrections

$$\Delta_{rs}^{\text{comp}}$$

from OD, DO, DD and any refined extra slots.

So the microscopic frontier is now a literal six-row finite overlap audit.

CXXV. FINAL SUMMARY

The six independent second-response coefficients

$$H_{AA}^{(2)}, \quad H_{EE}^{(2)}, \quad H_{OO}^{(2)}, \quad H_{AE}^{(2)}, \quad H_{AO}^{(2)}, \quad H_{EO}^{(2)}$$

are now written explicitly.

Their common leading working-branch form is

$$H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[\dots] + \Delta_{rs}^{\text{comp}},$$

where the bracket splits into:

$$\text{direct second variation} \quad + \quad \text{bilinear first variation.}$$

For the diagonal entries:

$$H_{rr}^{(2)} \sim J_{rr} K_0^\dagger + J_0 K_{rr}^\dagger + K_{rr} J_0^\dagger + K_0 J_{rr}^\dagger + 2J_r K_r^\dagger + 2K_r J_r^\dagger.$$

For the off-diagonal entries:

$$H_{rs}^{(2)} \sim J_{rs} K_0^\dagger + J_0 K_{rs}^\dagger + K_{rs} J_0^\dagger + K_0 J_{rs}^\dagger + J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger.$$

Thus the corrected odd transverse coefficient

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3)$$

is now reduced to a literal six-row microscopic overlap table.

No structural ambiguity remains in $H_{rs}^{(2)}$. The remaining frontier is only the finite evaluation of the rows of that table.

CXXVI. SIX-COEFFICIENT CLOSURE FOR THE CORRECTED ODD TRANSVERSE RESPONSE

CXXVII. GOAL

We now close all six symmetric second-response coefficients at once:

$$H_{AA}^{(2)}, \quad H_{EE}^{(2)}, \quad H_{OO}^{(2)}, \quad H_{AE}^{(2)}, \quad H_{AO}^{(2)}, \quad H_{EO}^{(2)}.$$

The corrected seed is frozen as

$$\delta_{\text{corr}} = (\delta_A, \delta_E, \delta_O) = (0.0096, -0.0288, 0.0096). \quad (294)$$

The odd transverse reduced coefficient is

$$\beta^{\text{corr}} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}) + O(\delta^3), \quad (295)$$

where

$$Q_2(\delta) = \frac{1}{2} \sum_{r,s \in \{A,E,O\}} H_{rs}^{(2)} \delta_r \delta_s. \quad (296)$$

CXXVIII. 1. EXACT SIX-ROW FORMULAS

The common exact working-branch definition is

$$H_{rs}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{rs} K_0^\dagger + J_r K_s^\dagger + J_s K_r^\dagger + J_0 K_{rs}^\dagger + K_{rs} J_0^\dagger + K_r J_s^\dagger + K_s J_r^\dagger + K_0 J_{rs}^\dagger \right] + \Delta_{rs}^{\text{comp}}. \quad (297)$$

Hence, explicitly:

$$H_{AA}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{AA} K_0^\dagger + 2J_A K_A^\dagger + J_0 K_{AA}^\dagger + K_{AA} J_0^\dagger + 2K_A J_A^\dagger + K_0 J_{AA}^\dagger \right] + \Delta_{AA}^{\text{comp}}, \quad (298)$$

$$H_{EE}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{EE} K_0^\dagger + 2J_E K_E^\dagger + J_0 K_{EE}^\dagger + K_{EE} J_0^\dagger + 2K_E J_E^\dagger + K_0 J_{EE}^\dagger \right] + \Delta_{EE}^{\text{comp}}, \quad (299)$$

$$H_{OO}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{OO} K_0^\dagger + 2J_O K_O^\dagger + J_0 K_{OO}^\dagger + K_{OO} J_0^\dagger + 2K_O J_O^\dagger + K_0 J_{OO}^\dagger \right] + \Delta_{OO}^{\text{comp}}, \quad (300)$$

$$H_{AE}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{AE} K_0^\dagger + J_A K_E^\dagger + J_E K_A^\dagger + J_0 K_{AE}^\dagger + K_{AE} J_0^\dagger + K_A J_E^\dagger + K_E J_A^\dagger + K_0 J_{AE}^\dagger \right] + \Delta_{AE}^{\text{comp}}, \quad (301)$$

$$H_{AO}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{AO} K_0^\dagger + J_A K_O^\dagger + J_O K_A^\dagger + J_0 K_{AO}^\dagger + K_{AO} J_0^\dagger + K_A J_O^\dagger + K_O J_A^\dagger + K_0 J_{AO}^\dagger \right] + \Delta_{AO}^{\text{comp}}, \quad (302)$$

$$H_{EO}^{(2)} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_{EO} K_0^\dagger + J_E K_O^\dagger + J_O K_E^\dagger + J_0 K_{EO}^\dagger + K_{EO} J_0^\dagger + K_E J_O^\dagger + K_O J_E^\dagger + K_0 J_{EO}^\dagger \right] + \Delta_{EO}^{\text{comp}}. \quad (303)$$

CXXIX. 2. MINIMAL FAST WORKING APPROXIMATION

If one wants the fastest executable approximation consistent with the current stage, one may drop direct second variations and completion terms:

$$J_{rs} \approx 0, \quad K_{rs} \approx 0, \quad \Delta_{rs}^{\text{comp}} \approx 0. \quad (304)$$

Then the whole six-row system collapses to

$$H_{rs}^{(2)} \approx -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger \right]. \quad (305)$$

So explicitly:

$$H_{AA}^{(2)} \approx -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_A K_A^\dagger + K_A J_A^\dagger \right], \quad (306)$$

$$H_{EE}^{(2)} \approx -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[J_E K_E^\dagger + K_E J_E^\dagger \right], \quad (307)$$

$$H_{OO}^{(2)} \approx -\frac{2\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O[J_O K_O^\dagger + K_O J_O^\dagger], \quad (308)$$

$$H_{AE}^{(2)} \approx -\frac{\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O[J_A K_E^\dagger + J_E K_A^\dagger + K_A J_E^\dagger + K_E J_A^\dagger], \quad (309)$$

$$H_{AO}^{(2)} \approx -\frac{\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O[J_A K_O^\dagger + J_O K_A^\dagger + K_A J_O^\dagger + K_O J_A^\dagger], \quad (310)$$

$$H_{EO}^{(2)} \approx -\frac{\alpha_X\beta_X}{M_X^2}\mathfrak{P}_O[J_E K_O^\dagger + J_O K_E^\dagger + K_E J_O^\dagger + K_O J_E^\dagger]. \quad (311)$$

CXXX. 3. HEAVY-EIGENBASIS REALIZATION FOR ALL SIX COEFFICIENTS

Let $\{\chi_\mu\}$ be the heavy eigenbasis with

$$\mathbb{O}_H \chi_\mu = E_\mu \chi_\mu, \quad E_\mu > 0.$$

Define the light-heavy first-variation matrix elements

$$K_{a\mu}^{(r)} := \langle \phi_a, V_{LH}^{(r)} \chi_\mu \rangle, \quad r \in \{A, E, O\}.$$

Then the fast second-order heavy-mediated part of all six coefficients is

$$H_{rs}^{(2),\text{fast}} = -\mathfrak{P}_O \left[\sum_\mu \frac{K_{\cdot\mu}^{(r)} K_{\mu\cdot}^{(s)\dagger} + K_{\mu\cdot}^{(s)} K_{\cdot\mu}^{(r)\dagger}}{E_\mu} \right]. \quad (312)$$

Thus all six coefficients are determined once the three first-variation channels

$$K^{(A)}, \quad K^{(E)}, \quad K^{(O)}$$

are known.

CXXXI. 4. FULL CORRECTED ODD TRANSVERSE INSERTION

The corrected odd transverse quadratic form is

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2)} + 4.1472 \times 10^{-4} H_{EE}^{(2)} + 4.608 \times 10^{-5} H_{OO}^{(2)} - 2.7648 \times 10^{-4} H_{AE}^{(2)} + 9.216 \times 10^{-5} H_{AO}^{(2)} - 2.7648 \times 10^{-4} H_{EO}^{(2)} \quad (313)$$

Substituting the fast formulas gives the one-line executable form

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 \left[4.608 \times 10^{-5} H_{AA}^{(2),\text{fast}} + 4.1472 \times 10^{-4} H_{EE}^{(2),\text{fast}} + 4.608 \times 10^{-5} H_{OO}^{(2),\text{fast}} - 2.7648 \times 10^{-4} H_{AE}^{(2),\text{fast}} + 9.216 \times 10^{-5} H_{AO}^{(2),\text{fast}} - 2.7648 \times 10^{-4} H_{EO}^{(2),\text{fast}} \right] \quad (314)$$

CXXXII. 5. WHAT IS ACTUALLY LEFT

At this point there is no structural ambiguity left.
The remaining microscopic input is exactly:

1. the three first-variation overlap channels

$$J_A, J_E, J_O, \quad K_A, K_E, K_O,$$

or equivalently the three light-heavy matrices

$$K^{(A)}, \quad K^{(E)}, \quad K^{(O)};$$

2. if higher accuracy is needed, the direct second variations

$$J_{rs}, \quad K_{rs},$$

and completion terms $\Delta_{rs}^{\text{comp}}$.

Thus the six-row audit is now fully closed as an executable finite-overlap computation.

CXXXIII. FINAL SUMMARY

All six second-response coefficients are now fixed at once.
The exact working formulas are:

$$H_{AA}^{(2)}, H_{EE}^{(2)}, H_{OO}^{(2)}, H_{AE}^{(2)}, H_{AO}^{(2)}, H_{EO}^{(2)}$$

with each coefficient written explicitly in terms of

$$J_{rs}K_0^\dagger, \quad J_rK_s^\dagger, \quad J_sK_r^\dagger, \quad J_0K_{rs}^\dagger$$

and the Hermitian-conjugate partners, plus completion terms.

Under the fast practical approximation

$$J_{rs} \approx K_{rs} \approx \Delta_{rs}^{\text{comp}} \approx 0,$$

all six coefficients collapse to the uniform formula

$$H_{rs}^{(2)} \approx -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_rK_s^\dagger + J_sK_r^\dagger + K_rJ_s^\dagger + K_sJ_r^\dagger].$$

Therefore the entire corrected odd transverse shift is reduced to the three first-variation channels

$$(J_A, K_A), \quad (J_E, K_E), \quad (J_O, K_O),$$

or equivalently the three light-heavy matrices

$$K^{(A)}, \quad K^{(E)}, \quad K^{(O)}.$$

So the six-row problem is no longer structurally open. It is now a finite overlap computation.

CXXXIV. COMPLETION OF J_i^a AND K_i^a IN THE EXPLICIT WORKING BRANCH AND THE RESULTING COMPUTABLE EFFECTIVE PDE

CXXXV. HONEST STATUS

The notes explicitly distinguish two levels:

1. the unreduced final-complete TECT expressions for $J_i^a[\Psi]$ and $K_i^a[P, \psi]$ in the physically relevant defect background are still honestly open;
2. however, inside the strongest explicit working branch / concrete one-mediator truncation, J_i^a is exact and K_i^a is explicit enough to be completed into a fully computable working formula.

So in what follows we complete J and K at the strongest explicit working-branch level, without pretending that this is already the final unconditional complete TECT action.

CXXXVI. 1. WORKING MICROSCOPIC BRANCH

The strongest explicit working branch is

$$\mathcal{L}_{\text{mic}} = \mathcal{L}_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \Psi]. \quad (315)$$

In the concrete one-mediator truncation, the notes give

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T_a \Psi + \Psi^\dagger T_a (\partial_i \Psi), \quad (316)$$

$$K_i^a[P, \psi] = \text{Tr}(T_a \partial_i P) n^\mu D_\mu \psi. \quad (317)$$

CXXXVII. 2. COMPLETING J_i^a

Define the adjoint-density field

$$m^a := \Psi^\dagger T^a \Psi. \quad (318)$$

Then (316) is exactly

$$J_i^a = \partial_i m^a. \quad (319)$$

So J_i^a is already completely explicit.

CXXXVIII. 3. COMPLETING K_i^a

Write

$$\rho := \Psi^\dagger \Psi, \quad z := \frac{\Psi}{\sqrt{\rho}}, \quad P = z z^\dagger = \frac{\Psi \Psi^\dagger}{\rho}. \quad (320)$$

Define the normalized adjoint projector coordinate

$$q^a := \text{Tr}(T^a P) = z^\dagger T^a z = \frac{m^a}{\rho}. \quad (321)$$

Also define the worldvolume derivative of the light field

$$\chi := n^\mu D_\mu \Psi. \quad (322)$$

Since

$$\text{Tr}(T_a \partial_i P) = \partial_i \text{Tr}(T_a P) = \partial_i q^a,$$

the current (317) becomes

$$K_i^a = (\partial_i q^a) \chi. \quad (323)$$

Now use $q^a = m^a / \rho$. Then

$$\partial_i q^a = \frac{\partial_i m^a}{\rho} - \frac{m^a}{\rho^2} \partial_i \rho. \quad (324)$$

Therefore the completed explicit working-branch formula for K_i^a is

$$K_i^a = \left(\frac{\partial_i m^a}{\rho} - \frac{m^a}{\rho^2} \partial_i \rho \right) \chi. \quad (325)$$

Thus both currents are now fully explicit in terms of Ψ (equivalently (ρ, z)) and the covariant worldvolume derivative.

CXXXIX. 4. HEAVY-MEDIATOR ELIMINATION

The heavy carrier equation is

$$X_i^a = -\frac{1}{M_X^2} (\alpha_X J_i^a + \beta_X K_i^a). \quad (326)$$

Substituting (319) and (325),

$$X_i^a = -\frac{1}{M_X^2} \left[\alpha_X \partial_i m^a + \beta_X \left(\frac{\partial_i m^a}{\rho} - \frac{m^a}{\rho^2} \partial_i \rho \right) \chi \right]. \quad (327)$$

Hence the exact tree-level working-branch effective interaction is

$$\mathcal{L}_{\text{eff}}^{(X)} = -\frac{1}{2M_X^2} \sum_{i,a} \left[\alpha_X \partial_i m^a + \beta_X \left(\frac{\partial_i m^a}{\rho} - \frac{m^a}{\rho^2} \partial_i \rho \right) \chi \right]^2. \quad (328)$$

This is the completed computable $J+K$ effective Class-II correction.

CXL. 5. LINEARIZATION AROUND THE FROZEN BRANCH

Let

$$\Psi = \bar{\Psi} + \psi, \quad \rho = \bar{\rho} + \delta\rho, \quad m^a = \bar{m}^a + \delta m^a, \quad q^a = \bar{q}^a + \delta q^a, \quad \chi = \bar{\chi} + \delta\chi,$$

with

$$\bar{\rho} = \bar{\Psi}^\dagger \bar{\Psi}, \quad \bar{m}^a = \bar{\Psi}^\dagger T^a \bar{\Psi}, \quad \bar{q}^a = \frac{\bar{m}^a}{\bar{\rho}}, \quad \delta\chi = n^\mu D_\mu \psi.$$

Then

$$\delta\rho = \bar{\Psi}^\dagger \psi + \psi^\dagger \bar{\Psi}, \quad (329)$$

$$\delta m^a = \bar{\Psi}^\dagger T^a \psi + \psi^\dagger T^a \bar{\Psi}, \quad (330)$$

and since $q^a = m^a/\rho$,

$$\delta q^a = \frac{1}{\bar{\rho}} (\delta m^a - \bar{q}^a \delta\rho). \quad (331)$$

Therefore

$$\delta J_i^a = \partial_i \delta m^a, \quad (332)$$

and

$$\delta K_i^a = (\partial_i \delta q^a) \bar{\chi} + (\partial_i \bar{q}^a) \delta\chi. \quad (333)$$

Substituting (331),

$$\delta K_i^a = \partial_i \left[\frac{1}{\bar{\rho}} (\delta m^a - \bar{q}^a \delta\rho) \right] \bar{\chi} + (\partial_i \bar{q}^a) n^\mu D_\mu \psi. \quad (334)$$

So the exact first variations needed in the six-row audit are now completely explicit.

CXLI. 6. FRÉCHET-OPERATOR FORM

The mixed Hessian block is defined by

$$H_{\Psi X} = \alpha_X \mathbb{J} + \beta_X \mathbb{K}.$$

The corresponding Fréchet operators are:

$$\boxed{\mathbb{J}_i^a[\bar{\Psi}] \psi = (\partial_i \psi^\dagger) T^a \bar{\Psi} + \bar{\Psi}^\dagger T^a (\partial_i \psi) + (\partial_i \bar{\Psi}^\dagger) T^a \psi + \psi^\dagger T^a (\partial_i \bar{\Psi}).} \quad (335)$$

$$\boxed{\mathbb{K}_i^a[\bar{P}, \bar{\Psi}] \psi = \partial_i \left[\frac{1}{\bar{\rho}} (\bar{\Psi}^\dagger T^a \psi + \psi^\dagger T^a \bar{\Psi} - \bar{q}^a (\bar{\Psi}^\dagger \psi + \psi^\dagger \bar{\Psi})) \right] \bar{\chi} + (\partial_i \bar{q}^a) n^\mu D_\mu \psi.} \quad (336)$$

This is the completed explicit pair $(\mathbb{J}, \mathbb{K})$ for the present working branch.

CXLII. 7. IMMEDIATE COMPUTATION CONSEQUENCE

At the current fast six-row level,

$$H_{rs}^{(2), \text{fast}} \approx -\frac{\alpha_X \beta_X}{M_X^2} \Re[J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger].$$

Since J_r and K_r are now explicit by (332) and (334), all six coefficients

$$H_{AA}^{(2)}, H_{EE}^{(2)}, H_{OO}^{(2)}, H_{AE}^{(2)}, H_{AO}^{(2)}, H_{EO}^{(2)}$$

are reduced to finite overlap calculations of the orbit variations

$$\psi = \Phi_A, \quad \Phi_E, \quad \Phi_O.$$

CXLIII. 8. WHAT IS ACTUALLY COMPUTED AND WHAT REMAINS OPEN

Computed exactly inside the explicit working branch:

1. $J_i^a = \partial_i (\Psi^\dagger T^a \Psi)$;
2. $K_i^a = (\partial_i q^a) \chi$, with $q^a = (\Psi^\dagger T^a \Psi) / (\Psi^\dagger \Psi)$;
3. the heavy-mediator elimination (327);
4. the full tree-level computable effective correction (328);
5. the explicit first variations (332), (334);
6. the explicit Fréchet operators (335), (336).

Still honestly open at the full complete TECT level:

1. whether the final complete TECT action keeps exactly this K -sector or deforms it by extra microscopic slots;
2. the final absolute coefficient extraction on the actual defect background.

CXLIV. FINAL SUMMARY

Inside the strongest explicit working branch, both Class-II currents are now fully completed. The longitudinal current is exactly

$$J_i^a = \partial_i m^a, \quad m^a = \Psi^\dagger T^a \Psi.$$

The projector-coupled current is explicitly

$$K_i^a = (\partial_i q^a) \chi, \quad q^a = \frac{m^a}{\rho}, \quad \rho = \Psi^\dagger \Psi, \quad \chi = n^\mu D_\mu \Psi,$$

hence

$$K_i^a = \left(\frac{\partial_i m^a}{\rho} - \frac{m^a}{\rho^2} \partial_i \rho \right) \chi.$$

Eliminating the heavy mediator gives the explicit computable effective correction

$$\mathcal{L}_{\text{eff}}^{(X)} = -\frac{1}{2M_X^2} \sum_{i,a} \left[\alpha_X \partial_i m^a + \beta_X \left(\frac{\partial_i m^a}{\rho} - \frac{m^a}{\rho^2} \partial_i \rho \right) \chi \right]^2.$$

Linearizing around the frozen branch yields

$$\delta J_i^a = \partial_i \delta m^a, \quad \delta m^a = \bar{\Psi}^\dagger T^a \psi + \psi^\dagger T^a \bar{\Psi},$$

and

$$\delta K_i^a = \partial_i \left[\frac{1}{\bar{\rho}} (\delta m^a - \bar{q}^a \delta \rho) \right] \bar{\chi} + (\partial_i \bar{q}^a) n^\mu D_\mu \psi,$$

with

$$\delta \rho = \bar{\Psi}^\dagger \psi + \psi^\dagger \bar{\Psi}.$$

Therefore the three orbit-direction current variations

$$(J_A, K_A), \quad (J_E, K_E), \quad (J_O, K_O)$$

are now explicit, and the fast six-row coefficients

$$H_{AA}^{(2)}, H_{EE}^{(2)}, H_{OO}^{(2)}, H_{AE}^{(2)}, H_{AO}^{(2)}, H_{EO}^{(2)}$$

are reduced to finite overlaps of these explicit variations.

So, within the current working branch, J and K are no longer open. What remains open is only their final realization inside the ultimate complete TECT action.

CXLV. EXPLICIT ORBIT-CURRENT COMPLETION FOR $\psi = \Phi_A, \Phi_E, \Phi_O$

CXLVI. GOAL

We now insert explicitly

$$\psi = \Phi_A, \quad \Phi_E, \quad \Phi_O$$

into the completed working-branch currents J_i^a and K_i^a , and derive the explicit orbit-current forms

$$(J_A, K_A), \quad (J_E, K_E), \quad (J_O, K_O).$$

This is the actual current-level completion needed for the six-row second-response audit.

CXLVII. 1. BACKGROUND NOTATION

Let the frozen background be

$$\bar{\Psi} := \Psi_{\text{corr}} = \Psi_{\text{BCC}} + 0.0096 \Phi_A - 0.0288 \Phi_E + 0.0096 \Phi_O. \quad (337)$$

Define the associated background scalars

$$\bar{\rho} := \bar{\Psi}^\dagger \bar{\Psi}, \quad \bar{m}^a := \bar{\Psi}^\dagger T^a \bar{\Psi}, \quad \bar{q}^a := \frac{\bar{m}^a}{\bar{\rho}}, \quad \bar{\chi} := n^\mu D_\mu \bar{\Psi}. \quad (338)$$

For each orbit direction $r \in \{A, E, O\}$, define

$$\chi_r := n^\mu D_\mu \Phi_r. \quad (339)$$

CXLVIII. 2. TWO BASIC ORBIT-RESPONSE SCALARS

For each $r \in \{A, E, O\}$, define the scalar density response

$$N_r := \bar{\Psi}^\dagger \Phi_r + \Phi_r^\dagger \bar{\Psi} = 2 \Re(\bar{\Psi}^\dagger \Phi_r), \quad (340)$$

and the adjoint-density response

$$M_r^a := \bar{\Psi}^\dagger T^a \Phi_r + \Phi_r^\dagger T^a \bar{\Psi} = 2 \Re(\bar{\Psi}^\dagger T^a \Phi_r). \quad (341)$$

Then

$$\delta_r \rho = N_r, \quad \delta_r m^a = M_r^a. \quad (342)$$

Since $q^a = m^a / \rho$, the orbit-direction variation of q^a is

$$Q_r^a := \delta_r q^a = \frac{1}{\bar{\rho}} (M_r^a - \bar{q}^a N_r). \quad (343)$$

This Q_r^a is the exact orbit-direction projector-coordinate response.

CXLIX. 3. EXPLICIT ORBIT-CURRENT FORMULAS

The working-branch currents are

$$J_i^a = \partial_i m^a, \quad K_i^a = (\partial_i q^a) \chi.$$

Therefore, in orbit direction r ,

$$J_{r,i}^a = \delta_r J_i^a = \partial_i M_r^a. \quad (344)$$

Likewise,

$$K_{r,i}^a = \delta_r K_i^a = (\partial_i Q_r^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_r. \quad (345)$$

Using (343), this becomes

$$K_{r,i}^a = \partial_i \left[\frac{M_r^a - \bar{q}^a N_r}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_r. \quad (346)$$

This is the exact completed orbit-current formula valid for $r = A, E, O$.

CL. 4. THE THREE EXPLICIT PAIRS

A. A-direction

$$N_A = \bar{\Psi}^\dagger \Phi_A + \Phi_A^\dagger \bar{\Psi}, \quad M_A^a = \bar{\Psi}^\dagger T^a \Phi_A + \Phi_A^\dagger T^a \bar{\Psi}, \quad (347)$$

$$J_{A,i}^a = \partial_i M_A^a, \quad (348)$$

$$K_{A,i}^a = \partial_i \left[\frac{M_A^a - \bar{q}^a N_A}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_A. \quad (349)$$

B. E-direction

$$N_E = \bar{\Psi}^\dagger \Phi_E + \Phi_E^\dagger \bar{\Psi}, \quad M_E^a = \bar{\Psi}^\dagger T^a \Phi_E + \Phi_E^\dagger T^a \bar{\Psi}, \quad (350)$$

$$J_{E,i}^a = \partial_i M_E^a, \quad (351)$$

$$K_{E,i}^a = \partial_i \left[\frac{M_E^a - \bar{q}^a N_E}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_E. \quad (352)$$

C. O-direction

$$N_O = \bar{\Psi}^\dagger \Phi_O + \Phi_O^\dagger \bar{\Psi}, \quad M_O^a = \bar{\Psi}^\dagger T^a \Phi_O + \Phi_O^\dagger T^a \bar{\Psi}, \quad (353)$$

$$J_{O,i}^a = \partial_i M_O^a, \quad (354)$$

$$K_{O,i}^a = \partial_i \left[\frac{M_O^a - \bar{q}^a N_O}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_O. \quad (355)$$

CLI. 5. FOURIER FORM FOR DIRECT COMPUTATION

If the frozen background and orbit modes are expanded on the reciprocal shell/basis as

$$\bar{\Psi}(x) = \sum_G \bar{A}_G e^{iG \cdot x}, \quad \Phi_r(x) = \sum_G B_G^{(r)} e^{iG \cdot x}, \quad (356)$$

then the Fourier coefficients of N_r and M_r^a are exact finite convolutions:

$$\hat{N}_r(Q) = \sum_G \left(\bar{A}_G^\dagger B_{G+Q}^{(r)} + B_G^{(r)\dagger} \bar{A}_{G+Q} \right), \quad (357)$$

$$\widehat{M}_r^a(Q) = \sum_G \left(\bar{A}_G^\dagger T^a B_{G+Q}^{(r)} + B_G^{(r)\dagger} T^a \bar{A}_{G+Q} \right). \quad (358)$$

Hence

$$\widehat{J}_{r,i}^a(Q) = iQ_i \widehat{M}_r^a(Q). \quad (359)$$

For the K -current, define

$$\widehat{Q}_r^a(Q) = \widehat{\rho}^{-1} * \left(\widehat{M}_r^a - \widehat{q}^a * \widehat{N}_r \right)(Q), \quad (360)$$

so that

$$\widehat{K}_{r,i}^a(Q) = iQ_i \widehat{Q}_r^a(Q) \bar{\chi} + (\partial_i \bar{q}^a) \widehat{\chi}_r(Q), \quad (361)$$

where

$$\widehat{\chi}_r(Q) = n^\mu \widehat{D}_\mu \Phi_r(Q).$$

Thus the orbit currents are fully reduced to finite Fourier convolutions.

CLII. 6. FAST SIX-ROW INSERTION

At the fast current stage,

$$H_{rs}^{(2),\text{fast}} \approx -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O [J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger]. \quad (362)$$

Substituting the explicit orbit currents (344) and (346), all six coefficients are now explicit functionals of the triplets

$$(N_A, M_A^a, \chi_A), \quad (N_E, M_E^a, \chi_E), \quad (N_O, M_O^a, \chi_O).$$

In particular, the corrected odd transverse coefficient becomes

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 \left[4.608 \times 10^{-5} H_{AA}^{(2),\text{fast}} + 4.1472 \times 10^{-4} H_{EE}^{(2),\text{fast}} + 4.608 \times 10^{-5} H_{OO}^{(2),\text{fast}} - 2.7648 \times 10^{-4} H_{AE}^{(2),\text{fast}} + 9.216 \times 10^{-5} H_{EO}^{(2),\text{fast}} \right] \quad (363)$$

CLIII. 7. WHAT REMAINS

Nothing structural is left at the current working-branch level.

The remaining task is only to insert the actual frozen branch harmonics $\bar{A}_G$ and orbit-mode harmonics $B_G^{(A)}, B_G^{(E)}, B_G^{(O)}$ into (357)–(361), and evaluate the six odd-projected overlaps.

CLIV. FINAL SUMMARY

For each orbit direction $r \in \{A, E, O\}$, define

$$N_r = \bar{\Psi}^\dagger \Phi_r + \Phi_r^\dagger \bar{\Psi}, \quad M_r^a = \bar{\Psi}^\dagger T^a \Phi_r + \Phi_r^\dagger T^a \bar{\Psi}, \quad \chi_r = n^\mu D_\mu \Phi_r.$$

Then the completed working-branch orbit currents are exactly

$$J_{r,i}^a = \partial_i M_r^a,$$

$$K_{r,i}^a = \partial_i \left[\frac{M_r^a - \bar{q}^a N_r}{\bar{\rho}} \right] \bar{\chi} + (\partial_i \bar{q}^a) \chi_r.$$

Therefore the three pairs

$$(J_A, K_A), \quad (J_E, K_E), \quad (J_O, K_O)$$

are now fully explicit.

In Fourier space they are finite convolutions:

$$\hat{N}_r(Q) = \sum_G \left(\bar{A}_G^\dagger B_{G+Q}^{(r)} + B_G^{(r)\dagger} \bar{A}_{G+Q} \right),$$

$$\hat{M}_r^a(Q) = \sum_G \left(\bar{A}_G^\dagger T^a B_{G+Q}^{(r)} + B_G^{(r)\dagger} T^a \bar{A}_{G+Q} \right),$$

$$\hat{J}_{r,i}^a(Q) = iQ_i \hat{M}_r^a(Q),$$

and

$$\hat{K}_{r,i}^a(Q) = iQ_i \hat{Q}_r^a(Q) \bar{\chi} + (\partial_i \bar{q}^a) \hat{\chi}_r(Q).$$

Hence, within the current strongest explicit working branch, both J and K are completely finished. What remains is only the finite overlap evaluation using the actual harmonics of the frozen corrected branch.

CLV. FULL OVERLAP-INTEGRAL FORM OF ALL SIX FAST COEFFICIENTS

CLVI. GOAL

We now substitute the completed orbit currents

$$(J_A, K_A), \quad (J_E, K_E), \quad (J_O, K_O)$$

directly into the fast six-row formula and reduce all six coefficients to explicit overlap expressions.

CLVII. 1. COMPLETED ORBIT CURRENTS

For each orbit direction $r \in \{A, E, O\}$, define

$$N_r = \bar{\Psi}^\dagger \Phi_r + \Phi_r^\dagger \bar{\Psi}, \quad M_r^a = \bar{\Psi}^\dagger T^a \Phi_r + \Phi_r^\dagger T^a \bar{\Psi}, \quad (364)$$

$$Q_r^a = \frac{M_r^a - \bar{q}^a N_r}{\bar{\rho}}, \quad \chi_r = n^\mu D_\mu \Phi_r, \quad (365)$$

with

$$\bar{\rho} = \bar{\Psi}^\dagger \bar{\Psi}, \quad \bar{q}^a = \frac{\bar{\Psi}^\dagger T^a \bar{\Psi}}{\bar{\rho}}, \quad \bar{\chi} = n^\mu D_\mu \bar{\Psi}.$$

Then the completed orbit currents are

$$J_{r,i}^a = \partial_i M_r^a, \quad K_{r,i}^a = (\partial_i Q_r^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_r. \quad (366)$$

CLVIII. 2. COMMON FAST SIX-ROW MASTER FORMULA

The fast second-response coefficients are

$$H_{rs}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O [J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger], \quad r, s \in \{A, E, O\}. \quad (367)$$

Substituting (366), we obtain the exact fast overlap form

$$H_{rs}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_i M_r^a ((\partial_i Q_s^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_s)^\dagger + \partial_i M_s^a ((\partial_i Q_r^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_r)^\dagger + ((\partial_i Q_r^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_r) (\partial_i M_s^a)^\dagger + ((\partial_i Q_s^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_s) (\partial_i M_r^a)^\dagger \right]$$

This single formula already contains all six coefficients.

CLIX. 3. DIAGONAL THREE COEFFICIENTS

For $r = s$, the expression simplifies to

$$H_{AA}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_i M_A^a ((\partial_i Q_A^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_A)^\dagger + ((\partial_i Q_A^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_A) (\partial_i M_A^a)^\dagger \right], \quad (368)$$

$$H_{EE}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_i M_E^a ((\partial_i Q_E^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_E)^\dagger + ((\partial_i Q_E^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_E) (\partial_i M_E^a)^\dagger \right], \quad (369)$$

$$H_{OO}^{(2),\text{fast}} = -\frac{2\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_i M_O^a ((\partial_i Q_O^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_O)^\dagger + ((\partial_i Q_O^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_O) (\partial_i M_O^a)^\dagger \right]. \quad (370)$$

CLX. 4. OFF-DIAGONAL THREE COEFFICIENTS

For the mixed pairs we get:

$$H_{AE}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_i M_A^a ((\partial_i Q_E^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_E)^\dagger + \partial_i M_E^a ((\partial_i Q_A^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_A)^\dagger + \text{h.c.} \right], \quad (371)$$

$$H_{AO}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_i M_A^a ((\partial_i Q_O^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_O)^\dagger + \partial_i M_O^a ((\partial_i Q_A^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_A)^\dagger + \text{h.c.} \right], \quad (372)$$

$$H_{EO}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_i M_E^a ((\partial_i Q_O^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_O)^\dagger + \partial_i M_O^a ((\partial_i Q_E^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_E)^\dagger + \text{h.c.} \right]. \quad (373)$$

CLXI. 5. NATURAL SPLIT: PROJECTOR-RESPONSE VERSUS WORLDVOLUME-RESPONSE

Each K_r contains two distinct microscopic mechanisms:

$$K_{r,i}^a = (\partial_i Q_r^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_r.$$

Therefore every $H_{rs}^{(2),\text{fast}}$ splits naturally as

$$H_{rs}^{(2),\text{fast}} = H_{rs}^{(2),Q} + H_{rs}^{(2),\chi}, \quad (374)$$

where

$$H_{rs}^{(2),Q} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O [\partial_i M_r^a (\partial_i Q_s^a \bar{\chi})^\dagger + \partial_i M_s^a (\partial_i Q_r^a \bar{\chi})^\dagger + \text{h.c.}], \quad (375)$$

$$H_{rs}^{(2),\chi} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O [\partial_i M_r^a ((\partial_i \bar{q}^a) \chi_s)^\dagger + \partial_i M_s^a ((\partial_i \bar{q}^a) \chi_r)^\dagger + \text{h.c.}]. \quad (376)$$

This isolates:

- the pure projector-density response through Q_r^a ,
- the explicit worldvolume derivative response through $\chi_r = n^\mu D_\mu \Phi_r$.

CLXII. 6. FOURIER-CONVOLUTION FORM

If

$$\bar{\Psi}(x) = \sum_G \bar{A}_G e^{iG \cdot x}, \quad \Phi_r(x) = \sum_G B_G^{(r)} e^{iG \cdot x},$$

then

$$\widehat{M}_r^a(Q) = \sum_G \left(\bar{A}_G^\dagger T^a B_{G+Q}^{(r)} + B_G^{(r)\dagger} T^a \bar{A}_{G+Q} \right), \quad (377)$$

and

$$\widehat{J}_{r,i}^a(Q) = iQ_i \widehat{M}_r^a(Q). \quad (378)$$

Likewise

$$\widehat{Q}_r^a(Q) = \widehat{\rho}^{-1} * \left(\widehat{M}_r^a - \widehat{q}^a * \widehat{N}_r \right)(Q), \quad (379)$$

so

$$\widehat{K}_{r,i}^a(Q) = iQ_i \widehat{Q}_r^a(Q) \bar{\chi} + (\partial_i \bar{q}^a) \widehat{\chi}_r(Q). \quad (380)$$

Therefore each fast coefficient becomes a finite Fourier sum of the generic form

$$H_{rs}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\sum_Q \widehat{J}_{r,i}^a(Q) \widehat{K}_{s,i}^a(-Q)^\dagger + \sum_Q \widehat{J}_{s,i}^a(Q) \widehat{K}_{r,i}^a(-Q)^\dagger + \text{h.c.} \right]. \quad (381)$$

CLXIII. 7. DIRECT INSERTION INTO β^{corr}

For the frozen corrected seed

$$\delta_{\text{corr}} = (0.0096, -0.0288, 0.0096),$$

the odd transverse quadratic form is

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2),\text{fast}} + 4.1472 \times 10^{-4} H_{EE}^{(2),\text{fast}} + 4.608 \times 10^{-5} H_{OO}^{(2),\text{fast}} - 2.7648 \times 10^{-4} H_{AE}^{(2),\text{fast}} + 9.216 \times 10^{-5} H_{AO}^{(2),\text{fast}} \quad (382)$$

Hence

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}). \quad (383)$$

This is now a fully explicit finite overlap computation.

CLXIV. 8. PRACTICAL COMPUTATIONAL ORDER

The actual executable order is now:

$$\boxed{\bar{A}_G, B_G^{(A)}, B_G^{(E)}, B_G^{(O)} \longrightarrow \hat{N}_r, \hat{M}_r, \hat{Q}_r, \hat{\chi}_r \longrightarrow \hat{J}_r, \hat{K}_r \longrightarrow H_{rs}^{(2),\text{fast}} \longrightarrow \beta^{\text{corr}}.} \quad (384)$$

No structural ambiguity remains in the fast six-row sector.

CLXV. FINAL SUMMARY

All six fast coefficients are now fully substituted.

For each orbit direction $r \in \{A, E, O\}$,

$$J_{r,i}^a = \partial_i M_r^a, \quad K_{r,i}^a = (\partial_i Q_r^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_r,$$

with

$$N_r = \bar{\Psi}^\dagger \Phi_r + \Phi_r^\dagger \bar{\Psi}, \quad M_r^a = \bar{\Psi}^\dagger T^a \Phi_r + \Phi_r^\dagger T^a \bar{\Psi}, \quad Q_r^a = \frac{M_r^a - \bar{q}^a N_r}{\bar{\rho}}.$$

Hence the entire fast six-row system is reduced to the single common formula

$$H_{rs}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\partial_i M_r^a ((\partial_i Q_s^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_s)^\dagger + \partial_i M_s^a ((\partial_i Q_r^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_r)^\dagger + \text{h.c.} \right].$$

Equivalently, in Fourier space,

$$H_{rs}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\sum_Q \hat{J}_{r,i}^a(Q) \hat{K}_{s,i}^a(-Q)^\dagger + \sum_Q \hat{J}_{s,i}^a(Q) \hat{K}_{r,i}^a(-Q)^\dagger + \text{h.c.} \right].$$

Thus all six coefficients are now explicit finite overlap computations of the actual harmonics

$$\bar{A}_G, B_G^{(A)}, B_G^{(E)}, B_G^{(O)}.$$

So, within the current strongest explicit working branch, the six-row fast sector is fully closed. What remains is only numerical or symbolic evaluation of the finite overlaps.

CLXVI. FIRST-SHELL MINIMAL COMPUTATION OF J AND K : EXACT VANISHING OF THE TRANSVERSE SECTOR IN THE RIGID-INTERNAL ORBIT ANSATZ

CLXVII. GOAL

We now perform the actual first-shell computation of the completed working-branch currents J_i^a and K_i^a under the minimal corrected orbit ansatz.

The corrected local branch is frozen as

$$\boxed{\Psi_{\text{corr}} = \Psi_{\text{BCC}} + 0.0096 \Phi_A - 0.0288 \Phi_E + 0.0096 \Phi_O.} \quad (385)$$

The question is whether the three orbit modes Φ_A, Φ_E, Φ_O , by themselves, already produce a nontrivial transverse K -sector at first-shell level.

CLXVIII. 1. MINIMAL RIGID-INTERNAL FIRST-SHELL TRUNCATION

The corrected orbit seed is an orbit-amplitude correction of the BCC shell branch. The minimal first-shell reading of this ansatz is that the internal direction remains locked:

$$\boxed{\Psi(x) = \phi(x) z_0, \quad z_0^\dagger z_0 = 1,} \quad (386)$$

with

$$\boxed{\phi(x) = \phi_{\text{BCC}}(x) + 0.0096 \varphi_A(x) - 0.0288 \varphi_E(x) + 0.0096 \varphi_O(x),} \quad (387)$$

and

$$\Phi_r(x) = \varphi_r(x) z_0, \quad r \in \{A, E, O\}.$$

This is the minimal first-shell truncation consistent with the frozen corrected orbit seed.

CLXIX. 2. EXACT COMPUTATION OF P , q^a , AND K_i^a

By definition,

$$\rho = \Psi^\dagger \Psi = |\phi|^2, \quad P = \frac{\Psi \Psi^\dagger}{\rho} = z_0 z_0^\dagger.$$

Hence P is exactly constant in space:

$$\boxed{\partial_i P = 0.} \quad (388)$$

Now

$$m^a = \Psi^\dagger T^a \Psi = |\phi|^2 z_0^\dagger T^a z_0.$$

Define the constant internal adjoint coordinate

$$\boxed{q_0^a := z_0^\dagger T^a z_0.} \quad (389)$$

Then

$$\boxed{m^a = \rho q_0^a, \quad q^a = \frac{m^a}{\rho} = q_0^a = \text{constant.}} \quad (390)$$

Therefore

$$\boxed{\partial_i q^a = 0.} \quad (391)$$

But the completed explicit working-branch formula is

$$K_i^a = (\partial_i q^a) \chi, \quad \chi = n^\mu D_\mu \Psi.$$

So we obtain

$$\boxed{K_i^a = 0 \quad \text{exactly in the rigid-internal first-shell truncation.}} \quad (392)$$

This is a full exact computation, not a heuristic statement.

CLXX. 3. EXACT COMPUTATION OF J_i^a

Using $m^a = \rho q_0^a$, the completed current

$$J_i^a = \partial_i m^a$$

becomes

$$\boxed{J_i^a = q_0^a \partial_i \rho.} \quad (393)$$

Hence J survives, but is purely scalar-density driven in this truncation.

CLXXI. 4. ORBIT-DIRECTION VARIATIONS

Take an orbit-direction perturbation

$$\psi = \Phi_r = \varphi_r z_0.$$

Then

$$N_r = \bar{\Psi}^\dagger \Phi_r + \Phi_r^\dagger \bar{\Psi} = \bar{\phi}^* \varphi_r + \varphi_r^* \bar{\phi} = 2\Re(\bar{\phi}^* \varphi_r),$$

and

$$M_r^a = \bar{\Psi}^\dagger T^a \Phi_r + \Phi_r^\dagger T^a \bar{\Psi} = q_0^a N_r.$$

Thus

$$\boxed{M_r^a = q_0^a N_r.} \quad (394)$$

So the orbit-direction J -variation is

$$\boxed{J_{r,i}^a = \partial_i M_r^a = q_0^a \partial_i N_r.} \quad (395)$$

Now the exact orbit-direction projector response is

$$Q_r^a = \frac{M_r^a - \bar{q}^a N_r}{\bar{\rho}}.$$

But $\bar{q}^a = q_0^a$, so

$$\boxed{Q_r^a = 0 \quad \forall r \in \{A, E, O\}.} \quad (396)$$

Hence

$$K_{r,i}^a = (\partial_i Q_r^a) \bar{\chi} + (\partial_i \bar{q}^a) \chi_r = 0,$$

because both Q_r^a and $\bar{q}^a$ are constant. Therefore

$$\boxed{K_{A,i}^a = K_{E,i}^a = K_{O,i}^a = 0.} \quad (397)$$

So the three completed orbit-current pairs are

$$\boxed{(J_A, K_A) = (q_0^a \partial_i N_A, 0),} \quad (398)$$

$$\boxed{(J_E, K_E) = (q_0^a \partial_i N_E, 0),} \quad (399)$$

$$\boxed{(J_O, K_O) = (q_0^a \partial_i N_O, 0).} \quad (400)$$

CLXXII. 5. IMMEDIATE CONSEQUENCE FOR ALL SIX FAST COEFFICIENTS

The fast six-row formula is

$$H_{rs}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O[J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger].$$

Since $K_A = K_E = K_O = 0$, every term vanishes identically:

$$\boxed{H_{AA}^{(2),\text{fast}} = H_{EE}^{(2),\text{fast}} = H_{OO}^{(2),\text{fast}} = H_{AE}^{(2),\text{fast}} = H_{AO}^{(2),\text{fast}} = H_{EO}^{(2),\text{fast}} = 0.} \quad (401)$$

Therefore

$$\boxed{Q_2(\delta_{\text{corr}}) = 0 \quad \text{in the minimal rigid-internal first-shell truncation.}} \quad (402)$$

Hence the corrected odd transverse coefficient becomes

$$\boxed{\beta^{\text{corr}} = 0 \quad \text{in the minimal rigid-internal first-shell truncation.}} \quad (403)$$

CLXXIII. 6. WHAT THIS ACTUALLY MEANS

This is a strong and useful result.

It proves that the three-orbit corrected residual seed

$$\Psi_{\text{corr}} = \Psi_{\text{BCC}} + 0.0096 \Phi_A - 0.0288 \Phi_E + 0.0096 \Phi_O$$

does *not* by itself generate a transverse Class-II response if it is interpreted as a pure shell-amplitude deformation with fixed internal direction z_0 .

Therefore any nonzero actual transverse coefficient must come from genuinely additional microscopic structure, namely at least one of:

1. non-rigid internal/projector texture so that P is no longer constant,
2. a defect background with $\partial_i \bar{q}^a \neq 0$,
3. completion-level slots beyond the minimal fast branch,
4. higher-shell or non-first-shell contributions.

So the current bottleneck is now sharply localized:

transverse transport cannot be generated by orbit-amplitude corrections alone.

CLXXIV. 7. LONGITUDINAL SIDE REMAINS ALIVE

Note that the vanishing of K does not kill the longitudinal current. The surviving current is

$$J_i^a = q_0^a \partial_i \rho.$$

So the JJ -driven longitudinal sector can still survive in this truncation. What disappears is specifically the mixed transverse JK route.

Thus the rigid-internal first-shell truncation yields:

longitudinal potentially alive, transverse identically dead. (404)

CLXXV. 8. HONEST NEXT STEP

Therefore the correct next move is not to keep squeezing the three-orbit amplitude ansatz. The next move must add one genuinely transverse microscopic ingredient, for example:

1. a nontrivial internal deformation $\delta z \perp z_0$,
2. a sign-resolved projector texture on the defect background,
3. or the explicit $TD + DT$ block from the full background-dependent Hessian.

Only then can $K_i^a \neq 0$ and hence $H_{rs}^{(2)} \neq 0$.

CLXXVI. FINAL SUMMARY

In the minimal first-shell truncation

$$\Psi(x) = \phi(x) z_0,$$

the projector is constant:

$$P = z_0 z_0^\dagger, \quad \partial_i P = 0.$$

Hence

$$q^a = \text{Tr}(T^a P) = z_0^\dagger T^a z_0 = q_0^a$$

is constant, so

$$\partial_i q^a = 0, \quad K_i^a = 0.$$

At the same time,

$$m^a = \rho q_0^a, \quad J_i^a = \partial_i m^a = q_0^a \partial_i \rho.$$

For each orbit direction $r \in \{A, E, O\}$,

$$M_r^a = q_0^a N_r, \quad Q_r^a = 0, \quad K_r = 0, \quad J_r = q_0^a \partial_i N_r.$$

Therefore all six fast transverse coefficients vanish identically:

$$H_{AA}^{(2),\text{fast}} = H_{EE}^{(2),\text{fast}} = H_{OO}^{(2),\text{fast}} = H_{AE}^{(2),\text{fast}} = H_{AO}^{(2),\text{fast}} = H_{EO}^{(2),\text{fast}} = 0.$$

Hence

$$\beta^{\text{corr}} = 0$$

in the minimal rigid-internal first-shell orbit-amplitude truncation.

So the actual conclusion is sharp:

$$\text{orbit-amplitude corrections alone cannot generate the transverse Class-II response.}$$

Any nonzero transverse coefficient must come from non-rigid projector/internal structure, defect-gradient structure, completion-level slots, or higher-shell input.

CLXXVII. SMALLEST NON-RIGID INTERNAL MODE AND THE FULL REVIVAL OF THE K -SECTOR

CLXXVIII. GOAL

We now add the smallest internal deformation orthogonal to the rigid locked direction z_0 , and show explicitly that this revives the transverse Class-II current K_i^a , hence making the six coefficients

$$H_{AA}^{(2),\text{fast}}, H_{EE}^{(2),\text{fast}}, H_{OO}^{(2),\text{fast}}, H_{AE}^{(2),\text{fast}}, H_{AO}^{(2),\text{fast}}, H_{EO}^{(2),\text{fast}}$$

generically nonzero.

CLXXIX. 1. MINIMAL NON-RIGID INTERNAL ANSATZ

Let

$$z(x) = \sqrt{1 - |\eta(x)|^2} z_0 + \eta(x) \xi, \quad (405)$$

with

$$z_0^\dagger z_0 = 1, \quad \xi^\dagger \xi = 1, \quad \xi^\dagger z_0 = 0. \quad (406)$$

Then

$$\Psi(x) = \phi(x) z(x). \quad (407)$$

To linear order in the internal amplitude η ,

$$z = z_0 + \eta \xi + O(\eta^2), \quad (408)$$

hence

$$P = z z^\dagger = z_0 z_0^\dagger + \eta \xi z_0^\dagger + \bar{\eta} z_0 \xi^\dagger + O(\eta^2). \quad (409)$$

CLXXX. 2. REVIVAL OF THE PROJECTOR-COORDINATE RESPONSE

Define

$$q^a := z^\dagger T^a z.$$

At the rigid base point,

$$q_0^a := z_0^\dagger T^a z_0.$$

Now define the tangent adjoint coefficient

$$\boxed{c^a := \xi^\dagger T^a z_0.} \quad (410)$$

Then

$$\boxed{q^a = q_0^a + \eta c^a + \bar{\eta} \bar{c}^a + O(\eta^2).} \quad (411)$$

Therefore

$$\boxed{\partial_i q^a = c^a \partial_i \eta + \bar{c}^a \partial_i \bar{\eta} + O(\eta \partial \eta).} \quad (412)$$

This is the precise place where the rigid-internal obstruction is removed.

CLXXXI. 3. COMPLETED J AND K CURRENTS WITH THE MINIMAL INTERNAL MODE

The working-branch completed currents are

$$J_i^a = \partial_i m^a, \quad K_i^a = (\partial_i q^a) \chi, \quad \chi = n^\mu D_\mu \Psi.$$

Write

$$\rho = |\phi|^2, \quad m^a = \rho q^a.$$

Using (411),

$$\boxed{m^a = \rho q_0^a + \rho(\eta c^a + \bar{\eta} \bar{c}^a) + O(\rho \eta^2).} \quad (413)$$

Hence

$$\boxed{J_i^a = q_0^a \partial_i \rho + (\partial_i \rho)(\eta c^a + \bar{\eta} \bar{c}^a) + \rho(c^a \partial_i \eta + \bar{c}^a \partial_i \bar{\eta}) + O(\eta^2, \eta \partial \eta).} \quad (414)$$

Likewise

$$\boxed{K_i^a = (c^a \partial_i \eta + \bar{c}^a \partial_i \bar{\eta}) \chi + O(\eta \partial \eta).} \quad (415)$$

So K_i^a is now generically nonzero as soon as

$$c^a \neq 0 \quad \text{and} \quad \partial_i \eta \neq 0.$$

CLXXXII. 4. GUARANTEED CHOICE OF A NONZERO INTERNAL TANGENT COEFFICIENT

Choose

$$\boxed{\xi_b := \frac{\Pi_\perp T^b z_0}{\|\Pi_\perp T^b z_0\|}, \quad \Pi_\perp := \mathbf{1} - z_0 z_0^\dagger,} \quad (416)$$

provided $\Pi_\perp T^b z_0 \neq 0$.

Then

$$\boxed{c_b^b = \xi_b^\dagger T^b z_0 = \|\Pi_\perp T^b z_0\| > 0.} \quad (417)$$

Therefore a minimal internal tangent mode can always be chosen so that at least one adjoint channel is guaranteed to couple.

CLXXXIII. 5. ORBIT-DIRECTION PERTURBATIONS

Now let the orbit perturbation in direction $r \in \{A, E, O\}$ be

$$\boxed{\psi_r = \varphi_r z_0 + \bar{\phi} \eta_r \xi.} \quad (418)$$

Then

$$\bar{\Psi} = \bar{\phi} z_0$$

at the rigid base point, and the density variation is

$$\boxed{N_r = \bar{\Psi}^\dagger \psi_r + \psi_r^\dagger \bar{\Psi} = \bar{\phi}^* \varphi_r + \varphi_r^* \bar{\phi},} \quad (419)$$

because the internal piece is orthogonal:

$$z_0^\dagger \xi = 0.$$

The adjoint-density variation is

$$\begin{aligned} M_r^a &= \bar{\Psi}^\dagger T^a \psi_r + \psi_r^\dagger T^a \bar{\Psi} \\ &= q_0^a N_r + \bar{\rho}(\eta_r c^a + \bar{\eta}_r \bar{c}^a). \end{aligned} \quad (420)$$

Hence

$$\boxed{M_r^a = q_0^a N_r + \bar{\rho}(\eta_r c^a + \bar{\eta}_r \bar{c}^a).} \quad (421)$$

Therefore

$$\boxed{Q_r^a = \frac{M_r^a - q_0^a N_r}{\bar{\rho}} = \eta_r c^a + \bar{\eta}_r \bar{c}^a.} \quad (422)$$

Now the completed orbit currents are

$$\boxed{J_{r,i}^a = \partial_i M_r^a = q_0^a \partial_i N_r + (\partial_i \bar{\rho})(\eta_r c^a + \bar{\eta}_r \bar{c}^a) + \bar{\rho}(c^a \partial_i \eta_r + \bar{c}^a \partial_i \bar{\eta}_r),} \quad (423)$$

and

$$\boxed{K_{r,i}^a = (\partial_i Q_r^a) \bar{\chi} = (c^a \partial_i \eta_r + \bar{c}^a \partial_i \bar{\eta}_r) \bar{\chi}.} \quad (424)$$

So the three pairs are

$$\boxed{(J_A, K_A) = \left(q_0^a \partial_i N_A + (\partial_i \bar{\rho})(\eta_A c^a + \bar{\eta}_A \bar{c}^a) + \bar{\rho}(c^a \partial_i \eta_A + \bar{c}^a \partial_i \bar{\eta}_A), (c^a \partial_i \eta_A + \bar{c}^a \partial_i \bar{\eta}_A) \bar{\chi} \right),} \quad (425)$$

$$\boxed{(J_E, K_E) = \left(q_0^a \partial_i N_E + (\partial_i \bar{\rho})(\eta_E c^a + \bar{\eta}_E \bar{c}^a) + \bar{\rho}(c^a \partial_i \eta_E + \bar{c}^a \partial_i \bar{\eta}_E), (c^a \partial_i \eta_E + \bar{c}^a \partial_i \bar{\eta}_E) \bar{\chi} \right),} \quad (426)$$

$$\boxed{(J_O, K_O) = \left(q_0^a \partial_i N_O + (\partial_i \bar{\rho})(\eta_O c^a + \bar{\eta}_O \bar{c}^a) + \bar{\rho}(c^a \partial_i \eta_O + \bar{c}^a \partial_i \bar{\eta}_O), (c^a \partial_i \eta_O + \bar{c}^a \partial_i \bar{\eta}_O) \bar{\chi} \right).} \quad (427)$$

CLXXXIV. 6. FULL REVIVAL OF THE SIX FAST COEFFICIENTS

Substitute these into the fast six-row formula:

$$H_{rs}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O [J_r K_s^\dagger + J_s K_r^\dagger + K_r J_s^\dagger + K_s J_r^\dagger].$$

Then

$$H_{rs}^{(2),\text{fast}} = -\frac{\alpha_X \beta_X}{M_X^2} \mathfrak{P}_O \left[\mathcal{J}_r^a (\mathcal{K}_s^a \bar{\chi})^\dagger + \mathcal{J}_s^a (\mathcal{K}_r^a \bar{\chi})^\dagger + (\mathcal{K}_r^a \bar{\chi}) \mathcal{J}_s^{a\dagger} + (\mathcal{K}_s^a \bar{\chi}) \mathcal{J}_r^{a\dagger} \right],$$

where

$$\mathcal{J}_r^a := q_0^a \partial_i N_r + (\partial_i \bar{\rho})(\eta_r c^a + \bar{\eta}_r \bar{c}^a) + \bar{\rho}(c^a \partial_i \eta_r + \bar{c}^a \partial_i \bar{\eta}_r), \quad (428)$$

$$\mathcal{K}_r^a := c^a \partial_i \eta_r + \bar{c}^a \partial_i \bar{\eta}_r. \quad (429)$$

Thus all six coefficients are now generically nonzero whenever the internal tangent mode is spatially varying and odd-projectable.

CLXXXV. 7. CORRECTED ODD TRANSVERSE COEFFICIENT

Therefore

$$Q_2(\delta_{\text{corr}}) = 4.608 \times 10^{-5} H_{AA}^{(2),\text{fast}} + 4.1472 \times 10^{-4} H_{EE}^{(2),\text{fast}} + 4.608 \times 10^{-5} H_{OO}^{(2),\text{fast}} \quad (430)$$

$$-2.7648 \times 10^{-4} H_{AE}^{(2),\text{fast}} + 9.216 \times 10^{-5} H_{AO}^{(2),\text{fast}} - 2.7648 \times 10^{-4} H_{EO}^{(2),\text{fast}}, \quad (431)$$

and

$$\beta^{\text{corr}} \approx -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa A_0 Q_2(\delta_{\text{corr}}). \quad (432)$$

So the transverse coefficient is revived by the minimal internal tangent texture.

CLXXXVI. 8. HONEST CONCLUSION

The previous vanishing result

$$K = 0, \quad H_{rs}^{(2),\text{fast}} = 0, \quad \beta^{\text{corr}} = 0$$

was not a no-go theorem for TECT. It was a theorem for the overly restrictive rigid-internal truncation.

Once the smallest non-rigid internal mode

$$\delta z = \eta \xi, \quad \xi \perp z_0$$

is admitted, the projector coordinate q^a varies,

$$\partial_i q^a \neq 0,$$

hence

$$K_i^a \neq 0,$$

and the six fast coefficients revive generically.

Therefore the correct minimal route to a nonzero transverse Class-II response is now fully identified.

CLXXXVII. FINAL SUMMARY

The rigid-internal truncation

$$\Psi = \phi z_0$$

forced

$$q^a = z_0^\dagger T^a z_0 = \text{constant}, \quad K_i^a = 0.$$

The smallest non-rigid internal deformation is

$$z = \sqrt{1 - |\eta|^2} z_0 + \eta \xi, \quad \xi^\dagger z_0 = 0.$$

With

$$c^a = \xi^\dagger T^a z_0,$$

the projector coordinate becomes

$$q^a = q_0^a + \eta c^a + \bar{\eta} \bar{c}^a + O(\eta^2),$$

so

$$\partial_i q^a = c^a \partial_i \eta + \bar{c}^a \partial_i \bar{\eta} + O(\eta \partial \eta).$$

Hence the completed Class-II current revives:

$$K_i^a = (\partial_i q^a) \chi = (c^a \partial_i \eta + \bar{c}^a \partial_i \bar{\eta}) \chi + O(\eta \partial \eta).$$

For orbit-direction perturbations

$$\psi_r = \varphi_r z_0 + \bar{\phi} \eta_r \xi, \quad r \in \{A, E, O\},$$

the completed orbit currents are

$$J_{r,i}^a = q_0^a \partial_i N_r + (\partial_i \bar{\rho})(\eta_r c^a + \bar{\eta}_r \bar{c}^a) + \bar{\rho}(c^a \partial_i \eta_r + \bar{c}^a \partial_i \bar{\eta}_r),$$

$$K_{r,i}^a = (c^a \partial_i \eta_r + \bar{c}^a \partial_i \bar{\eta}_r) \bar{\chi}.$$

Therefore the six fast coefficients

$$H_{AA}^{(2),\text{fast}}, H_{EE}^{(2),\text{fast}}, H_{OO}^{(2),\text{fast}}, H_{AE}^{(2),\text{fast}}, H_{AO}^{(2),\text{fast}}, H_{EO}^{(2),\text{fast}}$$

are no longer forced to vanish. They revive generically once the smallest internal tangent mode is included.

So the actual minimal microscopic source of the transverse Class-II response is now completely identified:

$$\text{a spatially varying internal tangent mode } \delta z = \eta \xi, \quad \xi \perp z_0.$$

CLXXXVIII. DIRECT COMPLETION OF THE TRANSVERSE SECTOR IN THE MINIMAL CANONICAL SPLIT

CLXXXIX. WHY WE CAN NOW STOP CHASING THE SIX-ROW ROUTE

The submission-ready canonical split already fixes

$$u_E = u_O = 0, \quad v_{\parallel} = 0, \tag{433}$$

so that the shell-relevant coefficients reduce to

$$c_{\parallel} = \alpha_X m_{\parallel} u_{\parallel}, \quad c_E = \beta_X \xi_E v_E, \quad c_O = \beta_X \xi_O v_O. \tag{434}$$

Hence the transverse coefficients are carried entirely by the K -sector:

$$\alpha = \beta_X \xi_E v_E I_{\theta}, \quad \beta = \beta_X \xi_O v_O I_{\theta}. \tag{435}$$

So the fastest mathematically correct route is:

$$\delta z \perp z_0 \implies K_i^a \neq 0 \implies (v_E, v_O) \implies (\alpha, \beta).$$

CXC. 1. MINIMAL NON-RIGID INTERNAL MODE

Take

$$z = z_0 + \eta \xi + O(\eta^2), \quad \xi^\dagger z_0 = 0, \quad \xi^\dagger \xi = 1. \quad (436)$$

Then

$$q^a = z^\dagger T^a z = q_0^a + \eta c^a + \bar{\eta} \bar{c}^a + O(\eta^2), \quad c^a := \xi^\dagger T^a z_0.$$

Therefore

$$K_i^a = (\partial_i q^a) \chi = (c^a \partial_i \eta + \bar{c}^a \partial_i \bar{\eta}) \chi + O(\eta \partial \eta). \quad (437)$$

Define the real tangent-adjoint coefficient

$$\Gamma_r^a := c^a \varepsilon_r + \bar{c}^a \bar{\varepsilon}_r = 2\Re(\varepsilon_r c^a). \quad (438)$$

CXCI. 2. PURE EVEN HARMONIC ANSATZ

Choose the smallest even transverse internal mode

$$\eta_E(x) = \varepsilon_E \varphi_E(x), \quad \eta_A = \eta_O = 0. \quad (439)$$

Then

$$K_{E,i}^a = \Gamma_E^a (\partial_i \varphi_E) \bar{\chi}. \quad (440)$$

Projecting onto the even tensor E^{ia} ,

$$v_E = \mathcal{N}_E \int d\xi f_+ f_- \theta'(\xi) \int d^2 y E^{ia} \Gamma_E^a \partial_i \varphi_E \bar{\chi}, \quad (441)$$

for some nonzero normalization factor $\mathcal{N}_E$.

Hence

$$\alpha = \beta_X \xi_E I_\theta v_E, \quad \beta = 0. \quad (442)$$

So a pure η_E harmonic produces an even transverse Dirac source and nothing odd.

CXCII. 3. PURE ODD HARMONIC ANSATZ

Choose the smallest odd transverse internal mode

$$\eta_O(x) = \varepsilon_O \varphi_O(x), \quad \eta_A = \eta_E = 0. \quad (443)$$

Then

$$K_{O,i}^a = \Gamma_O^a (\partial_i \varphi_O) \bar{\chi}. \quad (444)$$

Projecting onto the odd tensor J^{ia} ,

$$v_O = \mathcal{N}_O \int d\xi f_+ f_- \theta'(\xi) \int d^2 y J^{ia} \Gamma_O^a \partial_i \varphi_O \bar{\chi}, \quad (445)$$

for some nonzero normalization factor $\mathcal{N}_O$.

Hence

$$\beta = \beta_X \xi_O I_\theta v_O, \quad \alpha = 0. \quad (446)$$

So a pure η_O harmonic produces an odd transverse Dirac source and nothing even.

CXCIII. 4. NONZERO CRITERION

Since

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

the pure odd harmonic gives

$$\beta = -\frac{\sqrt{3}\pi}{16} \beta_X \xi_O \theta_0 \kappa v_O. \quad (447)$$

Therefore the exact nonzero criterion is

$$\beta \neq 0 \iff \beta_X \neq 0, \quad \theta_0 \kappa \neq 0, \quad \xi_O \neq 0, \quad v_O \neq 0. \quad (448)$$

A sufficient microscopic condition for $v_O \neq 0$ is

$$J^{ia} \Gamma_O^a \partial_i \varphi_O \neq 0 \quad \text{and} \quad \bar{\chi} \neq 0. \quad (449)$$

Likewise, for the even channel,

$$\alpha \neq 0 \iff \beta_X \neq 0, \quad \theta_0 \kappa \neq 0, \quad \xi_E \neq 0, \quad v_E \neq 0. \quad (450)$$

CXCIV. 5. SIMPLEST EXPLICIT HARMONIC CHOICE

To make the nonzero condition concrete, take

$$\varphi_O(y) = e^{ik \cdot y}, \quad \partial_i \varphi_O = ik_i e^{ik \cdot y}. \quad (451)$$

Then

$$v_O \propto i J^{ia} \Gamma_O^a k_i \int d\xi f_+ f_- \theta'(\xi) \bar{\chi} \int d^2 y e^{ik \cdot y}. \quad (452)$$

Thus, after the standard shell projection to the selected k -mode,

$$v_O \propto J^{ia} \Gamma_O^a k_i. \quad (453)$$

So a single projected odd plane wave already gives

$$\beta \propto -\frac{\sqrt{3}\pi}{16} \beta_X \xi_O \theta_0 \kappa (J^{ia} \Gamma_O^a k_i). \quad (454)$$

Exactly the same reasoning for the even mode yields

$$\alpha \propto -\frac{\sqrt{3}\pi}{16} \beta_X \xi_E \theta_0 \kappa (E^{ia} \Gamma_E^a k_i). \quad (455)$$

CXCV. 6. FINAL SHARP CONSEQUENCE

The rigid-internal three-orbit amplitude correction could never produce $K \neq 0$. But the smallest internal tangent harmonic does produce it.

Moreover, in the minimal canonical split, this already *completes the transverse sector directly*, without any need to continue through the slower six-row $H_{rs}^{(2)}$ route.

Thus the corrected transverse sector closes immediately once one adds the smallest non-rigid internal mode:

$$\eta_E \neq 0 \implies \alpha \neq 0, \quad \beta = 0, \quad (456)$$

$$\eta_O \neq 0 \implies \beta \neq 0, \quad \alpha = 0. \quad (457)$$

CXCVI. FINAL SUMMARY

In the minimal canonical split,

$$c_E = \beta_X \xi_E v_E, \quad c_O = \beta_X \xi_O v_O, \quad \alpha = c_E I_\theta, \quad \beta = c_O I_\theta.$$

So the transverse sector is carried entirely by the K -current.

The smallest non-rigid internal deformation is

$$z = z_0 + \eta \xi + O(\eta^2), \quad \xi^\dagger z_0 = 0,$$

which gives

$$q^a = q_0^a + \eta c^a + \bar{\eta} \bar{c}^a + O(\eta^2), \quad c^a = \xi^\dagger T^a z_0,$$

hence

$$\boxed{K_i^a = (c^a \partial_i \eta + \bar{c}^a \partial_i \bar{\eta}) \chi + O(\eta \partial \eta)}.$$

For a pure even harmonic

$$\eta_E = \varepsilon_E \varphi_E, \quad \eta_A = \eta_O = 0,$$

one gets

$$\alpha = \beta_X \xi_E I_\theta v_E, \quad \beta = 0.$$

For a pure odd harmonic

$$\eta_O = \varepsilon_O \varphi_O, \quad \eta_A = \eta_E = 0,$$

one gets

$$\beta = \beta_X \xi_O I_\theta v_O, \quad \alpha = 0.$$

Using

$$I_\theta = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa,$$

the odd coefficient is

$$\boxed{\beta = -\frac{\sqrt{3}\pi}{16} \beta_X \xi_O \theta_0 \kappa v_O}.$$

So the exact nonzero criterion is

$$\boxed{\beta \neq 0 \iff \beta_X \neq 0, \quad \theta_0 \kappa \neq 0, \quad \xi_O \neq 0, \quad v_O \neq 0}.$$

Thus the transverse sector is completed by the smallest internal tangent harmonic. The rigid-internal obstruction is fully removed.

CXCVII. FINAL MINIMAL PDE ANSATZ CARRYING $\lambda_\parallel$, α , AND β SIMULTANEOUSLY

CXCVIII. WHERE THE PROJECT NOW STANDS

At the current strongest explicit working-branch level, the remaining microscopic task is no longer to invent a new mechanism. It is to combine:

1. the Branch-B longitudinal primitive seed,
2. the smallest non-rigid internal even mode,
3. the smallest non-rigid internal odd mode,

into one minimal computable PDE ansatz.

The notes already fix the microscopic routing:

$$OO \rightarrow \lambda_c, \quad TD + DT \rightarrow (\Lambda_E, \Lambda_O),$$

and the reduced Dirac-type coefficient point is read from these transport overlaps.

CXCIX. 1. MINIMAL FIELD CONTENT

The smallest field content that can carry all three reduced coefficients at once is:

$$\boxed{\Psi(x) = \phi(x) z(x), \quad z^\dagger z = 1, \quad P = z z^\dagger, \quad X_i^a.} \quad (458)$$

We keep the same Class-II mediator X_i^a , but the internal unit vector z is no longer rigid.

CC. 2. MINIMAL INTERNAL COMPLETION

Choose an orthonormal internal frame

$$z_0, \quad \xi_E, \quad \xi_O, \quad z_0^\dagger \xi_E = z_0^\dagger \xi_O = \xi_E^\dagger \xi_O = 0.$$

Then the smallest normalized internal deformation carrying both even and odd transverse modes is

$$\boxed{z(x) = \sqrt{1 - |\eta_E(x)|^2 - |\eta_O(x)|^2} z_0 + \eta_E(x) \xi_E + \eta_O(x) \xi_O.} \quad (459)$$

To first order in the internal amplitudes,

$$\boxed{z = z_0 + \eta_E \xi_E + \eta_O \xi_O + O(\eta^2).} \quad (460)$$

Define the background adjoint charges

$$\boxed{q_0^a := z_0^\dagger T^a z_0, \quad c_E^a := \xi_E^\dagger T^a z_0, \quad c_O^a := \xi_O^\dagger T^a z_0.} \quad (461)$$

Then

$$\boxed{q^a := z^\dagger T^a z = q_0^a + \eta_E c_E^a + \bar{\eta}_E \bar{c}_E^a + \eta_O c_O^a + \bar{\eta}_O \bar{c}_O^a + O(\eta^2).} \quad (462)$$

Therefore

$$\boxed{\partial_i q^a = c_E^a \partial_i \eta_E + \bar{c}_E^a \partial_i \bar{\eta}_E + c_O^a \partial_i \eta_O + \bar{c}_O^a \partial_i \bar{\eta}_O + O(\eta \partial \eta).} \quad (463)$$

CCI. 3. MINIMAL WORKING MICROSCOPIC PDE

The strongest explicit working branch is

$$\boxed{\mathcal{L}_{\text{mic}} = \mathcal{L}_{\Psi}^{\text{ref}}[\Psi, P] + \frac{M_X^2}{2} X_i^a X_i^a + \alpha_X X_i^a J_i^a[\Psi] + \beta_X X_i^a K_i^a[P, \Psi].} \quad (464)$$

Inside this branch, the completed explicit currents are

$$\boxed{J_i^a = \partial_i m^a, \quad m^a := \Psi^\dagger T^a \Psi,} \quad (465)$$

and

$$\boxed{K_i^a = (\partial_i q^a) \chi, \quad \chi := n^\mu D_\mu \Psi.} \quad (466)$$

Using $\Psi = \phi z$, $\rho = |\phi|^2$, $m^a = \rho q^a$, one gets

$$\boxed{J_i^a = q^a \partial_i \rho + \rho \partial_i q^a.} \quad (467)$$

Substituting (462) and (463), the completed leading-order currents become

$$\boxed{J_i^a = q_0^a \partial_i \rho + (\partial_i \rho) \left(\eta_E c_E^a + \bar{\eta}_E \bar{c}_E^a + \eta_O c_O^a + \bar{\eta}_O \bar{c}_O^a \right) + \rho \left(c_E^a \partial_i \eta_E + \bar{c}_E^a \partial_i \bar{\eta}_E + c_O^a \partial_i \eta_O + \bar{c}_O^a \partial_i \bar{\eta}_O \right) + O(\eta^2, \eta \partial \eta),} \quad (468)$$

and

$$\boxed{K_i^a = \left(c_E^a \partial_i \eta_E + \bar{c}_E^a \partial_i \bar{\eta}_E + c_O^a \partial_i \eta_O + \bar{c}_O^a \partial_i \bar{\eta}_O \right) \chi + O(\eta \partial \eta).} \quad (469)$$

This is the minimal completed PDE-level current system carrying all three reduced channels.

CCII. 4. HEAVY-MEDIATOR ELIMINATION

The heavy field is algebraic:

$$X_i^a = -\frac{1}{M_X^2} (\alpha_X J_i^a + \beta_X K_i^a). \quad (470)$$

Hence the tree-level effective interaction is

$$\mathcal{L}_{\text{eff}}^{(X)} = -\frac{1}{2M_X^2} \sum_{i,a} (\alpha_X J_i^a + \beta_X K_i^a)^2. \quad (471)$$

This is now the actual minimal PDE one should solve.

CCIII. 5. LONGITUDINAL / TRANSVERSE SPLIT

The longitudinal seed is still the Branch-B primitive one:

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta^3} \lambda_c. \quad (472)$$

The notes fix the transport routing:

$$OO \rightarrow \lambda_c, \quad TD + DT \rightarrow (\Lambda_E, \Lambda_O).$$

Thus the reduced coefficients are

$$\alpha = I_{\theta} \Lambda_E, \quad \beta = I_{\theta} \Lambda_O, \quad (473)$$

with

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (474)$$

In the minimal canonical split, transverse transport is carried entirely by K :

$$\Lambda_E = \beta_X \xi_E v_E, \quad \Lambda_O = \beta_X \xi_O v_O, \quad (475)$$

so that

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \xi_E \theta_0 \kappa v_E, \quad (476)$$

$$\beta = -\frac{\sqrt{3}\pi}{16} \beta_X \xi_O \theta_0 \kappa v_O. \quad (477)$$

CCIV. 6. SMALLEST HARMONIC ANSATZ THAT ALREADY WORKS

The smallest harmonic completion is:

$$\eta_E(x) = \varepsilon_E \varphi_E(x), \quad \eta_O(x) = \varepsilon_O \varphi_O(x), \quad (478)$$

with φ_E, φ_O shell/worldvolume harmonics.

Then

$$v_E \propto E^{ia} (c_E^a \partial_i \eta_E + \bar{c}_E^a \partial_i \bar{\eta}_E) \bar{\chi}, \quad (479)$$

$$v_O \propto J^{ia} (c_O^a \partial_i \eta_O + \bar{c}_O^a \partial_i \bar{\eta}_O) \bar{\chi}. \quad (480)$$

Therefore:

$$\eta_E \neq 0 \implies \alpha \neq 0 \text{ generically}, \quad (481)$$

$$\eta_O \neq 0 \implies \beta \neq 0 \text{ generically}. \quad (482)$$

CCV. 7. FINAL MINIMAL PDE THAT ONE CAN NOW SOLVE

Collecting everything, the minimal explicit PDE system is:

$$\Psi = \phi z, \quad z = \sqrt{1 - |\eta_E|^2 - |\eta_O|^2} z_0 + \eta_E \xi_E + \eta_O \xi_O, \quad (483)$$

$$J_i^a = q_0^a \partial_i \rho + (\partial_i \rho) \left(\eta_E c_E^a + \bar{\eta}_E \bar{c}_E^a + \eta_O c_O^a + \bar{\eta}_O \bar{c}_O^a \right) + \rho \left(c_E^a \partial_i \eta_E + \bar{c}_E^a \partial_i \bar{\eta}_E + c_O^a \partial_i \eta_O + \bar{c}_O^a \partial_i \bar{\eta}_O \right), \quad (484)$$

$$K_i^a = \left(c_E^a \partial_i \eta_E + \bar{c}_E^a \partial_i \bar{\eta}_E + c_O^a \partial_i \eta_O + \bar{c}_O^a \partial_i \bar{\eta}_O \right) \chi, \quad \chi = n^\mu D_\mu(\phi z), \quad (485)$$

$$X_i^a = -\frac{1}{M_X^2} (\alpha_X J_i^a + \beta_X K_i^a), \quad (486)$$

$$\mathcal{L}_{\text{eff}}^{(X)} = -\frac{1}{2M_X^2} \sum_{i,a} (\alpha_X J_i^a + \beta_X K_i^a)^2, \quad (487)$$

together with the reduced coefficient sheet

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \lambda_c, \quad \alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \xi_E \theta_0 \kappa v_E, \quad \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \xi_O \theta_0 \kappa v_O. \quad (488)$$

This is the final minimal PDE ansatz that simultaneously carries the longitudinal seed and both transverse channels.

CCVI. FINAL SUMMARY

Very little remains at the strongest explicit working-branch level.

The rigid-internal truncation killed the transverse sector because q^a was constant and hence $K_i^a = 0$.

The smallest successful completion is the internal tangent ansatz

$$z = \sqrt{1 - |\eta_E|^2 - |\eta_O|^2} z_0 + \eta_E \xi_E + \eta_O \xi_O,$$

which yields

$$q^a = q_0^a + \eta_E c_E^a + \bar{\eta}_E \bar{c}_E^a + \eta_O c_O^a + \bar{\eta}_O \bar{c}_O^a,$$

and therefore

$$K_i^a = \left(c_E^a \partial_i \eta_E + \bar{c}_E^a \partial_i \bar{\eta}_E + c_O^a \partial_i \eta_O + \bar{c}_O^a \partial_i \bar{\eta}_O \right) \chi.$$

This revives the transverse sector directly.

The final minimal reduced coefficients are

$$\lambda_{\parallel} = \frac{1}{2} I_{\theta'^3} \lambda_c,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \xi_E \theta_0 \kappa v_E, \quad \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \xi_O \theta_0 \kappa v_O.$$

So the minimal PDE that must now be solved is:

$$\Psi = \phi z, \quad J_i^a = \partial_i (\Psi^\dagger T^a \Psi), \quad K_i^a = (\partial_i q^a) \chi, \quad X_i^a = -\frac{1}{M_X^2} (\alpha_X J_i^a + \beta_X K_i^a),$$

with the internal mode completion (η_E, η_O) .

At this point, the strongest explicit working-branch program is structurally closed. What remains is actual numerical or symbolic insertion of the overlap scalars

$$\lambda_c, \quad v_E, \quad v_O.$$

CCVII. FINAL ONE-LINE CRITERION FOR THE WORKING-BRANCH DIRAC/WEYL WITNESS

CCVIII. 1. GENERAL CANONICAL COEFFICIENT SHEET

At the current strongest explicit working-branch level, the canonical reduced coefficients are

$$\lambda_{\parallel} = \frac{1}{2} (\alpha_X m_{\parallel} u_{\parallel} + \beta_X \xi_{\parallel} v_{\parallel}) I_{\theta'^3}, \quad (489)$$

$$\alpha = (\alpha_X m_{\parallel} u_E + \beta_X \xi_E v_E) I_{\theta}, \quad \beta = (\alpha_X m_{\parallel} u_O + \beta_X \xi_O v_O) I_{\theta}. \quad (490)$$

The exact overlap factors are

$$I_{\theta} = -\frac{\sqrt{3}\pi}{16} \theta_0 \kappa, \quad I_{\theta'^3} = -\frac{3\sqrt{3}\pi}{256} \theta_0^3 \kappa^3. \quad (491)$$

CCIX. 2. MINIMAL CANONICAL SPLIT

In the minimal canonical split,

$$u_E = u_O = 0, \quad v_{\parallel} = 0,$$

so the coefficient sheet simplifies to

$$\lambda_{\parallel} = \frac{1}{2} \alpha_X m_{\parallel} u_{\parallel} I_{\theta'^3}, \quad \alpha = \beta_X \xi_E v_E I_{\theta}, \quad \beta = \beta_X \xi_O v_O I_{\theta}. \quad (492)$$

Substituting (491),

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512} \alpha_X m_{\parallel} u_{\parallel} \theta_0^3 \kappa^3, \quad (493)$$

$$\alpha = -\frac{\sqrt{3}\pi}{16} \beta_X \xi_E v_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16} \beta_X \xi_O v_O \theta_0 \kappa. \quad (494)$$

CCX. 3. FINAL YES/NO CRITERION

The exact graded nondegeneracy criterion is

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0).$$

In the minimal canonical split this becomes

$$\boxed{\theta_0 \kappa \neq 0, \quad m_{\parallel} u_{\parallel} \neq 0, \quad (\xi_E v_E, \xi_O v_O) \neq (0, 0).} \quad (495)$$

Equivalently, in the transport notation of the full shell witness,

$$\boxed{\lambda_c \neq 0, \quad (\Lambda_E, \Lambda_O) \neq (0, 0).} \quad (496)$$

CCXI. 4. SIGN CRITERION

From (493) and (494), the signs are

$$\boxed{\text{sgn } \lambda_{\parallel} = -\text{sgn}(\alpha_X m_{\parallel} u_{\parallel} \theta_0^3 \kappa^3),} \quad (497)$$

$$\boxed{\text{sgn } \alpha = -\text{sgn}(\beta_X \xi_E v_E \theta_0 \kappa), \quad \text{sgn } \beta = -\text{sgn}(\beta_X \xi_O v_O \theta_0 \kappa).} \quad (498)$$

Thus the sign problem is completely reduced to the microscopic overlap signs

$$m_{\parallel} u_{\parallel}, \quad \xi_E v_E, \quad \xi_O v_O.$$

CCXII. 5. EXPLICIT PURE-EVEN CANONICAL BRANCH

On the explicit pure-even canonical branch,

$$\gamma_E = 1, \quad \gamma_O = 0, \quad C_{\parallel} = C_E = C_O = 1,$$

the practical coefficient sheet is

$$\boxed{\lambda_{\parallel} = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad \alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad \beta = 0.} \quad (499)$$

Hence

$$\boxed{\alpha \neq 0 \iff \beta_X \theta_0 \kappa \neq 0,} \quad (500)$$

and, since $M_X^2 > 0$,

$$\boxed{\alpha > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0.} \quad (501)$$

CCXIII. 6. FAST PRACTICAL READING

So the fastest practical test is:

$$\boxed{\text{Longitudinal pass} \iff \alpha_X m_{\parallel} u_{\parallel} \theta_0^3 \kappa^3 \neq 0,} \quad (502)$$

$$\boxed{\text{Transverse pass} \iff \beta_X \theta_0 \kappa \neq 0 \quad \text{and} \quad (\xi_E v_E, \xi_O v_O) \neq (0, 0).} \quad (503)$$

Equivalently,

$$\boxed{\text{Full local Dirac/Weyl pass} \iff \theta_0 \kappa \neq 0, \quad m_{\parallel} u_{\parallel} \neq 0, \quad (\xi_E v_E, \xi_O v_O) \neq (0, 0).} \quad (504)$$

CCXIV. FINAL SUMMARY

At the current strongest explicit working-branch level, the reduced coefficient sheet is already closed. In the minimal canonical split it is

$$\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512}\alpha_X m_{\parallel} u_{\parallel} \theta_0^3 \kappa^3,$$

$$\alpha = -\frac{\sqrt{3}\pi}{16}\beta_X \xi_E v_E \theta_0 \kappa, \quad \beta = -\frac{\sqrt{3}\pi}{16}\beta_X \xi_O v_O \theta_0 \kappa.$$

Therefore the final yes/no criterion is

$$\boxed{\theta_0 \kappa \neq 0, \quad m_{\parallel} u_{\parallel} \neq 0, \quad (\xi_E v_E, \xi_O v_O) \neq (0, 0).}$$

The sign criterion is

$$\boxed{\operatorname{sgn} \lambda_{\parallel} = -\operatorname{sgn}(\alpha_X m_{\parallel} u_{\parallel} \theta_0^3 \kappa^3),}$$

$$\boxed{\operatorname{sgn} \alpha = -\operatorname{sgn}(\beta_X \xi_E v_E \theta_0 \kappa), \quad \operatorname{sgn} \beta = -\operatorname{sgn}(\beta_X \xi_O v_O \theta_0 \kappa).}$$

On the explicit pure-even canonical branch, this reduces further to

$$\boxed{\alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad \beta = 0,}$$

so

$$\boxed{\alpha > 0 \iff \frac{\beta_X}{M_X^2} \theta_0 \kappa > 0.}$$

Thus, within the current strongest explicit working branch, the whole problem is reduced to the microscopic overlap data

$$m_{\parallel} u_{\parallel}, \quad \xi_E v_E, \quad \xi_O v_O.$$

No structural ambiguity remains in the reduced PDE.

CCXV. STOP THE SIGN AUDIT AND MOVE TO THE ACTUAL REMAINING OBSTRUCTION

CCXVI. 1. WHY THE SIGN AUDIT IS NO LONGER THE BOTTLENECK

At the current strongest explicit working-branch level, the carrier-side witness problem is already effectively closed. In the explicit pure-even canonical branch, the notes already give

$$\lambda_{\parallel} \neq 0, \quad \alpha \neq 0, \quad \beta = 0, \quad \det M \neq 0,$$

under the explicit witness conditions. So the existence of a nondegenerate local reduced carrier is no longer the true frontier.

Therefore the sign of the coefficients is not the bottleneck for moving forward. For carrier existence, only

$$\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0)$$

matters. The sign only matters later for branch orientation, constitutive matching, or one-defect accept/reject conventions.

CCXVII. 2. WHAT IS ALREADY CLOSED

The explicit pure-even canonical coefficient sheet is already given as

$$\lambda_{\parallel} = \frac{3\sqrt{3}\pi}{512} \frac{\alpha_X \beta_X}{M_X^2} \theta_0^3 \kappa^3, \quad \alpha = \frac{\sqrt{3}\pi}{16} \frac{\beta_X}{M_X^2} \theta_0 \kappa, \quad \beta = 0.$$

Hence the completed practical working-branch infrared PDE is already fixed:

$$\left[i\partial_t - \lambda_{\parallel} \sigma_3(-i\partial_{\parallel}) - \alpha(\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) - \beta(-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1)) \right] \chi = 0.$$

So the carrier-side working branch is not what is preventing progress.

CCXVIII. 3. THE ACTUAL REMAINING OBSTRUCTION

The corrected backend residual stage fixes

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096),$$

and the residual-based projection extractor has already been run.

The notes explicitly state that the remaining obstruction is no longer branch search, but the failure of the current quartic reduced kernel closure.

Indeed, even after the corrected local branch refinement, the simple quartic shell kernel

$$(u, v, \kappa)$$

fails strongly, with

$$\Delta_{\text{ker}} = 32976.132300215555.$$

Therefore the actual frontier is:

replace the failed (u, v, κ) truncation by the next enlarged quartic reduced shell kernel.

CCXIX. 4. IMMEDIATE NEXT REDUCED OBJECT

The correct reduced object is not a naive reduced free energy, but the projected residual map

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix}, \quad \Psi(\delta) = \Psi_{\text{BCC}} + \delta_A \Phi_A + \delta_E \Phi_E + \delta_O \Phi_O.$$

Thus the right next task is not another sign audit, but:

$$U_{\alpha\beta}^{\text{failed}}(u, v, \kappa) \longrightarrow U_{\alpha\beta}^{\text{enlarged}}$$

followed by

$$\Psi_{\text{corr}} \implies U_{\alpha\beta}^{\text{enlarged}} \implies \text{corrected projected coefficients.}$$

CCXX. 5. FASTEST MATHEMATICALLY CORRECT MOVE

The fastest next move is to adopt the four-class exact shell kernel

$$U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_{\perp} P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})},$$

instead of forcing the incorrect three-parameter compression.

This is the smallest enlarged quartic shell kernel that can represent the extracted corrected class means

$$(U_d, U_{\text{opp}}, U_{\perp}, U_{\text{obl}})$$

without imposing the false constraint built into the (u, v, κ) ansatz.

CCXXI. 6. CORRECT STATUS STATEMENT

So the correct research status is:

carrier-side working-branch closure: effectively done,

corrected residual quartic closure: still open,

and therefore

the next real step is enlarged quartic residual closure, not more sign polishing.

CCXXII. FINAL SUMMARY

The sign audit is not the bottleneck anymore.

At the strongest explicit working-branch level, the carrier-side witness is already effectively closed:

$$\lambda_{\parallel} \neq 0, \quad \alpha \neq 0, \quad \beta = 0, \quad \det M \neq 0$$

on the explicit pure-even canonical branch.

So the project is not blocked by the carrier sign problem.

The actual remaining obstruction is the corrected residual quartic closure: even after freezing

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096),$$

the simple quartic shell kernel (u, v, κ) still fails strongly.

Therefore the correct next frontier is

construct the next enlarged quartic reduced shell kernel beyond (u, v, κ) ,

and rerun the corrected residual projection on that enlarged basis.

So the working-branch carrier problem is no longer what is stopping progress. The real unresolved problem is the enlarged residual quartic kernel closure.

CCXXIII. PROJECT: TECT — MASTER CONTENTS FOR CONTINUATION (CORRECTED RESIDUAL QUARTIC CLOSURE STAGE)

CCXXIV. CURRENT STAGE

The project has now reached the following precise stage:

carrier-side working-branch admissibility is effectively closed, (505)

while the actual remaining frontier is the corrected residual quartic kernel closure. (506)

Equivalently,

working-branch carrier witness: closed enough to move on, corrected residual quartic closure: open. (507)

The next conversation must therefore *not* restart sign-polishing or broad branch searches. It must begin directly from the enlarged quartic reduced shell kernel problem.

CCXXV. FROZEN OPERATIONAL BACKEND

The operational backend point is frozen as

$$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62). \quad (508)$$

This point should be treated as fixed unless a theorem-level reason appears to change it.

CCXXVI. CORRECT REDUCED OBJECT

The correct reduced object is the projected residual map, not a naive reduced free-energy critical point. Let

$$\Psi(\delta) = \Psi_{\text{BCC}} + \delta_A \Phi_A + \delta_E \Phi_E + \delta_O \Phi_O, \quad (509)$$

and define

$$G(\delta) = \begin{pmatrix} G_A(\delta) \\ G_E(\delta) \\ G_O(\delta) \end{pmatrix} := \begin{pmatrix} \Re \langle \Phi_A, R[\Psi(\delta)] \rangle_\Omega \\ \Re \langle \Phi_E, R[\Psi(\delta)] \rangle_\Omega \\ \Re \langle \Phi_O, R[\Psi(\delta)] \rangle_\Omega \end{pmatrix}. \quad (510)$$

This residual-based reduced object is primary.

CCXXVII. FINAL CORRECTED LOCAL SEED

The final geometry-preserving corrected seed is frozen as

$$\delta^{\text{final}} = (0.0096, -0.0288, 0.0096). \quad (511)$$

So the corrected local branch is operationally defined by

$$\Psi_{\text{corr}} = \Psi_{\text{BCC}} + 0.0096 \Phi_A - 0.0288 \Phi_E + 0.0096 \Phi_O. \quad (512)$$

The corresponding corrected diagnostics are

$$\|G(\delta^{\text{final}})\| \approx 0.03791852545979427, \quad (513)$$

$$\text{rel_residual}(\delta^{\text{final}}) \approx 0.036459908397927156, \quad (514)$$

$$\text{shell_fraction}(\delta^{\text{final}}) \approx 0.9156153162713265, \quad \text{orbit_balance}(\delta^{\text{final}}) \approx 0.8860846137625883. \quad (515)$$

CCXXVIII. PROJECTION EXTRACTOR STATUS

A residual-based projection extractor has already been implemented and executed. Its extraction chain is

$$\Psi \implies h_\alpha \implies (\sigma_0, \eta), \quad (516)$$

$$\Psi \implies U_{\alpha\beta} \implies (U_d, U_{\text{opp}}, U_\perp, U_{\text{obl}}), \quad (517)$$

and optionally

$$(U_d, U_{\text{opp}}, U_\perp, U_{\text{obl}}) \implies (u, v, \kappa, \Delta_{\text{ker}}). \quad (518)$$

These quantities must be interpreted honestly as residual-based effective reduced coefficients of the current solver family, not yet theorem-level coefficients of a fully closed microscopic energy functional.

CCXXIX. CORRECTED PROJECTION RESULTS

After inserting δ^{final} , the corrected projection gives

$$\sigma_0 = -109.41971030446824, \quad \eta = -6.502583100163014. \quad (519)$$

Orbit means:

$$\sigma_A = -115.81015796064617, \quad \sigma_E = -105.83201242243113, \quad \sigma_O = -108.719022860933. \quad (520)$$

Quartic class averages:

$$U_d = 2493.1087739398513, \quad U_{\text{opp}} = 14874.171297437068, \quad U_{\perp} = 9027.17290917827, \quad U_{\text{obl}} = 2244.8894311890813. \quad (521)$$

If these are compressed to the simple quartic kernel

$$U_{\alpha\beta} = u\delta_{\alpha\beta} + v + \kappa(\hat{k}_{\alpha} \cdot \hat{k}_{\beta})^2,$$

then one gets

$$u = -12381.062523497216, \quad v = 9027.17290917827, \quad \kappa = -27129.133911956756, \quad (522)$$

but the exact consistency defect becomes

$$\Delta_{\text{ker}} = 32976.132300215555. \quad (523)$$

Therefore the simple quartic (u, v, κ) truncation fails strongly even on the corrected local branch.

CCXXX. EXACT MEANING OF THE CURRENT OBSTRUCTION

The (u, v, κ) kernel predicts the consistency relation

$$U_{\text{opp}} = 4U_{\text{obl}} - 3U_{\perp}. \quad (524)$$

Hence the exact truncation defect is

$$\Delta_{\text{ker}} = U_{\text{opp}} - 4U_{\text{obl}} + 3U_{\perp}. \quad (525)$$

If $\Delta_{\text{ker}} = 0$, the quartic shell matrix is exactly representable by the (u, v, κ) kernel. If $\Delta_{\text{ker}} \neq 0$, the quartic reduced kernel basis must be enlarged.

Thus the next step is not another backend search. It is the construction of the next enlarged quartic reduced shell kernel.

CCXXXI. WHAT IS ALREADY EFFECTIVELY CLOSED

1. The broad search for the operational backend point is finished.
2. A best confirmed actual BCC branch exists numerically.
3. A corrected local seed preserving shell geometry has been extracted.
4. The reduced free-energy closure route is ruled out as the correct microscopic reduction.
5. The reduced residual route is identified as the correct reduced object.
6. The simple quartic (u, v, κ) truncation is decisively falsified.
7. The strongest explicit working-branch carrier witness is already sufficient to move on.

CCXXXII. WORKING-BRANCH CARRIER STATUS (DO NOT REOPEN UNLESS NEEDED)

On the explicit canonical witness branch, the notes already provide a nondegenerate carrier witness with

$$\boxed{\lambda_{\parallel} \neq 0, \quad (\alpha, \beta) \neq (0, 0), \quad \det M \neq 0,} \quad (526)$$

and the isolated doubled shell block has principal form

$$\boxed{H_D^{\text{pr}}(p) = v_{\perp} p_1' \Gamma^1 + v_{\perp} p_2' \Gamma^2 + \lambda_{\parallel} p_{\parallel} \Gamma^3, \quad v_{\perp} = \sqrt{\alpha^2 + \beta^2} > 0.} \quad (527)$$

This means the working-branch carrier problem is no longer the actual frontier. Do not spend the next conversation polishing signs unless a new contradiction appears.

CCXXXIII. WHAT REMAINS OPEN

1. The quartic reduced kernel basis must be enlarged beyond the simple (u, v, κ) form.
2. The next reduced closure must be built directly at the residual level, not from a naive reduced free-energy reconstruction.
3. After enlarging the quartic reduced basis, the corrected branch projection must be rerun on that enlarged basis.

CCXXXIV. IMMEDIATE NEXT TASK

The next conversation must begin from the following precise objective:

$$\boxed{\text{construct the next enlarged quartic reduced basis beyond } (u, v, \kappa),} \quad (528)$$

while keeping fixed:

$$\boxed{(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096).} \quad (529)$$

Equivalently:

$$\boxed{\text{replace the failed simple quartic kernel by the next exact/enlarged reduced shell kernel,}} \quad (530)$$

and then rerun the corrected projection on that enlarged basis.

CCXXXV. MINIMAL RECOMMENDED ENLARGEMENT

The fastest mathematically correct next step is the four-class exact shell kernel

$$\boxed{U_{\alpha\beta}^{(4c)} = u_d P_{\alpha\beta}^{(d)} + u_{\text{opp}} P_{\alpha\beta}^{(\text{opp})} + u_{\perp} P_{\alpha\beta}^{(\perp)} + u_{\text{obl}} P_{\alpha\beta}^{(\text{obl})},} \quad (531)$$

which directly represents the extracted corrected quartic class means

$$(U_d, U_{\text{opp}}, U_{\perp}, U_{\text{obl}})$$

without imposing the false (u, v, κ) consistency relation.

CCXXXVI. STRICT WORKING RULES FOR THE NEXT CONVERSATION

1. Do not reopen broad parameter scans over (μ^2, λ, γ) .
2. Keep the current operational backend point fixed unless a theorem-level reason appears to change it.
3. Keep the corrected seed δ^{final} fixed as the current corrected local branch.
4. Treat the residual-based reduced object as primary, not a naive reduced free-energy critical point.
5. Focus only on enlarging the quartic reduced kernel basis and extracting its corrected coefficients.
6. Do not return to carrier sign-polishing unless a direct contradiction appears in the enlarged residual system.

CCXXXVII. CURRENT STATUS / GOAL STACK

Current status: corrected residual quartic closure stage.

 (532)

Goal C: complete the corrected residual branch by an enlarged quartic shell kernel closure.

 (533)

Goal M: rerun the corrected projection on the four-class (or larger) exact shell kernel basis.

 (534)

Goal S: write the enlarged residual system directly in $(U_d, U_{\text{opp}}, U_{\perp}, U_{\text{obl}})$ instead of (u, v, κ) .

 (535)

CCXXXVIII. ONE-LINE CONTINUATION PROMPT

Start from the frozen backend $(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62)$ and corrected seed $\delta^{\text{final}} = (0.0096, -0.0288, 0.0096)$,

 (536)

then construct the next enlarged quartic reduced shell kernel beyond the failed (u, v, κ) truncation.

 (537)

$(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62), \quad \delta^{\text{final}} = (0.0096, -0.0288, 0.0096),$

(u, v, κ) truncation is decisively dead ($\Delta_{\text{ker}} = 32976.132300215555$).

Next: construct the four-class exact shell kernel directly from $(U_d, U_{\text{opp}}, U_{\perp}, U_{\text{obl}})$ and rerun the corrected residual projection.